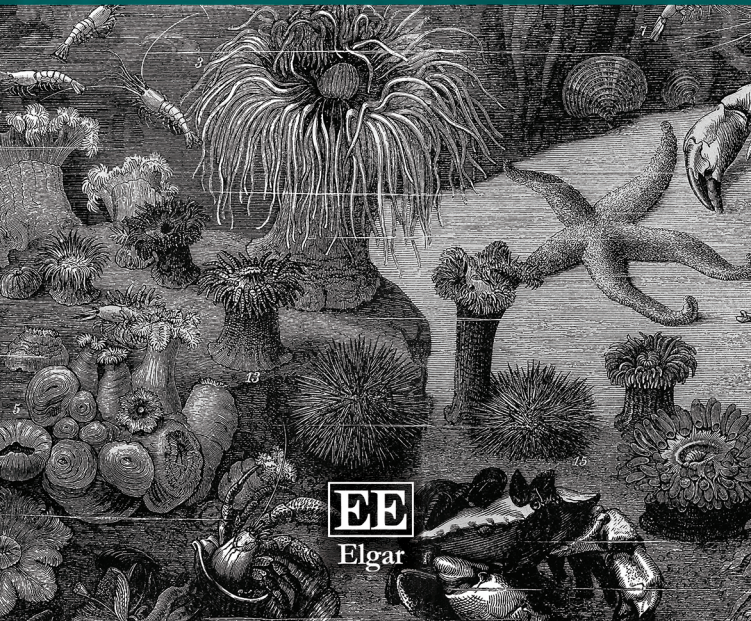
ENTREPRENEURSHIP FOOTPRINTS

ENTREPRENEURIAL ECOSYSTEMS

Theory, Practice and Futures

Ben Spigel



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Entrepreneurial Ecosystems

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Entrepreneurial Ecosystems

Theory, Practice and Futures

Ben Spigel

Chancellor's Fellow and Senior Lecturer in Entrepreneurship, University of Edinburgh Business School, UK

ENTREPRENEURSHIP FOOTPRINTS

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1. Introduction

1.1 ENTREPRENEURIAL ECOSYSTEMS AND ENTREPRENEURS

The central logic underlying what entrepreneurial ecosystems are and why they matter is that entrepreneurship is a team sport. We sometimes think of some founders, like Thomas Edison of General Electric, Steve Jobs of Apple, or Sara Blakely of Spanks as heroic geniuses who create something from nothing. But examining the history of any company shows the importance of a much wider network of supporters that were instrumental in helping entrepreneurs scale their new ideas into world-leading companies. These supporters include founders and early employees but also investors, government officials, early customers, mentors, and others who helped open doors that would have otherwise been closed. Interestingly, these crucial supporters tend to be located near to the founders and were often known by them before they started the venture. This proximity means that an entrepreneur is affected by their environment, which may help or hinder their entrepreneurial journey.

Entrepreneurial ecosystems do two things for entrepreneurs. First, they provide entrepreneurial resources. The various people and institutions in a place make available material resources like investment capital, skilled workers, and entrepreneurial knowledge that are more easily accessible to entrepreneurs than resources in other, more distant regions. These resources are crucial for ambitious entrepreneurs to create and grow innovative firms.

Second, entrepreneurial ecosystems create an environment in which these resources can be accessed. The culture of a place, the strength and density of social networks within it, the public policies that exist there, and its economic structure all influence not just the resources available but the extent to which entrepreneurs are able to gain access to them. Resources in ecosystems are not free-flowing but depend on the willingness of those within a community to provide them, either because they think they can profit from a transaction (for example, an investor) or because they think it is the right thing to do (for example, an entrepreneurial mentor).

Entrepreneurial ecosystems therefore provide a framework to understand the activities surrounding high-growth, high-ambition entrepreneurship. This brings together a number of strands on entrepreneurship, ranging from startup investment to entrepreneurial learning, to the geography of entrepreneurship. Just as importantly, it allows new perspectives about how entrepreneurs interact with their local economy and society and how this presents a path to encourage and support high-growth entrepreneurship. However, before this can be done it is important to develop a common view of what entrepreneurial ecosystems are and why they matter.

1.2 DEFINING ENTREPRENEURIAL ECOSYSTEMS

Without a proper definition of what an ecosystem is — and more importantly, what it is not — it is impossible to build a cohesive research community around the topic. Given that entrepreneurial ecosystems are a pre-paradigmatic field, there is not a single accepted definition of ecosystems. As shown in Table 1.1, different writers have used many different definitions. Some use terms besides entrepreneurial ecosystems, for example, Qian et al. (2013) discuss ‘regional systems of entrepreneurship’ but they imply the same phenomenon.

Table 1.1 Popular definitions of entrepreneurial ecosystems

Paper	Definition
Acs et al. 2017	The entrepreneurial ecosystem approach, just like strategy and regional development literatures, emphasizes the interdependence between actors and factors, but sees entrepreneurship (new value creation by agents) as the output of the entrepreneurial ecosystem. (p. 3)
Audrestch and Belitski 2017	We define systems of entrepreneurship (further ecosystem) as institutional and organizational as well as other systemic factors that interact and influence identification and commercialization of entrepreneurial opportunities. (p. 1031)
Bruns et al. 2017	The term entrepreneurial ecosystem was coined to refer to those elements in the entrepreneurs' environment that help them to succeed (or not) in their efforts to grow a new venture. (pp. 31–32)
Feld 2012	The Boulder thesis states that a prosperous ecosystem has four characteristics: (a) it is led by entrepreneurs; (b) it is inclusive where everyone is welcomed; (c) the involved people are committed long term (at least 20 years) to the ecosystem; and (d) there are many opportunities for gathering, that is, a lot of events. (pp. 25–28)
Isenberg 2010	This entrepreneurship ecosystem consists of a dozen or so elements (which we consolidate into six domains for convenience sake …) that, although they are idiosyncratic because they interact in very complex ways, are always present if entrepreneurship is self-sustaining. So although the combinations are always unique, in order for there to be self-sustaining entrepreneurship, you need conducive policy, markets, capital, human skills, culture, and supports. (p. 46)

Paper	Definition
Mason and Brown 2014	A set of interconnected entrepreneurial actors (both potential and existing), entrepreneurial organizations (e.g. firms, venture capitalists, business angels, and banks), institutions (universities, public sector agencies, and financial bodies), and entrepreneurial processes (e.g. the business birth rate, numbers of high growth firms, levels of 'blockbuster entrepreneurship,' number of serial entrepreneurs, degree of sell-out mentality within firms, and levels of entrepreneurial ambition) which formally and informally coalesce to connect, mediate and govern the performance within the local entrepreneurial environment. (p. 9)
Spigel 2017a	Entrepreneurial ecosystems are combinations of social, political, economic, and cultural elements within a region that support the development and growth of innovative start-ups and encourage nascent entrepreneurs and other actors to take the risks of starting, funding, and otherwise assisting high-risk ventures. (p. 50)
Stam 2015	A set of interdependent actors and factors coordinated in such a way that they enable productive entrepreneurship. (p. 1765)
Spilling 1996	The entrepreneurial system consists of a complexity and diversity of actors, roles, and environmental factors that interact to determine the entrepreneurial performance of a region or locality. (p. 91)

Of these, Stam and Spigel's (2018, p. 407) definition is the most general and widely accepted: entrepreneurial ecosystems are: "a set of interdependent actors and factors coordinated in such a way that they enable productive entrepreneurship within a particular territory." This definition has four key components: (1) 'interdependent actors and factors' (2) 'coordinated in such a way' (3) 'enable productive entrepreneurship' and (4) 'within a particular territory.' The first phrase refers to the types of actors involved in ecosystems, including entrepreneurs, investors, advisors, governments, non-governmental organisations (NGOs) and civil society groups. Ecosystems also involve factors beyond individuals and organisations and include social attributes and structures within a community such as cultures, networks, levels of trust, and investment capital. Implicit in the importance of 'actors and factors' is that the entrepreneurship process depends on more than the founders and their internal resources. This is in contrast to other work on entrepreneurship that has focused on individual personality traits (Frank et al. 2007), demographic backgrounds (Mungai and Velamuri 2011), or genetics (Nicolaou and Shane 2009) associated with successful entrepreneurship. Ecosystems research takes a post-heroic view of entrepreneurship (Braun et al. 2017), shifting the focus from the founder to the broader environment the founder and her organisation exists within. These various actors and factors will be discussed in more depth in Chapter 3.

The second part of the definition establishes that these actors and factors are interdependent. That is, an ecosystem is more than the sum of its parts. Spigel (2017a) argues that the operation and reproduction of ecosystems depend on the relational interactions between these different actors and factors, which support and reinforce one another in a cumulative and causal cycle. This means that ecosystems are more than the presence of a single element such as financial capital or a research university. Entrepreneurship is assisted by an array of different types of support, financing, and regional endow-

ments that, in turn, depend on one another to sustain them over time. However, this interdependence does not necessarily imply top-down coordination. Individual and organisational actors may have very different goals, such as an entrepreneur looking to build an innovative firm, an investor looking to maximise her profit, and a university administrator looking to attract public funding for a new incubation building. But through their actions, they contribute to a broader ecosystem that is beyond any single actor's control.

The third part of the definition, in which ecosystems 'enable productive entrepreneurship,' distinguishes ecosystems from previous work on the geography of entrepreneurship. Baumol (1990) famously distinguished between productive, unproductive, and destructive entrepreneurship. Productive entrepreneurship builds value both for the entrepreneur herself as well as to the broader society by introducing new technological innovations, increasing efficiency, or reducing barriers to markets. Unproductive entrepreneurship involves rent-seeking, which merely enriches the entrepreneur at the cost to society. Entrepreneurial ecosystems scrutinise the types of local environments that support productive entrepreneurship.

In the ecosystem literature, productive entrepreneurship is often seen as synonymous with high-growth ventures which are responsible for the majority of new job creation in developed economies (Brown and Mawson 2015). As will be discussed in Chapter 2, high-growth firms are those that achieve rapid growth in their revenues or employment through new product development or market expansion. The growth of these firms is often catalysed by venture capital or angel investment. We can be critical about how this might include forms of destructive entrepreneurship (such as the rise of the gig economy that has created a precarious labour force) and how it might exclude beneficial ventures like social enterprises. Nevertheless, the focus on productive or high-growth firms is a departure from earlier approaches to

the geography of entrepreneurship that has tended to focus on the total new firm births or the self-employment rate (e.g. Acs and Armington 2004; Hackler and Mayer 2008) or clusters and regional innovation systems which focus on the regional conditions that support larger anchor firms such as large manufacturing plants or corporate headquarters (Malmberg and Maskell 2002).

Finally, the fourth part of the definition refers to 'particular territories.' Entrepreneurial ecosystems are a geographic phenomenon rather than tied to a sector or industry. The focus on territorial views of entrepreneurship is an outcome of the socially embedded nature of the entrepreneurship process (Steyaert and Katz 2004). Cities and regions have distinctive cultures (Huggins and Thompson 2014), economic histories (Williams et al. 2013), labour markets (Massey 1995) and local policy initiatives (O'Connor et al. 2017) that led to fundamental differences in how entrepreneurship occurs. The focus on regions or other territories differentiates entrepreneurial ecosystems from other uses of the term 'ecosystem' in the management and policy literature such as innovation ecosystems (Iansiti and Levien 2004; Zahra and Nambisan 2011) or business ecosystems (Clarysse et al. 2014; Gawer 2014).

1.3 MODELS OF ENTREPRENEURIAL ECOSYSTEMS

The past decade has seen practitioners and researchers develop new models of entrepreneurial ecosystems. These models visualise the different elements of entrepreneurial ecosystems and how they interact in support of high-growth entrepreneurship. Four models have been particularly influential in how ideas of entrepreneurial ecosystem are communicated: The Isenberg Model of ecosystem domains (Isenberg 2010), the World Economic Forum's (2013) ecosystem pillars, Spigel's (2017a) ecosystem attributes, and Stam's (2015) systematic and framework conditions for ecosystems. While no model

is complete and all have their weaknesses, each emphasises different aspects of what entrepreneurial ecosystems are and how they work.

Isenberg's model (Figure 1.1) is the most well-known and cited model in both academic and policy work. It lists six different domains of entrepreneurial ecosystems: policy, finance, culture, supports, human capital, and markets. Each of these domains is linked to more specific elements. For example, the support element is made up of a region's physical infrastructure, its support professionals, and non-governmental institutions and agencies that provide various kinds of business support and advice. Isenberg's model is perhaps the most comprehensive of any of the popular ecosystem models, listing more than 50 components in its fullest form (Malecki 2018). Indeed, as Isenberg argued, while we can point to hundreds of different ecosystem elements, these can all be placed within the six main ecosystem domains.

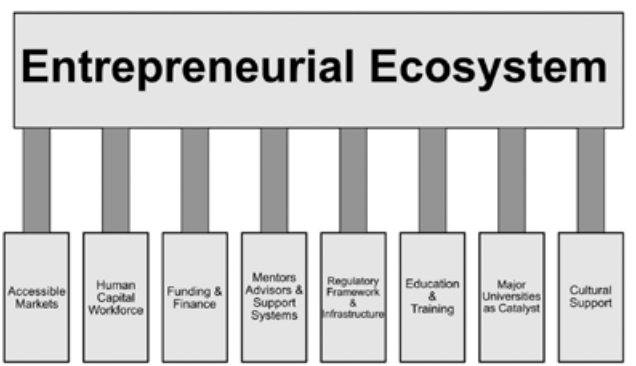
Building on Isenberg's ideas, the World Economic Forum (2013; 2014) surveyed global entrepreneurs to identify the local attributes and elements they thought were most important to enabling venture growth. From this, they identified eight ecosystem pillars: accessible markets, human capital, funding and financing, support systems, regulatory framework and infrastructure, education and training, major universities, and cultural supports (see Figure 1.2). Some of these pillars overlap with Isenberg's domains, but others such as research universities are unique to this model. The availability of these pillars can vary between regions, with some places like the Bay Area in the US having plentiful access to all pillars while other countries such as India and Mexico have far less access.

These two models identified the salient actors and factors that were supposed to make up entrepreneurial ecosystems, but they said very little about the relationships between these elements or how they produce positive entrepreneurial outcomes. The research models by Spigel and Stam both aim to better understand these relational processes. Spigel's model (Figure



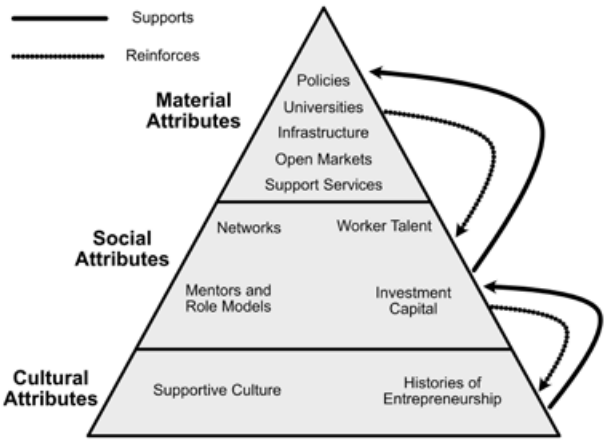
Source: After Isenberg 2010.

Figure 1.1 *Isenberg's model of entrepreneurial ecosystems*



Source: After World Economic Forum 2013.

Figure 1.2 *World Economic Forum's model of entrepreneurial ecosystems*



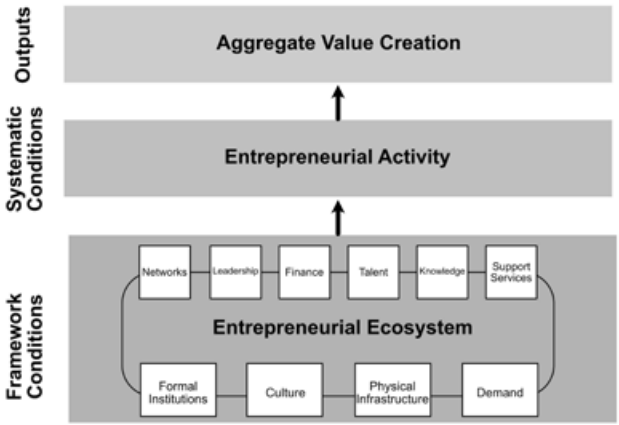
Source: After Spigel 2017a.

Figure 1.3 Spigel's model of entrepreneurial ecosystems

1.3) categorises different ecosystem attributes into three categories: cultural, social, and material. Cultural attributes are those linked to how entrepreneurship is understood within a region, including previous histories of entrepreneurship that produce narratives of risk and success and cultural support for ambitious entrepreneurial growth. Social attributes are aspects of the ecosystem that are accessed through or embedded in networks such as access to skilled workers and investment, mentorship, and business advice. Material attributes are those that have a physical presence in the region in some way, such as nearby customers, universities, support services, and business infrastructure as well as policies developed by local governments. This model emphasises the interconnectivity of different attributes in the ecosystem, each helping to reproduce other elements of the ecosystem over time.

Finally, Stam's model separates ecosystem elements into framework conditions made up of formal institutions, culture, infrastructure, and market demand and systematic conditions composed of networks, leadership, finance, talent, knowledge, and support services (see Figure 1.4). The framework conditions represent a region's institutional structure, the "fundamental causes of value creation in the ecosystem" (Stam 2015, p. 1767). The systematic conditions are the resources and support that high-growth entrepreneurs depend on to build and grow innovative new ventures. These systematic conditions depend on the framework conditions in order to develop. Importantly, Stam's model goes beyond just listing these framework and systematic conditions. He argues that their presence and interaction leads to outputs (high-growth entrepreneurial activity) which then leads to broader social outcomes with the creation of new aggregate value in a region or ecosystem. In this perspective, the purpose of ecosystems is more than increasing the rate of high-growth entrepreneurship: the point of ecosystems is to use entrepreneurship to create social value for the wider community.

These models are a useful starting point for understanding entrepreneurial ecosystems. Their primary function is to identify the types of people, resources, organisations, and institutions that are important to entrepreneurial ecosystems. These models provide a guide to what resources we should look for in analysing an entrepreneurial ecosystem and provide a basis for investigating the internal dynamics that operate and reproduce them over time. However, the models are simplifications. They do not discuss the messy relationships between these elements or, more importantly, what happens when one or several of these different elements are missing. To understand what happens in those cases, we need to delve more deeply in what ecosystems are and how they work.



Source: After Stam 2015.

Figure 1.4 Stam's model of entrepreneurial ecosystems

1.4 METAPHORS AND THEIR USES

None of the definitions or models discussed above make reference to biology or ecology. Entrepreneurial ecosystems have nothing to do with biological ecosystems — independent collections of biotic and abiotic components that stay in balance through energy flows and nutrient cycles. Instead, entrepreneurial ecosystems (and business ecosystems and innovation ecosystems) are a metaphor that emphasises the self-organising social and economic structures and networks of different actors surrounding a business phenomenon. Some authors such as Kuckertz (2019), Auerswald (2015) and Neumeyer and Corbett (2017) lean heavily on biological understandings of ecosystems when discussing entrepreneurial ones. Indeed, James Moore (1993), who first popularised the idea of ecosystems in strategic research, spoke directly

of an 'ecology' of competition that creates order out of the self-interested decisions of competing firms. However, others have warned about the "uses and abuses" of the ecosystem metaphor (Isenberg 2016, p. 564).

It is tempting to try to borrow ideas from natural ecosystems like fertiliser or keystone species. There is a seductive logic to seeing consultants who add little value but charge high fees as parasites or interpreting the flow of skilled workers away from failing ventures to new startups as being like a natural ecosystem's nitrogen cycle. However, it is dangerous to hew too close to a biological definition of ecosystems. Natural ecosystems tend to fall into equilibrium in which energy gained is balanced with energy lost. Natural ecosystems are resilient to change, returning to a stable state after a shock such as a flood or fire. But the goal of entrepreneurial ecosystems is to spur growth through the creation of new jobs and the attraction of new financial capital from outside the region. This is a directed goal, where policymakers, entrepreneurs and other ecosystem architects try to intelligently design the entrepreneurial ecosystem to make it the best possible environment for high-growth firms. This is a fundamental difference that cannot be ignored.

However, there are still useful lessons we can take from ecology when thinking about entrepreneurial ecosystems. As argued by Auerswald (2015), the introduction of the word 'ecosystem' by the British ecologist Arthur Tansley in 1935 moved the focus of natural history study away from the individual organisms to the transfer of energy between them and their environment. Similarly, the work of the pioneering geographer Alexander von Humboldt showed the importance of understanding the relationships between plants, animals, and their environment in addition to cataloguing new species. That is, the unit of analysis should be the linkages and flows between organisms rather than the organisms themselves. This is a clear lesson for entrepreneurship researchers. Entrepreneurial ecosystems are part of a similar shift in the

social sciences, seeking to understand the complex web of linkages between entrepreneurs, their new ventures, and the resources, actors, and institutions in their local context.

1.5 THE RISE OF ECOSYSTEM THINKING

Why then the term 'ecosystem'? Similar ideas have been introduced with similar names such as entrepreneurial systems, entrepreneurial environments, entrepreneurial milieus and entrepreneurial infrastructure (Malecki 2018). While early uses of the term ecosystem date to works by Bahrami and Evans (1995) and Spilling (1996), ecosystem terminology did not gain currency in the research or practice community until recently. Daniel Isenberg is the crucial player in popularising the current incarnation of entrepreneurial ecosystem thinking. However, its rise was not his work alone. He tapped in a larger zeitgeist around applying the idea of ecosystems to help explain economic growth and business strategy.

We see this most prominently in new thinking around business ecosystems: groups of companies and other users who create value for all parties through their mutual interactions (Adner 2016; Clarysse et al. 2014). Examples of business ecosystems include Facebook, a software platform that creates value by allowing users to connect and share content and then giving advertisers access to those users. Business ecosystems are different from complex supply chains like those operated by Toyota because power relationships in business ecosystems are not hierarchical. A business ecosystem might have a keystone player, such as the firm that creates and owns a central platform, but other users have their own power in the system, and the keystone actor must ensure that users are sufficiently rewarded to ensure they stay engaged (Iansiti and Levien 2004). This is a significant shift from the established dogma of strategic management based on creating the right combination of internal resources to achieve competitiveness.

Though business ecosystems and entrepreneurial ecosystems are very different things, they share a common interest in how factors outside an organisation predict success or failure. In a business ecosystem, managers try to set up social and technological systems that create value for all parties. This requires some control (social media platforms must remove abusive users and create a common way of using the site), but it also requires giving freedom to users to create new types of value. Entrepreneurial ecosystems have different goals but take a similar perspective: high-growth firms require coordination with a larger population made up of other entrepreneurs, investors, mentors, workers in the region as well as more diffuse forces such as culture, politics, and economies. No one person is in charge of a business or entrepreneurial ecosystem, but both emerge and are sustained by the collective activities of many individual actors all pursuing their own goals.

We can term these types of views 'ecosystem thinking,' even though the subjects of these different ecosystems range from a single firm to a region to a global community. Ecosystem thinking involves a shift in focus away from a single actor and towards the broader environment those actors experience. This is a new take on business strategy that requires a rethink about how firms and other actors see themselves. For both business and entrepreneurial ecosystems, everyone benefits from building a better community. Indeed, self-interested actions are detrimental to success because they damage the larger ecosystem and will eventually destroy the value being created. Technology has in some ways democratised power in these relationships, making it harder for one actor to dominate the creation and distribution of value. This has allowed new types of ecosystems to grow and thrive, including entrepreneurial ecosystems.

1.6 STRUCTURE OF BOOK

This book's purpose is to introduce the main ideas of entrepreneurial ecosystems and show how these fit into the larger fields of entrepreneurship research and economic geography. The goal is not to profile one or two ecosystems but rather to explore how the literature on entrepreneurial ecosystems has developed and what it tells us about the modern entrepreneurship process.

Chapter 2 looks at the intellectual foundations of entrepreneurial ecosystems, both in terms of the research traditions that have inspired current perspectives on ecosystems as well as the broader changes in society that have enabled their recent popularity. Entrepreneurial ecosystems draw on a deep heritage of thinking from different fields and one of its main advantages is its ability to bring together these different perspectives to create a more nuanced understanding of the regional embeddedness of high-growth entrepreneurship.

Chapter 3 reviews what we know about the actors and factors commonly associated with high-performing entrepreneurial ecosystems. The chapter explores the existing literature on these individual and organisational actors and local social, cultural, and material attributes to understand how they contribute to the overall ecosystem.

However, lists of actors and factors by themselves do little to explain what entrepreneurial ecosystems are or how they work. Therefore, Chapter 4 looks at how ecosystems work through the lens of practices and processes. Practices are the activities such as learning and networking that entrepreneurs and other actors use in the course of their regular lives that constitute the ecosystem and create the benefits associated with being in an ecosystem. Processes are the broader social flows and collective actions that produce and reproduce ecosystems over time. Understanding these practices and processes is key to explaining why ecosystems are so important for supporting high-growth entrepreneurship.

Chapter 5 looks at ecosystems outside of well-studied places like the United States and Western Europe. Entrepreneurial ecosystems in the Global South are not just resource-poor versions of their more developed Western counterparts but have their own unique functions and structures. The chapter critiques existing approaches to studying these more peripheral ecosystems and shows how the different economic contexts in places such as China, sub-Saharan Africa, and Eastern Europe led to ecosystems working in very different ways than has been previously assumed.

Chapter 6 addresses the most important critiques of entrepreneurial ecosystems. As it is a relatively new theory, there are many gaps that need to be addressed and areas that are not as well described as they could be. These include the lack of relationality between ecosystem elements and the lack of a theory that can drive testable predictions. Just as important are the more critical analyses of ecosystems which see them as part of a broader neo-liberal agenda which limits their impact on overall social welfare.

Finally, the book concludes with Chapter 7 by summarising the book's major arguments and makes a case for why entrepreneurial ecosystems matter to researchers, policymakers, and entrepreneurs themselves. The chapter ends with a suggestion of the most exciting future research avenues in studying entrepreneurial ecosystems.

2. The intellectual foundations of entrepreneurial ecosystems

2.1 THE NEED FOR FOUNDATIONS

The ideas underlying entrepreneurial ecosystems are not new, but the term's popularity is. Recent interest in ecosystems comes from business and entrepreneurship writers like Brad Feld (2012), Daniel Isenberg (2010) and Victor Hwang and Greg Horowitt (2012). These writers drew attention to the importance of the community surrounding the entrepreneurship process, which can catalyse the entrepreneurship process and generate sustainable economic growth. The loss of traditional anchors like large manufacturing plants has made entrepreneurial ecosystems an attractive prospect for many regional policymakers and local boosters. Policymakers and third-sector organisations in developed and developing economies are increasingly investing in ecosystem projects designed to unlock latent entrepreneurial potential. This raises the concern that policymakers are rushing into a new economic development fad not backed by rigorous research. As Stam (2015) warns, this can lead to research being led by policy rather than policy being led by research.

But the rising research interest in ecosystems is due to more than a response to policy attraction. Entrepreneurial ecosystems are positioned at the nexus of several ongoing research debates about the nature of entrepreneurship, the balance between structure and agency, and the socially embedded nature of business. Entrepreneurial ecosystems

draw on a diverse research base that includes work from geography, economics, business and entrepreneurship research, and sociology. This ranges from classical work in political economy by Alfred Marshall (1920) to cutting-edge research on social capital and entrepreneurial cultures. As shown in Figure 2.1, we can identify five main literatures that inform current understandings of entrepreneurial ecosystems: (1) high-growth entrepreneurship, (2) context, (3) embeddedness, (4) clusters and regional innovation systems, and (5) entrepreneurial environments. In addition to these research foundations, we can also identify three major practice trends that have informed ecosystem thinking: (1) lean business models, (2) digital affordances, and (3) policy devolution to cities.

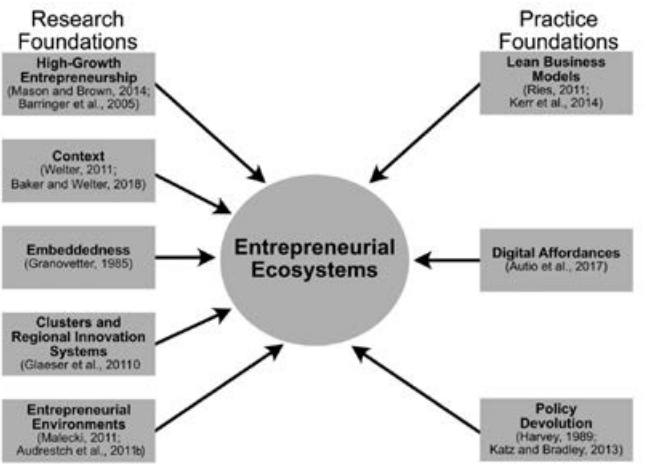


Figure 2.1 *The research and practice foundations of entrepreneurial ecosystems*

The remainder of the chapter discusses these different ideas and how they have influenced our current understanding of entrepreneurial ecosystems. The chapter concludes by distin-

guishing entrepreneurial ecosystems from allied theories such as clusters, RIS, and entrepreneurial environments.

2.2 RESEARCH FOUNDATIONS

2.2.1 High-Growth Entrepreneurship

The realisation that high-growth entrepreneurship is a key driver of job creation is one of the main reasons for the increasing interest in entrepreneurial ecosystems. Indeed, one of the main distinctions between ecosystems and other ways of studying the geography of entrepreneurship is the focus on this specific type of firm. Evidence from many economies suggest that high-growth entrepreneurship is responsible for the majority of new job creation (Henrekson and Johansson 2009; Morris and Török 2018; OECD 2010). High-growth entrepreneurship can be differentiated from other types of entrepreneurship such as low-growth entrepreneurship (firms that grow to a stable size and then cease growth), self-employment (non-incorporated firms that consist of only the entrepreneur, such as consultancies or gig economy work), lifestyle firms (firms whose aim is to provide a sufficient income to the founder while minimising risk and work) and subsistence entrepreneurship (engaging in entrepreneurship because of a lack of opportunities in the traditional labour market). As the name suggests, high-growth firms experience rapid growth for a set period. The OECD (2008) defines a high-growth firm as one that grows at least 20 per cent in terms of employees or revenues year-on-year for three years, while a 'gazelle' firm has these growth rates while being younger than five years old. Other organisations use different definitions, such as firms that have participated in an accelerator or received risk capital.

While studies from over two decades ago suggested that high-growth firms were essential to economic development, a wave of recent studies have confirmed this phenomenon

and shown its importance for regional economies (Brown et al. 2014; 2017; Henrekson and Johansson 2009). In essence, this stream of work argues that a small percentage of all new ventures — ranging from 2 per cent in Austria to 6 per cent in the UK — are responsible for a disproportionate proportion of new job creation (Anyadike-Danes et al. 2018). This is because these firms can scale through either external investments or high sales, requiring new employees. Even though other types of firms, such as large corporations or small lifestyle firms, employ more people, they are not net job creators.

There is a tendency to see entrepreneurship as an unalloyed good: politicians of all sides rarely miss a chance to say that small businesses are the backbone of the economy or that entrepreneurs are key members of their community. This has led to a proliferation of startup support programs and policies to help encourage entrepreneurship. But, many new ventures are of limited social and economic value and entrepreneurship itself is not always a positive thing. Compared to those in the traditional labour force, entrepreneurs and the self-employed have lower incomes (Hamilton 2000) and have higher levels of stress and worse health (Rietveld et al. 2014). Entrepreneurship rates increase in bad economic times as workers turn to self-employment in the absence of jobs in the traditional labour market. Most of the positive economic impact of entrepreneurship comes from a small share of high-growth entrepreneurial firms that can use technological and business innovation to open new markets and create wealth.

Fostering high-growth entrepreneurship is, therefore, an alternative economic development strategy than traditional approaches such as trying to support all forms of entrepreneurship through broad subsidies. The experiences of places like San Francesco and Berlin are examples of how high-growth entrepreneurship can create strong regional economic bases. However, supporting this type of entrepreneurship requires being able to define what a high-growth firm is

and identify the characteristics of a high-growth entrepreneur. As Barringer et al. (2005, p. 665) argue: the rapid growth of new ventures "is not a random or chance event, but is associated with specific firm attributes, behaviours, strategies, and decisions." Personal characteristics associated with the entrepreneur such as their industry experience, access to resources, and motivations for growth, and organisational characteristics of the venture such as levels of strategic planning, innovation practices, and employee training are associated with higher growth (Hermans et al. 2015). This does not mean that these characteristics cause high-growth entrepreneurship, but they create the conditions in which an entrepreneur can identify opportunities, develop a scalable business model, gather the resources necessary to exploit the opportunity and avoid barriers to growth caused by both business factors (such as an inability to access customers) and personal factors (such as the need to sell the company to pay for a child's education).

But defining a high-growth firm is rarely straightforward. Growth is not necessarily rapid or linear, with some firms experiencing little growth for years before a quick expansion, or a quick expansion levels off with little subsequent growth (Brown et al. 2017). A lifestyle firm may become a high-growth firm for a time before plateauing as a low-growth firm (Miozzo and DiVito 2016). Beyond the specifics of what percentage growth delineates high-growth from a low-growth one, we can question if certain types of firms such as social enterprises are excluded from high-growth policies even though they create substantial social and economic value. High-growth is often seen almost as a synonym for high-tech, even though many high-growth firms are not in technology sectors (Brown et al. 2017). Conversely, there are many high-tech, high-growth firms that create quality jobs but arguably have limited social value, such as gig economy startups. There may be an over-focus on firms that seek out venture capital and too little focus on firms that are able to

grow through re-investing their revenues into product innovation and market expansion (Aldrich and Ruef 2018).

While there has always been a vibrant literature on the geography of entrepreneurship (e.g. Acs and Armington 2004; Audretsch and Fritsch 1994; Audretsch et al. 2011b; Davidsson and Wiklund 1997; Davidsson et al. 1994; Hart and Gudgin 1994; Reynolds et al. 2004), this work rarely focuses on the smaller subset of high-growth entrepreneurial firms (with exceptions such as Qian (2016) and Bosma and Schutjens (2007)). But high-growth firms need different conditions and support than other types of entrepreneurs. High-growth firms are more likely to depend on the presence of nearby research universities for both technological breakthroughs and for producing skilled employees. They are also more likely to draw on local sources of financial capital and mentorship (Coad et al. 2016).

The importance of high-growth firms to economic development is one of the main motivating factors for entrepreneurial ecosystems. Distinguishing the characteristics of high-growth entrepreneurs and firms from their lower-growth counterparts allows for more targeted support. High-growth firms need and use different types of support than other, lower-growth firms. As research on the internal characteristics of high-growth firms expands, we gain a better understanding of how entrepreneurs and managers make use of their networks, skills, and local resources to innovate and grow. This informs how we understand the role of entrepreneurial ecosystems in either supporting or preventing this growth.

2.2.2 Context

Context is one of the most fertile areas of study in contemporary entrepreneurship research (Zahra et al. 2014). Contextual research on entrepreneurship seeks to understand the role of the different influences which surround the entrepreneurship process (Baker and Welter 2018). This view differs from

more positivist forms of entrepreneurship research that seek to control away contextual factors with the goal of producing generalisable findings of entrepreneurship that transcend temporal, social or political boundaries (Ucbasaran et al. 2001). Contextual approaches to entrepreneurship argue that such general understandings are not possible; that the entrepreneurship process is so deeply embedded in economic and social structures that there are few universal laws of venture creation and growth (Steyaert and Katz 2004).

Johns (2006) divides context into two types: discrete and omnibus (see also Welter 2011). Discrete contexts are individual variables and elements that affect entrepreneurship, such as age, gender, place, time, industry, or religion. Omnibus contexts are the gestalt combination of these discrete contexts that generate the overall environment surrounding and altering the entrepreneurship process. This provides two pathways to study entrepreneurship. The first is to study the impact of contextual factors, such as work that shows how an entrepreneur's gender affects their experiences as entrepreneurs (Elam 2008; Swail and Marlow 2018). This stream of research has constantly shown that there is no single entrepreneurial journey, but instead a complex landscape of how entrepreneurship is performed.

The second approach is to focus on the broader influence of omnibus contexts. These are large-scale institutional fields that profoundly affect the entrepreneurship in numerous ways. In their review of the context literature, Baker and Welter (2018) suggest that two types of context are particularly important: 'where' and 'when.' Where contexts are the place that entrepreneurship occurs within and how entrepreneurs' behaviours are embedded in place-based structural and institutional contexts. Beyond obvious place-based factors such as the presence of entrepreneurial resources and specific policies like taxes or property rights (which help constitute the overall entrepreneurial environment) place-based entrepreneurial cultures and informal institutions develop in particular locations

that help constitute the meaning and values of entrepreneurship (Thorton 1999). This leads to a varied geography of not just the quantity of entrepreneurship — how many and what type of firms are created — but also of the nature of the entrepreneurship process itself.

When contexts reflect the dynamism of the entrepreneurship along with the processes through which contexts are reproduced over time. For example, the context of a nascent firm that is little more than an entrepreneur with an idea is very different than a more established venture seeking series B financing. Entrepreneurs at these two stages will interact with the world and in the marketplace very differently despite that both are nominally engaged in entrepreneurship. But beyond the stage of entrepreneurship, when context points to the importance of history in understanding the entrepreneurship process. Entrepreneurs use their understanding of history (their own history, history of their people, history of their firm, history of their industry) to plot their paths forward (Wadhwani 2016).

Entrepreneurial ecosystems research is fundamentally contextual, concerned with understanding the impact of localised contexts on the entrepreneurship process and understanding how these contexts develop and the mechanisms through which their influence is enacted. Context informs ecosystems research in two ways. First, it emphasises that geographic context has a profound influence on the nature of the entrepreneurship context. The characteristics of a place, from its formal institutions and support mechanisms (Huggins and Williams 2011) to its industrial specialisation or diversification (Bhawe and Zahra 2019) to its local culture (Huggins and Thompson 2014; Spigel 2013), will all affect the nature and course of the entrepreneurship process. This is one of the key arguments of ecosystems research: that place matters for entrepreneurship. Second, it suggests that high-growth entrepreneurship is a specific entrepreneurial context, distinctive

from other types of entrepreneurship such as subsistence entrepreneurship or lifestyle entrepreneurship.

2.2.3 Embeddedness

Embeddedness is a social theory that provides a mechanism to understand how context shapes entrepreneurial action. From an ecosystems perspective, the concentration of high-growth entrepreneurship is not the result of endowments like plentiful natural resources or good transportation links. Instead, it is related to the place-based social, political, economic, and cultural structures an entrepreneur operates within. This perceptive builds on one of the most important sociological theories of the twentieth century: social embeddedness. Embeddedness refers to how people and organisations are entangled in complex webs of social relationships that both enable and contain potential economic actions (Granovetter 1985; 2005). This framework was a response to both under-socialised views of human activities common in neo-classical economic theory, where choices were modelled under conditions of economic rationality and nearly perfect information and to over-socialised views common in Marxist theory where class determines individual actions (Hayter 2004).

In developing the theory of embeddedness, Mark Granovetter argued that market behaviour was grounded in networks of social relations that influence the types of actions or choices that are possible. Economic choices are embedded in network-based relationships that enable and constrain potential decisions in ways that go beyond rational economic calculation. Network ties are generative as well as restrictive: it is not that social ties and obligations only constrain potential action, but they also create new opportunities and find new ways to accomplish goals (Dacin et al. 1999). Embeddedness, therefore, explains why all actions taken by actors are not economically rational, but still make sense given their broader social context.

Like every other economic activity, entrepreneurship is embedded in the social structures surrounding it (Anderson and Jack 2002). Structural embeddedness, the ties and relationship between an actor and their broader community, are the primary way in which embeddedness affects the entrepreneurship process (Zukin and DiMaggio, 1990). Networks are important to entrepreneurs, who use them to access resources such as investment, market knowledge, and customers (Hoang and Antoncic 2003; Partanen et al. 2018). Entrepreneurs' networks are sources of advice (MacKinnon et al. 2009) and of community support that helps entrepreneurs establish their identity (Pret and Carter 2017). Social norms and the risks of alienating friends and colleagues may prevent entrepreneurs from using particular business strategies that may damage these relations, no matter how profitable (Witt 2004). But potentially negative actions are not just avoided due to a cost/benefit calculation, embeddedness affects the norms of what is seen as acceptable behaviour, and types of behaviour are right to employ. Actors adhere to these social rules not just to avoid sanctions but rather because they want to continue to be seen as upstanding members of their community (James 2007). However, not everyone is embedded to the same degree: newcomers to a place or those who do not plan to engage in a community for a long time may be less constrained by social rules than those who have spent their lives in a particular social context (Kalantaridis and Bika 2006).

Entrepreneurs are territorially embedded in the localised networks and communities where they start and grow their firms. Territorial embeddedness refers to the social structures of a place that affect economic activity, ranging from informal social connections to more formal political governance and economic structures, which are rooted in particular places (Hess 2004). Entrepreneurs' social networks (Schutjens and Völker 2010), sources of financing (Christensen 2007), and business advisors (Mole and Capelleras 2017) tend to be local. Thus, while high-growth entrepreneurship is without

a doubt an international experience, many of the key resources entrepreneurs need to create and grow these firms will come from their local community. This means entrepreneurs depend on localised, trust-based ties with others in their ecosystem in order to get the resources they need. The importance of embeddedness to entrepreneurship is evidenced by the fact that entrepreneurs who are more embedded in their local community perform better than those who are less embedded because those with stronger local ties are better able to activate these connections to access important resources and support (Dahl and Sorenson 2012).

Embeddedness provides a mechanism to understand why aspects of the entrepreneurship process differ between contexts. Entrepreneurship is not a wholly economic phenomenon where strategic choices are made based on dispassionate economic logic. It is bound up in social conventions and norms that affect how entrepreneurship is viewed and performed. The norms of one place might encourage entrepreneurial actors embedded in it to be more ambitious in their growth plans, while the norms of another place may discourage such outlooks. While entrepreneurs can and do violate these norms, such behaviour might result in being excluded from networks and unable to build up the reciprocal trust required to get the resources and support she needs (de Clercq and Voronov 2009). But at the same time, entrepreneurship is not a wholly social phenomenon. Entrepreneurs and business owners can become over-embedded in their communities and networks in ways that would negatively affect their firms' survival or growth, for example by keeping more expensive suppliers because of their long-term relationships (Uzzi 1996).

This embeddedness mechanism is important for how we understand entrepreneurial ecosystems. Ideas of embeddedness connect the disparate elements of an ecosystem — the people, the resources, the institutions — with the daily practices of entrepreneurial actors. These factors are more than just potential resources for entrepreneurs. Their exist-

ence helps create the local norms of what it means to be an entrepreneur and what types of actions are appropriate for an entrepreneur to take. Embeddedness helps explain why place matters in entrepreneurship and why its impact can never be fully escaped.

2.2.4 Clusters and Regional Innovation Systems

Clusters and regional innovation systems (RIS) have been two of the most popular regional economic growth frameworks. Both theories examine how proximity between firms and other organisations in a region can increase competitive advantage and improve the overall economy. While both approaches have similar intellectual roots in early economic geography work of Alfred Marshall (1920) and observations of post-war Italian industrial districts (Piore and Sabel 1984), the sources of their advantages differ. Clusters investigate how co-location increases productivity while RIS study how knowledge spillovers and networks between nearby firms and universities promote innovation. But despite their differences, these two theories provide the bulk of the intellectual foundations for what ecosystems are and how they work, including the role of networks, knowledge spillovers, and cooperation between firms. Both push our attention away from the firm in isolation and towards the broader geographic environment it exists within for understanding the source of sustainable competitiveness.

The central insight of clusters is that firm competitiveness is enhanced through proximity with other firms. In Michael Porter's (1998b; 2000) early work on the topic, this focused on proximity between competitors in the same industry. Even though they compete in the marketplace, nearby firms can cooperate to create shared infrastructure that reduces their individual costs. These advantages are referred to as localisation economies: the cost savings produced by similar firms being able to share common resources (Beaudry and

Schiffauerova 2009). In manufacturing-focused clusters, this might be physical infrastructure such as a shared rail spur. In contrast, firms in a knowledge-based cluster might co-invest in a university research facility to help build a skilled workforce in the region. As more firms in a similar market or sector develop in a region, a specialised supplier base emerges to fill local needs. This reduces supply costs due to easing logistics challenges and it allows customers and suppliers to build up long-term relationships that drive down costs and improve quality (Glaeser and Kerr 2009). The presence of multiple firms that require similar types of employees (e.g. welders or programmers) encourages local workers to invest in their skills to enter these fields, and attracts already qualified workers to the region, improving the overall quality of the labour force.

Cluster benefits go beyond reduced costs through shared resources. Urbanisation economies are the advantages firms get through knowledge sharing and learning from nearby firms (Capello 2002). Firms in the same sector can purposefully share best practices to improve all of their operations, observe each other to identify market changes and new techniques, or knowledge can spread through movement of workers between firms (Casper 2007; Giuliani 2007). This allows for knowledge spillovers that increase firms' innovativeness and competitiveness. Urbanisation economies can enable more radical innovations by combining knowledge bases from different industries. These so-called Jacobs Externalities help the development of new products in diverse clusters by combining complementary knowledge and insights from a variety of markets (Beaudry and Schiffauerova 2009).

RIS differ from clusters in that the focus in on the localised institutions' structures and relationships between actors that support innovative activity. RIS emphasise the role of tacit knowledge — knowledge bound up in particular contexts and which cannot easily be transmitted globally — within the innovation process and the importance of frequent informal

collaborations between different parties to drive forward new ideas (Gertler 2003). RIS approaches explicitly recognise that innovation comes from a number of sources, including internal R&D activities in firms, basic and applied university research, and open innovation processes incorporating both users and producers. Innovation requires input from a number of actors, some within the same organisation but others outside of it. This input is fostered by close physical proximity and dense network connections between innovation actors (Cooke 2001).

Coordination of a RIS often falls to a single central figure, such as a major multinational firm, public research lab, or university. Coordinating actors help bring different organisations together to align goals and incentives and to build knowledge-sharing networks (Benneworth et al. 2017). But this coordination is impossible without institutional trust within the broader region that enables the potential sharing of proprietary markets and technological knowledge (Cooke 2001).

Early RIS theory left little room for entrepreneurship. Entrepreneurial firms largely lack the internal resources and absorptive capacity for the type of innovation RIS are supposed to support (Liao et al. 2003). Indeed, as Sternberg (2007, p. 656) argues: "the neglect of entrepreneurial activities by the literature on RIS is astonishing." Research on entrepreneurial regional innovation systems (ERIS) has sought to address this by exploring the role of highly innovative new ventures into RIS frameworks (Cooke 2007). Entrepreneurial spinout firms from both universities and larger corporate firms serve as key innovation nodes (Lawton Smith et al. 2014) in ERIS. But more broadly, this work looked at how ERIS differed from RIS because they lack a central powerful coordinating actor. Still, instead they have distributed nodes of knowledge production and sharing that are connected by dense social networks between founders, employees, and researchers (Ylinenpää 2009).

The central contribution of the sizeable cluster and (E)RIS literature to ecosystems research is that proximity to other organisations, be they in similar or different markets, can be a source of competitive advantage to entrepreneurial firms. This provides a way to explain how entrepreneurial firms are better off being located near to each other rather than in virgin territory. Ecosystems theory draws heavily on these two concepts for both the types of resources that are needed in an ecosystem and how a concentration of high-growth firms can create their own supportive environment. The advantages of co-location for high-growth take the form of efficiencies found through shared infrastructure or access to a specialised labour pool, the ability to capture knowledge spillovers from knowledge producers like universities or major firms, or the opportunity for radical innovation through serendipitous contact between diverse actors mediated by localised social networks. Clusters and RIS provide empirically validated frameworks for how firms benefit from co-location and proximity as well as insights into the underlying social processes that surround firm innovation and scaling.

But despite their importance to its intellectual foundations, ecosystems are a distinct concept from clusters and RIS. As shown in Table 2.1, we can draw out five major differences between ecosystems and these two ideas: the key actors, the role of the state, how benefits from proximity are accrued, the role of knowledge, and the impact of industry and sectors. These differences derive from ecosystems' specific focus on high-growth entrepreneurship and how these types of firms differ from both lower-growth new ventures and larger corporate firms. Entrepreneurial ecosystems have a very different focus than clusters and RIS, leading to different research questions and approaches.

Table 2.1 Differences between entrepreneurial ecosystems and cluster/RIS theory

Theme	Entrepreneurial Ecosystems	Clusters / RIS
Key actors	Entrepreneurs are key actors in an ecosystem, with the ability to identify challenges and help create structures to overcome common problems. Other actors, such as existing firms, who can draw on ecosystem resources to catalyse new growth, startup workers, mentors, advisors, and dealmakers are also crucial constituencies.	Large anchor firms, public agencies, and universities are the most important actors due to their large stocks of resources and ability to produce and exploit novel technological and market knowledge.
Role of the state	Ecosystems are primarily led by entrepreneurs, particularly around the creation of networking and support organisation and identifying critical needs. The state can supplement this role and help provide necessary resources.	The state plays a prominent role as the lead organiser for support programs and brings competing actors together in order to create public goods. Significant investments in research or coordination organisations.
Accessing regional resources and benefits	Focus on the specific difficulties and opportunities entrepreneurs and startups face in accessing localised resources such as liabilities of newness and lack of internal absorptive capacity.	Little differentiation between large firms and smaller startups and new ventures in how firms access local resources. Importance of absorptive capacity in internalising knowledge spillovers.

Theme	Entrepreneurial Ecosystems	Clusters / RIS
Role of knowledge	In addition to technical and market knowledge, the importance of entrepreneurial knowledge in supporting the formation and growth of new ventures and creation of an entrepreneurial culture. University knowledge spillovers are important but less so than their role as producers of skilled entrepreneurs and workers.	Focus on technical and market knowledge to drive incremental and radical innovation and help expansion into new markets. Frequently the role of knowledge producers like universities or research labs.
Impact of industry	Ecosystems focus less on industry or market and more on underlying technology (e.g. digital technology). Entrepreneurial knowledge largely transcends industry structures, and lack of direct competition encourages cooperation.	Importance of knowledge flows within industries to reduce costs and between industries to catalyse radical innovation.

Note: Adapted from Spigel and Harrison 2018.

2.2.5 Entrepreneurial Environments

Entrepreneurial environments refer to a diverse literature from economic geography, entrepreneurship research, and regional policy that seeks to understand why some places have persistently higher rates of entrepreneurship than others. This is a broad field of research that could equally be called entrepreneurial systems (Neck et al. 2004), infrastructure (Van de Ven 1993), milieu (Ritsilä 1999) or clusters (Delgado et al. 2010). But it is unified by an interest in what causes higher levels of entrepreneurship than can be explained by traditional factor endowments such as capital, labour, and innovation (Malecki 2009). This field provides a conceptual and empirical basis for understanding the role of more ephemeral forces such as culture and history in supporting (or preventing) high-growth entrepreneurship in a region.

Entrepreneurial environments can be thought of as the intangible aspects of a place that affect the supply of entrepreneurs and their ability to thrive. Supply of entrepreneurship — how many people desire to become entrepreneurs — is influenced by economic conditions: strong economic growth creates many opportunities for entrepreneurs to exploit, while bad economic times cause people to turn to self-employment to replace their lost income. But it is also affected by factors such as a culture that helps determine the social prestige of entrepreneurship. Other factors such as the institutional infrastructure of public support, networks of experienced business mentors, and the presence of investors then affect the potential of entrepreneurs to scale their firms (Vaillant and Lafuente 2007).

Culture is one of the most important influences on the entrepreneurship process (Malecki 2011; Spigel 2017b). Economic geographers have long argued that a place's cultural environment, built up over time through its social and industrial legacy, and demographic trends has a profound effect on the way an economy operates (Massey 1995). While cul-

tural differences are normally understood on national levels (see Hofstede 2001), it is clear that sub-national units like city-regions also have distinctive cultures that encourage or discourage particular economic activities (James 2005; Staber 2007).

Saxenian's (1994) comparative history of Silicon Valley and Boston is an instructive example of how culture can affect entrepreneurial development. Both regions hosted pioneering research in digital computing in the 1950s and 1960s and had early successes in this emerging industry of microcomputers. Over this time, Silicon Valley developed an open culture that encouraged knowledge sharing and entrepreneurship while Boston's culture was influenced by the city's patrician banking culture, which discouraged risk-taking. As a result, Silicon Valley was able to anticipate the transition from microcomputers (which were actually large, centralised servers) to the burgeoning personal computer market. In contrast, Boston's computer industry missed this shift and was decimated by the recessions of the 1980s.

Historically high (or low) levels of entrepreneurship build up a collective understanding of entrepreneurship which can normalise self-employment and business ownership as a career path or makes it an unusual thing that is only contemplated when there are no other options (Lafuente et al. 2007). These histories are hard to change. For example, regions with high levels of entrepreneurship in the East of Germany before the Second World War saw increased rates of entrepreneurship after the fall of the Berlin Wall, despite forty years of socialist economic policy (Wyrwich 2012). Resiliency was not due to existing endowments like factories or infrastructure but rather due to the entrepreneurial culture that had built up in those regions more than a century before.

The entrepreneurial environment literature is a critical foundation for modern work on entrepreneurial ecosystems. It argues that the heterogeneous landscape of entrepreneurship is not just the result of economic factors but is also deeply

bound up in local cultural and institutional differences. These more intangible factors do not only affect entrepreneurs, they also influence everyone involved in the entrepreneurship process, which leads to more tangible outcomes like how much support or investment is available to scaling firms. Ecosystems research has built on this to position culture as a core element of a strong entrepreneurial ecosystem by showing that cultural factors are crucial for the development of almost every other element of an ecosystem. Further, the entrepreneurial environments literature has also made a compelling argument for the importance of history, not just for producing culture but also for its impact on what types of resources and support are available and for whom.

2.3 PRACTICE FOUNDATIONS

2.3.1 Lean Business Models

One of the most significant developments in the practice of entrepreneurship over the past two decades has been the rise of the lean startup model. Popularised by business writers such as Eric Ries (2011) and Steve Blank (2003), this approach eschews purposeful business planning in favour of rapid prototyping, market tests, and business model innovation. Rather than investing in up-front R&D to perfect a new product, entrepreneurs instead develop a minimal set of features that allow the product to get to market, the minimum viable product (MVP). The MVP tests the market to make sure that it is solving customer problems and to identify new markets and solutions that were not originally anticipated. The lean model calls for experiments to test different product features or marketing plans to avoid confirmation bias. Entrepreneurs use this experimental data to shift the firm, undertaking 'pivots' in their product design, target market, or the underlying business model in response to their experiences in the market. The Lean Startup approach has become one of the dominant

frameworks through which entrepreneurship is taught and practised (Bortolini et al. 2018).

While the lean startup model is inherently practice-based, it can be understood through the lens of organisational learning (Levinthal and Contigiani 2018). By engaging in the marketplace through the quick release of MVPs and collecting data on customer use, firms generate knowledge about the market that allows them to interactively redesign their product offering or overall business model to better deliver value to customers (Sosna et al. 2010). In essence, the goal of the lean startup model is to shift Knightian uncertainty — the unknown associated with entrepreneurial action — to more calculable risk (Kerr et al. 2014). Market experiments create organisational knowledge that helps founders understand their product/market fit and the effectiveness of their business model. By avoiding the high fixed costs associated with up-front product development or back stock, lean startups can react quicker to what they learn in the marketplace and retain more strategic flexibility.

This popularity of lean startup models was aided by the introduction of the Business Model Canvas (BMC), a design tool that helps visualise different components of a firm's business model (Osterwalder and Pigneur 2010). While the BMC was far from the first attempt to create a typology of business model components (Teece 2010; Zott and Amit 2007), its easy to use and modular design of nine interlinked business model components has helped it become the dominant business model paradigm. This tool shifted business model development from an analytical function based on creating profitable financial models to a design-thinking function based on solving customer problems. The BMC is frequently used in the lean startup approach to encourage rapid business model innovation based on the new data obtained through MVP experiments.

Both the lean startup and BMC techniques create generic entrepreneurship models that can be applied across a variety

of different firm types, sectors, and contexts. This is a subtle shift from prior startup models that required far greater market-specific planning for different business domains. That is, a digital software firm can use lean startup strategies and tools like the BMC just as well as a service or retail entrepreneur can. This means that entrepreneurs from a variety of different sectors and backgrounds can share business knowledge and insights in ways that were not previously possible. As will be discussed throughout this book, this ability to learn from a large group of other entrepreneurs is crucial for the emergence of entrepreneurial ecosystems. Because so many entrepreneurs use the basic ideas of lean business models, it is possible to build large communities of entrepreneurs who can share their insight and expertise about the startup and scale-up process. Lean startup methods and tools like the BMC create a common ground that extends beyond industry or sectoral boundaries in a way that did not exist previously.

2.3.2 Digital Affordances

The lean business model methodology calls for entrepreneurs to engage in market and product experiments in order to refine their business model and product/market fit. New technical developments aided this approach over the past decade, which has drastically reduced the cost of these experiments (Kerr et al. 2014). Technologies such as cloud computing, open-source software, and affordable digital business platforms like Google Analytics mean that entrepreneurs no longer have to build their tools from scratch, reducing startup costs. For example, Ewens et al. (2018) found that the introduction of Amazon Web Services — a cloud computing product that allows customers to rent computer time — reduced the initial investment digital firms needed to raise by 20 per cent. Similarly, e-commerce platforms like Shopify or payment systems like Stripe allow new entrepreneurs to quickly build

on-line shops for a fraction of the cost that they would have faced a decade ago.

Autio et al. (2017) refer to these developments as 'digital affordances' which have shifted power in existing value chains, allowed for the disintermediation of existing middlemen (and the creation of new middlemen) and aided the coordination of dispersed actors. All of these shifts helped build new opportunities and niches that can be exploited by entrepreneurs. By lowering the cost of experiments, these affordances make lean startup approaches possible and reduce barriers to the creation of innovative, growth-oriented firms. While these new technologies are most relevant to digital firms, a similar suite of technologies has emerged that provide similar advantages for firms that make physical products. Flexible manufacturing technologies allow firms to organise smaller production runs at lower costs, allowing them to experiment with new designs. Newer 3D printing technologies may reduce such costs even further, allowing for easier individual customisation. For larger-scale production, improved logistics services and intermediaries such as Alibaba make it easier and cheaper for entrepreneurs around the world to access the lower-cost manufacturing facilities.

Just as importantly, most of these digital technologies are industry-agnostic. New software platforms help automate business processes common across all sectors such as sales, human resources, customer support and billing, marketing, document management, and compliance. These tools can be used by business-to-consumer product firms just as easily as business-to-business enterprise software firms and they can be adopted by firms who have a physical product as easily as by firms with a digital or service-based offering. While business platforms and marketing tools are not society-altering general purpose technologies like the internal combustion engine, they are sufficiently generic that they can be used by a wide array of different actors in a variety of different ways. In essence,

digital affordances create a new category of generic business know-how that entrepreneurs in many different fields can use.

Digital affordances widen the knowledge spillovers that entrepreneurs benefit from in an ecosystem. Entrepreneurs can now learn business and processes insights from a wide array of different companies rather than just those in their own narrow sector. These affordances make it easier for an ecosystem to build up a critical mass of entrepreneurial actors who can share relevant experiences and knowledge. This does not imply that specialised market and technological expertise is no longer necessary. Entrepreneurs still need deep insights in their market to identify opportunities and know-how to solve problems in the market. However, the growth of generic digital tools means that they can more easily draw on outside knowledge to solve common business problems such as how to evaluate the effectiveness of a marketing campaign or automate customer billing.

These generic business technologies and platforms create communities between businesses that in previous years would have very different technological bases. Indeed, it is becoming increasingly difficult to delineate 'high-tech firms' from firms that use technology (Nathan and Rosso 2015). This allows ecosystems of entrepreneurial actors to be able to help one another by sharing knowledge and insights about how to use these common digital technologies and techniques to power growth across a diverse range of industries. Without these digital affordances, few places would have the scale in one particular sector to create the same activity we see in modern entrepreneurial ecosystems.

2.3.3 Policy Devolution

The final practice trend driving interest in ecosystems is the emergence of the city-region as the locus of economic development activity. Regional governments have seen their authority and responsibility for economic development

increase over the past several decades due to political and social changes. These shifts have been noted by Marxist geographers who identified the growth of inter-city competition over investment and jobs as a new form of 'urban entrepreneurialism' (Harvey 1989) and researchers looking at shifts in power in an increasingly globalised economy (Swyngedouw 2010). Most recently, policy and legal scholars are looking to guide cities and states in a time of increasing political paralysis within national governments (Katz and Bradley 2013). Either by choice or because the responsibility was thrust upon them, city-regions have increased responsibility and power for economic development. Cities are expected to compete with each other for business and jobs by reducing taxes and regulation, investing in infrastructure, and helping build a talented regional labour pool. The effects of this shift are visible in major political initiatives such as the European Union's Smart Specialisation strategy (Foray 2014), the UK's 'Northern Powerhouse' policies (Lee 2016) or the neo-federalist movement in progressive American legal circles.

The scale and complexity of regional economic development policies have increased measurably over the past few decades. Early economic development efforts often involved little more than marketing and branding campaigns. But new approaches to economic development include mega-projects such as New York City's Cornell Tech program, which represent multi-billion dollar investments in new infrastructure. This illustrates a shift from somewhat passive economic development strategies that used levers of tax rates and incentives to attract in new employers to more active policies designed to build on a place's existing specialities through the promotion of entrepreneurship in tandem with attracting new firms to the region.

The most obvious way cities have embraced their new economic development responsibility is by creating public support programs that aim to provide training, networking opportunities, or other resources to potential high-growth

entrepreneurs. These range from support for small, informal networking groups to large, multi-year accelerator or venture financing programs (Larty et al. 2017). While the overall usefulness of these programs has been debated for decades (Lerner 2009), the trajectory is for more public interventions and programs financed or run by local governments with the aim of increasing entrepreneurial capacity. Traditionally, research has tended to focus on the efficacy of individual programs (e.g. Brown et al. 2015), but ecosystem perspectives have pushed researchers to look at the broader systems of support programs within communities (Motoyama and Knowlton 2016). Networks of overlapping programs create a deeper pool of resources and support that entrepreneurs can draw on based on their immediate needs (Spigel 2016). The ability of these networks of support programs to mesh together will depend on the ability of entrepreneurs to effectively lead their ecosystem and work with public officials to create the types of support they need (Harper-Anderson 2018).

Cities adopt ecosystem strategies as a way to build a more resilient economic base than their previous smokestack chasing strategies. Unlike their national counterparts, city governments can focus on their local specialities and take advantage of unique resources, such as a prominent business leader or a local historical feature that can be mobilised to bring the entrepreneurial community together. This discourse of ecosystems, with its focus on innovation, technology, and local leadership appeals to many policymakers looking to improve their economic base. Even more importantly, ecosystem development strategies require far less up-front investment in infrastructure and tax breaks than other approaches.

2.4 CONCLUSION: WHAT'S NEW ABOUT ENTREPRENEURIAL ECOSYSTEMS?

Entrepreneurial ecosystem concepts draw on decades of work from a variety of different fields and are informed by new trends in business and urban politics. This raises the question: what is new about entrepreneurial ecosystems? Are they just a reimagining of clusters or innovation systems or a new way of thinking about the context of entrepreneurial contexts? What is the difference between an entrepreneurial ecosystem and an entrepreneurial environment or an entrepreneurial RIS? Fundamentally, are entrepreneurial ecosystems simply old wine in new bottles (Scaringella and Radziwon 2017)?

While entrepreneurial ecosystems do not introduce any fundamentally new concepts to the discussion of place and entrepreneurship, it syntheses many different ideas together in new ways and gives these ideas a practice and policy relevance that did not previously exist. The most significant shift of ecosystems work is the specific focus on scale-up entrepreneurial firms (Brown and Mason 2017; Isenberg and Onyemah 2016). Inherent in ecosystems is the acknowledgement that these firms are affected by their context in different ways than other types of firms. High-growth entrepreneurs need different types of support than lower-growth ones, including both formal support like seed investment programs but also more informal support like business mentorship facilitated by social networks between entrepreneurs. This perspective allows us to envision new support regimes as well as acknowledge that high-growth entrepreneurs use their networks and local communities in very different ways than other types of entrepreneurs.

But ecosystems are more than a new philosophy on scale-up support. It brings new vibrancy to debates about how entrepreneurs are embedded in local contexts and cultures and just as importantly, their ability to affect these structures.

Entrepreneurial ecosystems are inherently about this reciprocal relationship between social structure and entrepreneurial agency. Ecosystems research demonstrates how local cultures, institutions, and social systems influence entrepreneurial action in a way that was not done in clusters or RIS research. But more importantly, more than seeing entrepreneurial actors as passive actors embedded in larger social systems, ecosystems approaches sees these actors possessing active agency to construct and change their situation. By positioning entrepreneurs at the centre of the ecosystem, they are seen as active architects of their communities. Entrepreneurs are not just influenced by context, but they are able to construct it as well. While entrepreneurs' (and other ecosystem participants) simultaneous role as actors in and architects of ecosystems has been acknowledged before (see Feldman 2014; Spigel 2013), ecosystems research brings this to the forefront.

Furthermore, as will be discussed in the next chapter, ecosystems take an integrative approach to the types of actors, factors, and resources that make up entrepreneurial ecosystems. The goal is not necessarily to isolate a single variable of interest to understand its impact of firm performance but rather to understand how the ecosystem as a whole produces and sustains the resources, networks, and environments that support high-growth entrepreneurship. These connections between elements are enabled by new developments in the practice and political economy of entrepreneurship, such as the growth of common lean startup models and shared platforms and business ecosystems that allow more crossover between entrepreneurs.

3. The actors and factors of entrepreneurial ecosystems

3.1 INTRODUCTION

Most work on entrepreneurial ecosystems to date has focused on identifying the actors and factors that make up thriving ecosystems. This has been done through observations of successful ecosystems (Feld 2012; Spigel 2017a), statistical analyses of multiple regions (Audretsch and Belitski 2017; Bruns et al. 2017; Spilling 1996) and reviews of existing entrepreneurship research (Mason and Brown 2014; Stam 2015) to suggest the different people, organisations, and social structures that are associated with increased levels of high-growth entrepreneurship. Identifying common attributes shared between successful ecosystems allows us to understand better how a supportive environment promotes high-growth entrepreneurship and allows policymakers to assess and address regional weaknesses.

While researchers have used different terms for these attributes, they can be condensed into two categories: actors and factors. Actors are individuals or organisations who possess agency within the ecosystem to in some way affect or contribute to high-growth entrepreneurship. They have their own goals and priorities and can work to enact change based on these. The level of agency differs: entrepreneurs have a great deal of autonomy to create new organisations, networks, and shift the culture of a place. Larger organisational actors like universities, large corporations, or public support organisa-

tions are limited in their choices due to corporate or political considerations. Factors are social and economic structures and characteristics that affect how the entrepreneurship process plays out in a particular place. For example, a strong entrepreneurial culture might encourage workers to spin out their own firms, while the absence of physical infrastructure such as high-speed internet might create a barrier to high-growth entrepreneurship. Table 3.1 summarises the most common types of actors and factors that are cited in 11 major papers on entrepreneurial ecosystems.

The most commonly listed attributes are revealing, but do not necessarily indicate which are the most critical or which are unnecessary. Besides entrepreneurs themselves, the most commonly cited actor is investors and factor is supportive public policy. This reflects both their practical importance in financing firm growth as well as the substantial research literature on entrepreneurial finance and policy (Fraser et al. 2015). But other elements are less frequently cited despite their importance to cities. The remainder of the chapter looks at each of these actors and factors to explore their role in supporting high-growth entrepreneurship.

Spigel and Harrison (2018) distinguish between the strength of an ecosystem, defined by the amount of entrepreneurial resources in an ecosystem and its functionality, the ease with which entrepreneurs are able to access them. Strength of resources is a function of an ecosystem's actors: they produce and provide the resources needed to create and scale innovative firms. The functionality of an ecosystem stems from an ecosystem's factors like its culture, policy environment, and infrastructure. Therefore, to create a strong, well-functioning ecosystem requires not just assembling the right actors; it also requires the appropriate factors to ensure that resources can circulate within the ecosystem.

Table 3.1 The actors and factors of entrepreneurial ecosystems

Attribute	Type	Count (out of 11)	Definition
Entrepreneurs	Actor	11	Entrepreneurs and other actors who can provide collective leadership to identify and address challenges facing the startup and scale-up community
Investors	Actor	10	Private and public sources of startup capital including angel investors, venture capitalists, and public granting agencies
Skilled workers	Actor	9	The presence of a highly skilled labour pool in the region that is accessible to startup and scale-up firms
Role models	Actor	3	Successful entrepreneurs who mentor and advise younger entrepreneurs
Support professions and dealmakers	Actor	5	Business service providers such as lawyers, accountants, and advisors that specialise in the unique needs of startup and scale-up firms
Universities	Organisational Actor	7	Universities conducting cutting-edge research that spills over to the local community
Anchor firms	Organisational Actor	5	Large organisations such as branch offices, corporate headquarters, or universities that create new opportunities for entrepreneurs and attract skilled migrants to the region

Attribute	Type	Count (out of 11)	Definition
Incubators, accelerators and support organisations	Organisational Actor	7	Public and private organisations that train and otherwise support startup and scale-up entrepreneurs, including incubators, accelerators, and networking groups
Culture	Factor	9	A localised culture that encourages entrepreneurship, innovation and risk-taking and creates a high social status for ambitious entrepreneurship
Network	Factor	7	Dense social ties between entrepreneurs, investors, supporters, and other entrepreneurial actors that allow the flow of knowledge and insights within the community
Public policy	Factor	11	A local government that supports high-growth entrepreneurship through policies, public investment, and leadership
Markets and early customers	Factor	6	The presence of early customers and low barriers of entry for new ventures to join local supply chains
Physical infrastructure	Factor	7	The presence of high-quality office space, transportation facilities, digital networks, manufacturing space, and other necessary infrastructure for high-growth entrepreneurship

Note: Papers reviewed were Brown and Mason 2017; Cohen 2005; Dubini 1989; Feld 2012; Isenberg 2010; Mack and Mayer 2016; Mason and Brown 2014; Neck et al. 2004; Spigel 2017a; Stam 2015; and World Economic Forum 2013.

3.2 THE ECOSYSTEM ACTORS AND THEIR ROLES

3.2.1 Entrepreneurs and Ecosystem Leaders

By definition, an entrepreneurial ecosystem does not exist without entrepreneurs. Entrepreneurs identify opportunities and gaps in the marketplace, design new products, services, and business models to fill them, and then bring together the capital, skills, connections, and other resources necessary to create a successful venture. Entrepreneurs are leaders of their own ecosystem because they and they alone have the best perspective on what kinds of support they need and what kinds of support are unnecessary.

The popular perception of high-growth entrepreneurs is that they are young college dropouts who pursue their goal of building a billion-dollar technology startup. While there are examples of this, such as Bill Gates (dropped out of Harvard at 20 to start Microsoft) and Mark Zuckerberg (dropped out of Harvard at 21 to start Facebook), they are the exception to the rule. The founders of high-growth firms are likely to be in their late 30s or early 40s, have a bachelor's or graduate degree, and have prior industry or entrepreneurial experience (Anyadike-Danes et al. 2018). One of the most telling characteristics of a high-growth entrepreneur is previous experience in the sector they are competing within (Siegel et al. 1993). The stereotypes of high-growth founders being male and white still hold true (Brown et al. 2018). However, the majority of high-growth entrepreneurs are not in the technology sector, they are mostly in services or knowledge-intensive business services (Brown et al. 2017). Indeed, many high-growth firms are not necessarily new ventures: firms often remain small for several years before a market change, shift in the founder's goal, or infusion of capital spur growth (Miozzo and DiVito 2016).

Regardless of their personal characteristics, the most important attribute that distinguishes a high-growth entrepreneur from a lower-growth one is the desire for growth (Barringer et al. 2005). That is, the entrepreneur must want to create a rapidly growing firm rather than simply be carried along in the tides of the market (Hermans et al. 2015). Work on entrepreneurial environments and geography suggest that these desires are not randomly distributed across the landscape. People with such ambitions tend to cluster in the same place. This can be because a place has a particularly entrepreneurial culture, institutional configuration, or history that encourages ambitious growth outlooks (Aoyama 2009; Fritsch and Storey 2014) or because the place's reputation or economy attracts people with these pre-existing goals (Florida et al. 2008; Lécuyer 2006). This is not to say people with high-growth ambitions will only be found in these places; successful high-growth entrepreneurship can and does happen anywhere (Mahroum 2016). However, as ecosystems need a critical mass of highly ambitious, growth-oriented entrepreneurs, this creates constraints on the ability of struggling regions to try to create a healthy ecosystem without high-growth entrepreneurs.

Feld (2012) argues that entrepreneurs, rather than the government, are the best leaders of their ecosystem. Entrepreneurs are skilled at identifying challenges or gaps in the marketplace and marshalling the resources needed to address them. This makes them good at identifying common challenges facing their community and creating new groups, organisations, or movements to address them. Flexible and agile entrepreneurs can address immediate issues quicker than public agencies that are bound by procedural and budgetary limitations. Entrepreneurial leadership in ecosystems takes many different forms. In some situations, it involves organisational actions like creating formal or informal groups to help build the community or lobby to address common challenges (Thompson et al. 2017). In other cases, it might involve leading by example

by modelling pro-community practices like when a successful entrepreneur mentors others (Spigel 2017b).

Nevertheless, it is wrong to expect that entrepreneurs will create and maintain their ecosystem. The organisational efforts that underlie an ecosystem such as leading a group or setting up a talk series is a tremendous amount of work. Ecosystem building can distract entrepreneurs from focusing on their own firms. Organising events, lobbying politicians, and building coalitions is hugely time-consuming and requires social skills that few possess. Ecosystems, therefore, present a free-rider problem: entrepreneurs who put little effort into organising the ecosystem benefit as much as those who invest a great deal of their time and energy in creating organisations and/or orchestrating events (Pitelis 2012; 2013).

The willingness of entrepreneurs to contribute to and lead their ecosystem is, therefore, often driven by culture. A local culture that creates a high social status to leaders in the business community will encourage leadership without the need to personally profit from the ecosystem development activities. This type of culture is particularly important for ensuring that successful entrepreneurs stay in the ecosystem as their firm grows, both as entrepreneurs but also as investors, role models, and dealmakers. This recycling (see Chapter 4) helps build ecosystems over time through the actions of entrepreneurs.

Regardless of their ecosystem building actions, entrepreneurs are the most essential aspect of an ecosystem because they are ultimately responsible for the high-growth ventures that ecosystems seek to support. The purpose of all the other elements of the ecosystem — from policy to investment to cultural change — are all oriented towards growing entrepreneurs' growth ambition and innovative ability. Without a sufficient population of entrepreneurs interested in building a community, any other ecosystem actor or factor will ultimately be ineffectual.

3.2.2 Investors

Investors are the most commonly cited actor (besides entrepreneurs) within the ecosystems literature. The importance of investors within ecosystems stems from the geographically concentrated nature of risk capital. Even in a time of globally mobile capital, investors tend to invest in nearby firms (Mason and Harrison 2003). As Sorenson and Stuart (2001, p. 1550) note: "venture capitalists exhibit highly localised investment patterns in both physical and industry space." This is due to the role of social networks in the investment process. Investors use their networks to find potential investments, appraise the character and competence of the founding team, and monitor their investments after the fact (Rosiello and Parris 2009). This begins long before a formal pitch or investment proposal, with informal connections between investors and entrepreneurs used to guide the firms as they grow to the stage where they are investable. Networks help reduce information asymmetry in the investment process, where the entrepreneur attempts to provide the best possible picture of their prospects while hiding more negative information from the investor (Shane and Cable 2002). These networks are critical: investors with stronger network ties tend to perform better than those with weaker networks (Hochberg et al. 2007).

Investors' networks are generally local. Investors, like everyone else, build their social networks through daily interactions with others, introductions from people they know, and serendipitous encounters. While travel, conferences, and social media allow us to forge global connections, these more distant ties are harder to build and maintain than local ones (Bathelt et al. 2004). Investors have a better knowledge of nearby opportunities than more distant ones. This is most true for more informal investors like angels or family and friends, but more formal venture capitalists have exhibited a 'one-hour rule' in which most of the firms they invest in tend to be within a one-hour drive of their office (Griffith et al. 2007).

Investors are tied to their ecosystem; they provide important financing for entrepreneurs in the ecosystem, but they also depend on their networks in the ecosystem to identify and evaluate investment opportunities. But their role in the ecosystem extends beyond their capital. The most effective investors add value to firms by providing strategic advice, mentorship, serving on the board of directors, and using their connections to help entrepreneurs to enter new markets or find new customers (Politis 2008). This increases the competitiveness of these firms, helping them avoid common barriers in the scale-up process. Investors also act as key nodes in an ecosystem, helping to build networks within their community by sponsoring networking events or talks. Their position in the ecosystems gives them influence and power to bring different communities together in order to solve common problems.

The effectiveness of an ecosystem is tied to the quantity and quality of investors within it. A lack of investors is often one of the most pressing issues facing an ecosystem. Though this capital gap can be addressed through government intervention or high net worth individuals acting as early seed investors, the absence of skilled investors who evaluate firms and provide guidance on how to grow is a more difficult gap to fill. This knowledge is critical for helping firms to grow and is difficult to replace in the absence of experienced investors.

Innovative firms will find it difficult to scale if they are in ecosystems which lack available pools of investment capital. Without early-stage sources of capital from family and friends or angel investors, it is difficult for a company to develop new products or enter new markets. Without local investors, entrepreneurs have to dedicate extra time and resources to building non-local networks to investors in other places. Many regional governments have looked to use public investment programs to compensate for weak local investment scenes, but these programs often add less value to firms than private investors (Luukkonen et al. 2013) and the usefulness of such programs is questionable (Lerner 2009).

3.2.3 Skilled Workers

Skilled workers are one of the most important but most overlooked elements of entrepreneurial ecosystems. Decades of research on economic development, industrial clusters, and innovation systems has shown that the most important factor for regional economic success is human capital (Kerr 2018). New ventures need highly skilled employees in order to develop new products and business models that can quickly scale into globally competitive firms (Stam 2013). Skilled workers drive innovation, productivity increases, and help overcome problems that would stand in the way of growth. While we often think of skilled workers in terms of technical skills, such as programming or welding, business and creative skills are also crucial to firm performance in knowledge-based economies.

The availability of skilled workers is therefore a key ecosystem resource that entrepreneurs use as they grow their firms. Without access to skilled workers, entrepreneurial firms are at a competitive disadvantage, requiring that they either slow their growth due to lack of skills, invest time and energy in training employees to improve their skills or outsource work to other regions. These workers can be created locally — by gaining experience through their employment or getting training at a university or college — or attracted to a region through its high quality of life, low cost of living, or job opportunities (Imeraj et al. 2017). Universities and large anchor firms also play a major role in attracting skilled migrants to a region by creating educational or employment opportunities (Larsson et al. 2017; Wolfe and Gertler 2004).

The movement of workers between entrepreneurial firms in the same ecosystem is an essential vector for knowledge transfer (Dahl and Pedersen 2004). Workers who develop new skills and insights at one employer carry these skills with them, increasing the effectiveness of their new employers (Tambe and Hitt 2014). This knowledge transfer can increase

overall levels of innovation by spreading knowledge of new technologies and changes in the market (Henry and Pinch 2001). Indeed, employees' social networks can become an essential resource for firms, who use them to source new market knowledge or find additional employees (Lengyel and Eriksson 2017).

Workers at entrepreneurial firms can also be seen as a pool of potential new entrepreneurs in ecosystems (Dahl and Sorenson 2014). Spinout firms, in which workers at a larger firm leave to start their own firms in a related sector, have often been key founding events for new ecosystems, as was the case in both Detroit and Silicon Valley (Klepper 2010; Lécuyer 2006). Spinouts are founded by workers after they have developed their experience and legitimacy at a larger corporate employer before a disagreement with management, they develop a desire to own their business which leads them to engage in entrepreneurship, or they are laid off from their jobs and have to turn to entrepreneurship to preserve their income (Klepper 2007).

However, we cannot assume that all workers are equally willing or able to work at high-growth ventures. Employment at scale-up firms is typically more precarious than at larger corporate firms. High-growth firms have little job security, poorly defined job roles, and working outside normal hours is generally expected (Cockayne 2016; Neff 2012). While the lower pay at many entrepreneurial ventures compared to larger firms can be balanced out by stock options, the majority of these never accumulate value. This reduces the attractiveness of new and growing firms for many potential employees (Moser et al. 2017). In response, smaller firms typically emphasise their more open culture that allows employees more input on product design and strategy or their 'coolness' through a fun work atmosphere (Neff et al. 2005). Many strong ecosystems such as Silicon Valley, Waterloo, Canada or Boulder, Colorado have developed cultural outlooks that attach substantial social prestige to working at a 'cool' startup

or scale-up (Spigel 2017a). Such a culture attracts local workers from larger firms to work at smaller, riskier startups, helping the smaller firms benefit from knowledge spillovers.

The provision of skilled workers in an ecosystem is therefore not just a function of how many educated people there are in the workforce. It is a function of both human capital — the knowledge and skills they possess — and of cultural capital, their ability to function effectively in a startup or scale-up environment. It does not matter how much human capital there is in a region if entrepreneurs cannot access it. If entrepreneurs cannot access workers and their skills, it will be difficult for entrepreneurs to develop new products and grow the firm, ultimately limiting the capacity of the ecosystem to support growth.

3.2.4 Role Models and Mentors

Role models and mentors are successful entrepreneurs and business people who share their knowledge and insight with younger founders. Role models typically do this from afar, being observed by other entrepreneurs based on their success while mentors are more hands-on, working with one or several younger entrepreneurs to provide their experience and advice. Their presence makes three contributions to entrepreneurial ecosystems: encouraging new entrepreneurs, teaching entrepreneurial skills, and helping create and reinforce entrepreneurial cultures. As such, they play an important role in how ecosystems support firm growth through their direct impact on the entrepreneurs they mentor and also by helping to set a broader culture of mutual support within the entrepreneurial community.

First, their example encourages nascent entrepreneurs to take on the risks of starting their own new venture. There is plentiful evidence that shows that interacting with an entrepreneur — be it a parent, classmate, friend, or colleague — increases entrepreneurial intentions and self-efficacy (Bosma

et al. 2012; Lafuente et al. 2007; Mungai and Velamuri 2011). Interacting with an entrepreneurial role creates an opportunity for a shift in a learner's identity, allowing for the emergence of a more entrepreneurial self (Cope 2005).

Second, through their teaching or just by being observed, role models and mentors spread entrepreneurial know-how within an ecosystem. Learning by example helps entrepreneurs avoid common mistakes and other painful lessons that would otherwise have to be learnt through (repeated) personal experience. Importantly, the types of lessons entrepreneurs learn from role models and mentors changes over the course of their entrepreneurial lifecycle (Aldrich and Yang 2014). At early stages, the example of role models such as entrepreneurial parents or close family friends will help legitimise an entrepreneurial career and help normalise being alert for opportunities. After they have founded their venture, entrepreneurs can find other entrepreneurs in a similar sector to mentor them in order to learn new business practices and techniques (Zozimo et al. 2017).

Finally, mentors and role models can help cause broader cultural shifts within a region to create a more entrepreneurial culture (Spigel 2013). This is less through individual mentorship and teaching but rather by others in the community observing their success and imitating their actions, which in turn normalises entrepreneurship and reduces cognitive barriers to firm formation (Vaillant and Lafuente 2007). Over time, the presence of highly visible role models in a community reinforces an entrepreneurial culture, which in turn helps aid the emergence of new successful entrepreneurs as role models (Wyrwich 2012).

Like other actors in an ecosystem, mentors and role models affect their context and in turn, are affected by it. The decision to be a mentor is a choice, one which has rewards (the social rewards of being a mentor and the potential for financial rewards if it enables an early-stage investment) but also costs in terms of the time and energy dedicated to mentorship. The

culture and institutional structure of the ecosystem helps determine how much pull there is for experienced entrepreneurs to become mentors and how much demand there is for mentorship from younger entrepreneurs (Spigel 2017b). While different organisations like mentoring groups or accelerators can work to connect mentors with entrepreneurs, the effectiveness of these programs ultimately depends on the cultural value of mentorship that is found within the ecosystem.

3.2.5 Support Professionals and Dealmakers

Like mentors, support professionals and dealmakers provide insight and know-how to entrepreneurs. However, they do this for a much broader population of an ecosystem. They build networks and support infrastructure within ecosystems, creating forums for entrepreneurs and other actors to meet, and share insights. In addition to giving expert advice, they use their social and financial capital to contribute to the regional ecosystem by sponsoring events, serving on boards and committees, and otherwise building coalitions of entrepreneurs and government officials to improve a place's overall entrepreneurial conditions.

Support professionals are specialised advisors such as lawyers, accountants, and investment bankers that specialise in the needs of high-growth startups (Kenney and Patton 2005). Scale-up firms face unique legal and accounting challenges, such as organising stock options, reviewing investment term sheets, or patent applications. While some of these issues can be dealt with by generalists, specialised challenges require specialised advisors who have experience in complex issues like corporate governance, IP law, or options valuation. There is a circular and cumulative effect in which the proliferation of high-growth startups in a region allows support professionals to specialise in their needs, which in turn helps even more high-growth startups to form and thrive (Patton and Kenney 2005).

Over time, the role of these support professionals may begin to move beyond their role as specialised legal or accountancy advisors to become more diversified business advisors. Suchman (2000) argues that the development of modern Silicon Valley involved local business lawyers becoming matchmakers, helping their clients find new business opportunities. This helped form dense local networks that helped entrepreneurs access new sources of financing market growth. Legal firms, accountancies, or consultancies may sponsor events that act as gathering places for entrepreneurs. These events act as the infrastructure for the ecosystem. They provide a forum for entrepreneurs to network and creates the potential for them to connect with new advisors or find a mentor to help them on their journey.

More broadly, support professionals make up a large proportion of what Feldman and Zoller (2012, p. 24) refer to as dealmakers: "individuals with valuable social capital … [who] act in the role of mediating relationships, making connections and facilitating new firm formation." This ranges from major billionaire philanthropists such as Ewing Marion Kauffman in Kansas City to professionals like bankers, lawyers, or successful entrepreneurs who want to act as stewards of their local community (Feldman 2014). In essence, dealmakers are successful and respected members of a region's business community who use their own social capital to support the regional business environment. While in years past this might have involved serving with the local Chamber of Commerce, more recently this is more likely to involve more direct activities such as serving as a board member for an entrepreneurial support organisation, founding new support programs to help specific groups, or proactively connecting with entrepreneurs and building new network connections between them (Storper et al. 2015). Dealmakers can leverage their social capital to help individual firms by serving as non-executive directors or as members of an advisory board. In this capacity, dealmakers use their extensive personal and business networks to help

firms connect with new customers, suppliers or investors. This increases firm performance and survival (Kemeny and Feldman 2014).

Like other sources of business advice, the impact of specialised support professionals and dealmakers is local (Kenney and Patton 2005). Entrepreneurs turn to local sources for business advice due to the ease of arranging a meeting and the difficulty of matching with a good advisor outside their local community. Similarly, dealmakers have a specific connection with their home community that makes them want to give back to it. Thus, the presence of specialised advisors and dealers constitute an ecosystem asset that local entrepreneurs benefit from, in turn creating the potential for the creation of more specialised advisors and dealmakers.

3.2.6 Universities

Knowledge-intensive research universities are almost universally regarded as an important element of successful entrepreneurial ecosystems. Histories of entrepreneurship in Boston (Saxenian 1994), Silicon Valley (Lécuyer 2006), and Cambridge (Huggins 2008) have pointed to universities' role in building a conducive ecosystem. Universities have shifted from being cloistered places of learning to economic development engines (Valero and Van Reenen 2018). They have the potential to perform four critical functions in entrepreneurial ecosystems: knowledge creators, promoters of academic entrepreneurship, talent producers, and ecosystem coordinators. These actions produce new entrepreneurs, skilled workers, and new technological knowledge that power an ecosystem's development and growth.

The most well-studied aspects of universities are their role in knowledge creation and spillovers. Universities produce new cutting-edge knowledge through their basic and applied research activities. This knowledge spills over to the surrounding region through formal and informal connections between

research labs and nearby firms (D'Este and Patel 2007). These connections range from formal knowledge-sharing agreements in which a private firm sponsors university research to more informal and contingent ties such as engineers from a local firm attending a research talk or graduates taking jobs at local firms (de Wit-de Vries et al. 2018). This knowledge can form the basis for new or improved products or processes that increase firm competitiveness or represent new entrepreneurial opportunities. As with other types of knowledge transfer, while this transfer can happen globally much of it happens locally through informal connections (Gertler and Levitte 2005). But while these knowledge spillovers have been shown to have beneficial effects for regional economies (Stuetzer et al. 2017), we do need to be a bit sceptical on how much university research benefits local entrepreneurial firms. There is an often myopic focus on the research universities produce with little understanding of how much of it is commercialised (Brown and Mawson 2019). Entrepreneurial firms have limited absorptive capacity, which is needed to incorporate complex external knowledge into firm operations (Qian and Acs 2011).

Universities are also important sources of entrepreneurial activity by faculty and students who commercialise their research (Shane 2004). Academic entrepreneurs can create new spinout firms to commercialise their research by finding outside investors to help finance the R&D required to transform basic scientific discoveries into sellable products. Because universities retain an equity stake in these spinouts, many offer entrepreneurial support such as providing initial seed capital, access to university facilities, and incubation programs to help early-stage firms develop their product and find initial customers (Markman et al. 2005). But even this has a limited effect given the small number of academic spinoffs that succeed in becoming scale-up entrepreneurial ventures.

This means that the role of universities in ecosystems research is not limited to new knowledge creation. Indeed,

one of universities' most vital roles in an ecosystem is producing and attracting highly skilled workers who start or join startups and scale-ups. They do this by developing students' skills by teaching them technical and critical thinking skills or attracting already skilled researchers as staff or faculty. But, as was argued earlier, human capital is not useful to entrepreneurs if they cannot access it. Therefore, universities in ecosystems must also help instil an entrepreneurial spirit in students and faculty. This involves both formal entrepreneurship education programs as well as creating a cultural ethos supportive of entrepreneurship. An entrepreneurial culture at a university promotes entrepreneurship as a legitimate career option compared to other paths such as academics or corporate employment (Bercovitz and Feldman 2008). This encourages staff and students to be more alert to entrepreneurial opportunities emerging out of their research, creating the potential for spinout formation. An entrepreneurial culture also shifts institutional priorities, allowing entrepreneurship and commercialisation to be seen as an important part of academic career progression rather than a distraction from its primary research mission (Colyvas 2007). Over time, successful university spinouts from an entrepreneurial university can help shift a region's cultural outlooks, making the entire place more entrepreneurial (Bathelt and Spigel 2011; Bramwell and Wolfe 2008). The effects of this can be substantial: MIT estimates that its alumni have founded more than 30,000 firms which together employ 4.6 million people and produce revenues of 1.9 trillion USD (Roberts et al. 2015).

Finally, universities can be important ecosystem coordinators. Their large institutional presence in a region, ability to operate programs such as incubators and accelerators and role as educators give universities a powerful platform to bring together disparate political and business actors to build new support programs. Universities like Carnegie Mellon in Pittsburgh and Cornel University in New York City have become leaders of wide-ranging programs designed to develop

local entrepreneurial ecosystems focused on cutting-edge technologies (Katz and Nowak 2018). Universities frequently host speaker series, entrepreneurship training programs, and other events that bring together a community's entrepreneurial community, allowing stronger network ties to form (Williams et al. 2016).

Universities have a multifaceted role in ecosystems as both producers of knowledge, entrepreneurs, and workers as well as being a hub of an entrepreneurial community. Not all universities have the capacity to take on all these roles. However, histories of thriving entrepreneurial ecosystems in the US and Europe have shown that universities are a common starting point for the creation of a strong ecosystem as well as a powerful engine for strengthening the ecosystem as it develops. Even in the absence of novel research and spinouts, universities' role as talent producers and ecosystem coordinators can serve as a keystone in smaller ecosystems in a variety of different contexts.

3.2.7 Anchor Firms

Much like universities, anchor firms — large corporate or public employers based in a region — are important actors in an ecosystem because they attract skilled workers (and potential entrepreneurs) to the region, support the local business community and serve as a source of entrepreneurial opportunities. While most anchor firms are large, multinational companies, public employers such as research labs, hospitals, or federal civil service centres can also play these roles (Feldman et al. 2005). Indeed, given the profile of the typical high-growth entrepreneur discussed in Chapter 2 (in their late 30s with several years of industry experience), anchor firms may be more important than universities in building a pool of potential high-growth entrepreneurs. Workers at anchor firms gain unique insights into niche market opportunities that can form the basis for entrepreneurial ventures. Their

association with anchors builds their reputation within their industry, which may help their ventures overcome the liability of newness that is often associated with new ventures (Delmar and Shane 2004).

Spinouts from anchor firms are an important mechanism for building a region's entrepreneurial community and culture. Studies of Detroit in the 1900s and Silicon Valley in the 1960s and 1970s have shown how disagreements between workers and management in existing firms led to spinout activities, which started larger entrepreneurial movements in those regions (Klepper 2007; 2010). The departing entrepreneurs are able to build on the processes and routines of their parent firms, driving forward innovation and creating cluster dynamics that improve the overall competitiveness of firms in the region (Martin and Sunley 2006). Disagreements between nascent entrepreneurs and management are not the only source of spinouts: large-scale layoffs or corporate restructurings are another way anchor firms can create entrepreneurial activity in a region (Spigel and Vinodrai 2020). The loss of a job reduces the opportunity cost of venture creation (Virick et al. 2013). Studies of Washington DC's bio-tech cluster (Auerswald and Dani 2017; Feldman 2001) and Ottawa's telecommunications cluster (Harrison et al. 2004) have demonstrated how layoffs of publicly employed scientists and engineers contributed to bursts of entrepreneurship in the region which seeded the development of strong entrepreneurial ecosystems.

Some anchor firms encourage spinouts because management believes that the entrepreneurs' innovations may ultimately add value to the parent firms while others have tried to ruthlessly block spinouts for fear of creating new competition (Saxenian 1994). In either case, anchor firms often have a profound effect on the business culture of their home region (Schoenberger 1997). The founders or senior executives of these firms become role models and prominent leaders of the local business community. The practices of anchor firms and their management team, such as their support for entre-

preneurial action, inter-firm cooperation, and openness, will diffuse into the broader economy and affect the state of the local culture.

Increasingly, some anchor firms have taken up the role of promoting entrepreneurship within ecosystems they are located in. One of the major ways this occurs is through corporate accelerators: intensive entrepreneurship support programs in which a large firm supports new ventures which may eventually develop new products and services that complement the anchor's activities (Kohler 2016). The accelerator provides training and investment in new ventures while at the same time helps to spur intrapreneurship at the corporate firm by exposing executives to new business models and technologies (Shankar and Shepherd 2018). An anchor firm's top management team may also become local dealmakers, working with politicians to improve the region's overall business environment or serving on the board of directors for local scale-up firms.

Finally, anchor firms can play a unique role in ecosystems by becoming a source of opportunities for local entrepreneurs (Gray et al. 1996). While high-growth entrepreneurship is often seen as the domain of 'born globals' who focus on global demand, anchor firms have large supply chains that create new opportunities for entrepreneurs that can be filled by spinouts or local entrepreneurs. This is particularly true of industrial anchor firms or those in sectors that outsource a great deal of technological development, such as the oil and gas industry or pharmaceutical development (Spigel 2017a). Local needs create opportunities that entrepreneurs can build upon to eventually reach global markets.

Anchor firms' importance in ecosystems stems from their scale. They are one of the few actors who can do things such as attracting new skilled workers to the region, support activities like corporate accelerators or spinoffs, and otherwise convene different parties together to build a stronger environment. Just as importantly, their scale allows them to create dif-

ferent entrepreneurial opportunities an ecosystem can develop around. Given the multi-varied roles they can have, the part they play in ecosystem development has differed from an active participant to more passive. However, their presence is an important footing on which ecosystems can develop on.

3.2.8 Incubators, Accelerators, and Support Organisations

Entrepreneurship support organisations (ESOs) are the public, private, or quasi-public organisations that provide different types of training and support to new and growing ventures. The purpose of ESOs is to help entrepreneurs correct for the liability of newness: the vulnerability new organisations face due to their lack of resources, market position, and legitimacy. Early-stage entrepreneurship is marked by uncertainty — the unknown unknowns of if a technology can be developed or a market entered — that make investment particularly hazardous (Nambisan 2018). This leads to the so-called valley of death in which innovative firms need early-stage capital to fund product and market development efforts but lack the legitimacy to get this investment.

ESOs help new and growing ventures funding gap through financing and training. Without such support, firms with potential may fail in infancy due to temporary resource constraints. Ideally, these programs identify early-stage entrepreneurs, give them support and training and with additional financing that helps them develop market-ready products, allowing the entrepreneurs to become commercially sustainable on their own. Public investment in these programs has become one of the central tools available to regional and national governments to promote entrepreneurship-led economic development and job-creation efforts (Bondonio and Greenbaum 2014; Feldman et al. 2016).

There is no clear typology of ESOs, but we can distinguish three major categories: incubators, accelerators, and other

support programs. Incubators provide subsidised office space, training, and funding for highly innovative ventures over a long period of time to help them commercialise new technologies. Such organisations date back to the 1980s and are often connected with research universities or science parks looking to support academic entrepreneurship (Amezcua et al. 2013). They are aimed at ventures which require long periods of R&D, product development, and regulatory approval which would otherwise make it difficult to find the resources from private sources.

Incubators provide three key resources. First, they provide subsidised infrastructure such as office space, laboratory equipment, and other materials, reducing the overheads facing scientific firms and limiting their need for initial seed capital. Second, being part of an incubator provides a degree of legitimacy, helping them forge collaborations and partnerships (Theodoraki et al. 2017). Finally, incubators generate proximity between technology entrepreneurs, encouraging the flow of entrepreneurial knowledge and insight between incubated firms regarding common challenges (Fernández et al. 2015). Generally, incubators do not put time limits on firms, allowing them to use the facility until the founders can survive without the incubator's support or come to the conclusion their venture is not viable. However, this support also creates the potential of perpetuating 'zombie firms' that would not survive without the incubator's support (Harrison and Leitch 2010).

Accelerators are a more recent development in entrepreneurial support. Like incubators, accelerators provide office space and other support services to firms with whom they work. However, this support is time-limited, often to less than a year, in order to encourage rapid product development and commercialisation (Drori and Wright 2018). Very early-stage firms are taken in cohorts and run through training programs based around lean startup methodologies so that the founders can quickly achieve product/market fit (Pauwels et al. 2016). The cohort finishes the accelerator after a specified period of

time by demoing their products to a group of investors. Unlike incubators which are typically run as non-profits connected with universities, accelerators are often for-profit firms that generate revenues by taking an equity stake in the firms they work with, so that they profit as the firm grows and exits (Hochberg 2016). This creates a variety of different accelerator forms, including private accelerators run by entrepreneurs, corporate accelerators designed to help a larger firm enter new markets or spur innovation in their own platform, as well as public accelerators designed to foster an entrepreneurial ecosystem and generate economic growth (Bliemel et al. 2018; Shankar and Shepherd 2018).

Other types of ESOs have different missions and goals, depending on their circumstances and context. Their functions often include training, funding, and building networks within the entrepreneurs' community and between entrepreneurs and other stakeholders such as universities, corporations, or investors (Motoyama and Watkins 2014). These organisations perform functions that have not been provided by the private sector due to a lack of capacity or market failure (Mole and Capelleras 2017). In addition to providing direct resources such as training or financing, ESOs can also function as intermediaries by helping forge connections between different actors and coordinate larger support networks within regions (Clayton et al. 2018). However, the efficacy of these programs has often been questioned, with their impact hard to determine, given the ambiguity of the entrepreneurship process (Brown et al. 2015).

ESOs of all types have been well studied in the literature as part of wider debates around the purpose and effectiveness of entrepreneurship policies (Lundstrom and Stevenson 2006). However, ESOs have typically been studied in isolation, with researchers looking at the impact of a single program or type of support (e.g. Bondonio and Greenbaum 2014; Larty et al. 2017). One of the major contributions of ecosystems perspectives to date has been to advance a more holistic view of

entrepreneurship support which examines the interactions and overlap between different forms of entrepreneurship support (Motoyama and Knowlton 2016; Spigel 2016). Instead of looking at the activities of a single program, ecosystem perspectives look at the array of different types of support available in the ecosystem, particularly in terms of the ability of programs to 'hand off' graduates to others so that there are continual sources of support available for scaling firms (Harrington 2016). This perspective also includes the role of more informal organisations and entrepreneur-led groups in addition to the formal programs funded by the state (Colombelli et al. 2017; Hanlon and Saunders 2007).

3.3 ECOSYSTEM FACTORS AND THEIR EFFECTS

3.3.1 Entrepreneurial Culture

Researchers have studied the relationship between culture — the shared meanings and understandings of the external world held by a community — and entrepreneurship for more than a century. Weber (1930) famously located the economic success of Northern European nations in a particular Protestant culture that encouraged entrepreneurship and frugality. This idea was built on by a series of researchers in the 1990s and early 2000s who compared the impact of national cultural outlooks on entrepreneurial activities (e.g. McGrath et al. 1992). Such work found differences in the size of personal networks (Drakopoulou Dodd and Patra 2002) and attitudes towards risk-taking and competitiveness (Lee and Peterson 2000). This work found that cultural beliefs about community, individuality, and wealth influence the willingness of individuals to engage in entrepreneurship and produce more subtle effects on how entrepreneurs interact with their community and find the resources they need to build innovative and high-growth firms (Hayton and Cacciotti 2013). This research aligned with

a broader shift in the social sciences at this time that tried to engage with culture not as a separate thing from the economy but seeing both as intertwined forces that produce the human experience (Lash and Urry 1994). Cultural outlooks were an important institutional force that helped produce economic development trajectories and even entire systems of capitalism (Putnam et al. 1993).

However, national cultures are often too heterogeneous to be useful for predicting economic outcomes. This has led other researchers to look at the impact of smaller-scale regional cultures on entrepreneurship and other economic activities (Aoyama 2009; Gertler 2010; Hayton and Cacciotti 2013). These are the cultural outlooks towards risk, innovation, cooperation, and entrepreneurship that build up in specific places over time due to its economic and social history (Gertler 2004). These cultural outlooks have a profound impact on not only entrepreneurial outcomes but also the specificities of how the entrepreneurship process plays out locally. AnnaLee Saxenian's (1994) history of Boston's Route 128 and Silicon Valley was one of the founding works in this tradition. She traced how cultural attitudes started in universities like Berkeley and Stanford that encouraged highly risky, innovative entrepreneurship and spinouts helped Silicon Valley emerge and navigate complex changes in the global economy in the 1970s and 1980s. At the same time, the more conservative, finance-led culture in Boston led to the downfall of the region's burgeoning microcomputer cluster in the 1980s. In her study, Saxenian showed how cultural norms of trust, reciprocity, innovation, and risk-taking developed (or did not develop) in each place over decades, which then informed the business practices of entrepreneurs and investors.

Local cultures develop over time though contingent historical events such as the legacy of a dominant industry, the actions of prominent individuals, as well as broader social trends such as demographic change. Regardless of its origins, a place's culture influences future economic and social tra-

jectories by developing collective understandings within the community that shape their practices and goals. Some cultural beliefs might enable activities associated with high-growth entrepreneurship, while other beliefs might discourage these actions. Cultural attitudes that support openness and cooperation between firms encourage knowledge sharing and cooperation while cultural outlooks focused on firm secrecy and zero-sum competition will act as barriers to these activities. In other words, business and economic practices are not constructed in a rational manner based on anticipated risk and reward. These practices are deeply embedded in localised cultural structures that influence what types of activities are seen as legitimate and accepted and which are illegitimate or taboo (James 2007).

Local cultures are durable and hard to change, even in the face of extreme circumstances. For example, areas of the former East Germany with higher rates of entrepreneurship before the Second World War maintained these rates after the fall of the Berlin Wall, despite fifty years of repression of entrepreneurship (Fritsch and Storey 2014; Wyrwich 2012). Shifting the culture of a place takes sustained effort throughout the community for years or even decades. Highlighting entrepreneurial success stories is an important agent of change because it helps normalise entrepreneurial action, showing others that it is an acceptable career path (Spigel 2013). Others have noted the importance of civic leaders developing new organisations and institutions to encourage trust within the business community (Storper et al. 2015).

Every major work on entrepreneurial ecosystems has identified local culture as a keystone of the broader ecosystem. Spigel (2017a) sees cultural factors as the foundation for other elements of the ecosystem because culture helps to orient these otherwise independent actors and factors towards a common goal of increasing high-growth entrepreneurship. In the absence of a supportive culture, even well-designed and well-meaning policies aimed at supporting scale-up entrepre-

neurship will likely fail because it will be difficult to generate the community buy-in necessary for these programs to work (Dennis 2011; Lundstrom and Stevenson 2006). A strong local entrepreneurial culture is associated with higher rates of entrepreneurial mentorship (Lafuente et al. 2007), increased trust between founders (Aoyama 2009), and leads to entrepreneurs being better at identifying opportunities (Stuetzer et al. 2014). Such characteristics, in turn, contribute to higher levels of opportunity-driven entrepreneurship and increase the survival of the firms that are created (Malecki 2009).

But while ecosystem research has extolled the importance of an entrepreneurial culture, there is little clarity over if and how such cultures can be purposefully developed. The emergence of a localised culture is a long-term process dependent on a place's economic and social legacy and contingent historical actions such as a prominent business person dedicating their social capital to developing the local startup community. Individual case studies of the entrepreneurial ecosystems such as Phoenix, Arizona (Mack and Mayer 2016) or San Diego, California (Walshok and Shragge 2014) have shown their entrepreneurial cultures developed organically over long periods of time in response to internal changes and external shocks. Organisational cultures from large corporate firms, public sector employers or universities can permeate through a region, creating local norms for entrepreneurial action (Harrison et al. 2004). Successful businesses and entrepreneurs play an important role in building local cultures, as their practices become legitimised in the community as a result of their success (Spigel 2013).

3.3.2 Networks

Networks connect entrepreneurs with their broader environment. Since the early 1980s, researchers have noted how founders use their social ties to identify opportunities and unmet needs in the marketplace, acquire insight about how to

pursue these opportunities, and find the financing and other resources they need to build new organisations (Birley et al. 1991). Work in this field has looked at the types of resources, such as financial capital or market knowledge found in entrepreneurs' networks (Keating et al. 2014), how the structure of their networks impacts their usefulness (Burt 2004), and how these networks are developed, maintained, and governed (Rasmussen et al. 2015). This contributed to a vibrant literature that decentres the study of entrepreneurship away from the founder, her firm, and its internal resources to the external connections with her broader environment (Hoang and Antoncic 2003). Indeed, within the field of entrepreneurship research, perhaps the only consistent finding across times, geographies, and contexts is that entrepreneurs with more network connections tend to perform better than those with fewer (Stam et al. 2014).

The importance of networks stems from their role as vectors for different resources entrepreneurs need to create and scale firms. From the resource-based view of the firm, success depends on the ability to bring together bundles of different resources that enable them to create value for customers (Kor et al. 2007). These resources range from intellectual property, knowledge of how to market to customers, key employees and partners, as well as the financing required to carry out a firm's key activities (Lichtenstein and Brush 2001). Entrepreneurs do not possess all these resources at the time of firm formation — if they did they would not be engaging in entrepreneurship. Rather, they draw on their existing social networks and build new connections to find the resources they need or the support they need to create new resources.

Entrepreneurs use their networks throughout the entrepreneurship process. At the very earliest stages, they use their networks to identify new ideas and opportunities (Arenius and de Clercq 2005; Tocher et al. 2015). This may involve flows of knowledge between a potential entrepreneur and industry colleagues that help identify an unmet need that can set off

an entrepreneurial journey. As the entrepreneurial journey continues, one of the most important uses of networks is the acquisition of entrepreneurial knowledge and know-how (Hanlon and Saunders 2007). Friends, colleagues, and prior contacts are used to test ideas and get advice on firm strategy (Audretsch et al. 2011a). Entrepreneurs use their connections to find potential mentors and advisors who can help them avoid common entrepreneurial mistakes (Zozimo et al. 2017).

As a firm develops, networks connect entrepreneurs with potential investors and customers. Investors such as early-stage angels and venture capitalists use their social connections to uncover potential investments and to appraise the quality of both the venture and the founding team (see section 3.2.2). The importance of networks to evaluate the venture stems from the inherent information asymmetry between the entrepreneur and the investor: the entrepreneur will present information on the firm in the best possible light and attempt to obscure negative information (Shane and Cable 2002).

However, network connections by themselves do nothing: entrepreneurial actors must purposefully use them if they are to give access to the resources and support they need. This requires trust between parties. Trust is important because as Dubini and Aldrich (1991, p. 306) explain: "Mobilizing resources also involves asking others to raise money, labor, and effort for a venture with an uncertain future." Those providing the resources must trust that the entrepreneur will reciprocate in the future. Trust is built over time after many repeated interactions and is aided by the presence of dense, overlapping connections between parties (Henn 2012).

As discussed previously, entrepreneurs' networks tend to be localised. While the internet and cheap international travel allow entrepreneurs to build connections across the globe, the bulk of their networks tend to manifest in the places they live (Schutjens and Völker 2010). This is evidenced by the superior performance of entrepreneurs who have lived in a place for several years before they start their firm compared to new

arrivals, as their localised networks are better positioned to help them access the resources they need (Dahl and Sorenson 2012). Knowledge and insight about the entrepreneurship process buzz throughout a local community while global pipelines to distant locations require more effort to build and maintain (Bathelt et al. 2004).

Locally based networks are key processes through which ecosystems operate (Spigel and Harrison 2018). Entrepreneurs' social and business connections link them and their firms with the broader community they work within, providing a means for them to access resources and insights outside of very formalised, de-personalised methods like bank loan applications. Ecosystems benefit from the presence of dense networks that connect entrepreneurs together and with other ecosystem actors like advisors, mentors, customers, and investors. Networks bind together an entrepreneurial community and act as a vector for the flow of entrepreneurial resources and know-how within a community. Particular individuals with particularly well-developed networks may emerge as dealmakers able to connect different parties in order to create more entrepreneurial value (Feldman and Zoller 2012). Networking events that bring together different parts of an entrepreneurial ecosystem are one of the most common support programs found in most ecosystems (Spigel 2016). Intermediary organisations are created to actively build new networks within an ecosystem to encourage entrepreneurship and knowledge spillovers (Clayton et al. 2018). Thus, almost every aspect of an ecosystem either requires networks to function or has the goal of building and improving local networks between entrepreneurial actors.

The nature of networks — their density, the trust between actors, and the types of resources that flow within networks — are linked with the broader cultural norms they were created within (Malecki 2012; Sorenson 2019). Different groups and places have different norms regarding the importance of networking, such as how much trust to put in professional col-

leagues, and the extent to which it is acceptable to ask social contacts for business resources (Coleman 1988; Portes 1998). The way entrepreneurs develop and use networks can be seen as one of the clearest manifestations of a culture. This means that poor networks in an ecosystem are likely not simply the result of too few networking programs or events, but rather reflect a local culture that discourages inter-personal trust. Thus, the promotion and normalisation of entrepreneurial networking is a key policy intervention for many nascent ecosystems.

3.3.3 Public Policy

Public policy is the use of government power and resources to bring about a particular social outcome, such as an increase in the number of entrepreneurs or the survival of early-stage firms (Hart 2009). Within the last fifty years, entrepreneurship and SME policy have become a major government priority: over 12 billion GBP is spent annually in the UK on supporting small businesses and entrepreneurs (Arshed et al. 2014) while Sweden spends more on small- and medium-sized business support than its police (Fotopoulos and Storey 2018). This spending is motivated by the belief that entrepreneurship is an economic engine for job creation and innovation, but particular market failures hurt entrepreneurial ventures (Dennis 2011). Both national and regional governments have increased their interventions to encourage entrepreneurship and support small and young businesses as part of these efforts. Such interventions range from small, tactical investments in accelerators or other ESOs to more fundamental shifts in taxation, immigration, government operations, and intellectual property regimes.

Governments have multiple policy options they can employ to achieve their entrepreneurship and economic development goals. Lundstrom and Stevenson (2006) developed a three-part framework of entrepreneurship policy in which

policy is aimed at improving entrepreneurial motivation (the willingness of existing residents to engage in entrepreneurship), opportunities (the resources and support available in the community for entrepreneurs) and skills (the business acumen of existing entrepreneurs). This then leads to five broad policy levers states can employ:

- Increasing the demand for entrepreneurship by supporting the development of a dynamic economy rich in opportunities.
- Increasing the supply of potential entrepreneurs through educational policy, immigration policies that allow in more entrepreneurs, and promoting a more entrepreneurial culture.
- Increasing the skills of existing entrepreneurs and the resources available to them through the provision of training, support for accelerators or incubators, and financing for R&D.
- Interventions to shape the broader entrepreneurial culture such as through media programs, educational support for entrepreneurship, and awards for entrepreneurs.
- Altering the risk and reward profiles of entrepreneurship through changes in tax law.

These five broad policy levers from governments create a universe of options to spur entrepreneurship, ranging from tax incentives for early-stage angel investors, business plan competitions, new mandates for primary and secondary schools to teach enterprise skills, or the development of intermediaries to encourage academic entrepreneurship. However, the overall effectiveness of policy interventions is difficult to isolate (Minniti 2008). Some policies, while well-meaning, end up crowding out private investment. Others simply cannot overcome long-term existing institutional and cultural attitudes that act as barriers to entrepreneurship. More perniciously, financial support for entrepreneurial firms may prevent inefficient firms from failing, creating ‘zombie’ firms

that devour scarce resources (Acs 2010). However, despite its flaws, policy is still important, particularly in regions missing important entrepreneurial resources or actors. In an ideal situation, supportive policy corrects a market failure and the entrepreneurial activity that is supported helps build a more sustainable entrepreneurial ecosystem that does not rely on continual public subsidy.

Research on entrepreneurial ecosystems has a bifurcated attitude towards the role of entrepreneurship policy. Many ecosystem practitioners like Feld (2012) are sceptical of top-down government interventions in building and maintaining entrepreneurial ecosystems. The legacy of failed policies to build new Silicon Valleys has shown the limitations of what public policy can achieve (Audretsch 2019). Many governments are quick to import perceived best practices from more prosperous regions only to see them fail because they do not align with local contextual factors (Harrison and Leitch 2010). While governments have resources and power, they often have little insight into the actual needs of high-growth entrepreneurs and instead invest in areas that do not need support. This leads to ineffective, unneeded support and wasted public funds.

However, on the other hand, researchers have identified governments as one of the few actors who can help break the status quo of a weak ecosystem to help improve conditions or help bring new resources to the ecosystem (Colombelli et al. 2017). Only governments can provide public goods such as education, infrastructure, and early-stage funding that are nearly impossible for the private sector to offer on its own (Pitelis 2013). Given the importance of universities and early-stage funding, the government will always have a role in cultivating and supporting ecosystems. However, and Autio and Levie (2017) argue, public policies for ecosystems must be based around deep stakeholder engagement to identify the needs of entrepreneurs rather than top-down solutions.

Despite the lack of consensus about the effectiveness of public policies, entrepreneurial ecosystem approaches represent two advances in how we understand the role of public policy in supporting entrepreneurship. The first is the differentiating between enterprise support and high-growth policy. Enterprise support represents how the government supports all forms of entrepreneurship, the bulk of which consists of sole traders and other low-growth firms. This may involve training on how to incorporate and file for taxes or help writing up a business plan. High-growth policies are focused on the specific needs of the smaller subset of entrepreneurs with growth ambition. They need different types of support both at the early stage in order to access financing to support risky R&D to later-stage support to access highly skilled workers and global markets. This is not to say that enterprise support policies are unnecessary but rather that high-growth support is a separate policy domain.

Second, ecosystems perspectives shift how we understand the relationship between different support policies. As with ESOs (see section 3.2.8), instead of looking at the impact of individual initiatives or policies, we instead see each policy and program as a node in a broader network of public and private support. Support typically targets particular sectors (e.g. life science or digital), types of entrepreneurs (e.g. young people or women) or phases of the firm lifecycle (pre-start or scale-up). Thus, entrepreneurs will not receive support from a program for their entire journey but will instead pass between multiple support programs as their needs change. For example, Spigel (2016) identifies more than 60 support programs aimed at technology entrepreneurs in Edinburgh. It is not one policy contributing to an ecosystem, but multiple different policies sponsored by several different government bodies at a variety of spatial scales.

Public policy is a crucial element of entrepreneurial ecosystems, but ultimately a limited one. There are few actors besides the State that have the resources and mandate to

bring together disparate actors in the ecosystem and provide resources such as early-stage risk capital that the private market cannot. Ecosystems arguments shift how we understand the role of policy, away from single programs designed to increase the overall firm formation rate towards networks of more specialised programs aimed at a smaller subset of high-growth entrepreneurship. But most importantly, ecosystems approaches acknowledge the limitations of policy. Top-down public intervention alone cannot build an ecosystem where the underlying attributes are absent nor can it shift cultural outlooks by itself. Instead, policymakers must work within existing structures and constraints to strengthen existing elements and work to cultivate weaker ones.

3.3.4 Local Markets and Early Customers

There is little doubt that regions with vibrant, growing economies have higher rates of opportunity-driven entrepreneurship than depressed ones (Glaeser et al. 2010). Economic growth creates new opportunities for entrepreneurs in both serving the expanding consumer market as well as potential to enter the supply chains of the industries that are driving the economic expansion. While early papers such as Malecki (1993) and Spilling (1996) mentioned the importance of having a dynamic local economy for supporting an entrepreneurial community, the contemporary importance of local markets in entrepreneurial ecosystems has been most forcefully made by practitioners such as Isenberg (2010) and economic development groups such as the World Economic Forum (2013; 2014). The WEF, in particular, stressed the importance of having open local markets that included large and small companies as well as governments as potential early-stage customers. This viewpoint aligns with the WEF's larger agenda of reducing barriers to business.

In some ways, the importance of local markets for high-growth entrepreneurship can be seen as an unex-

pected argument, given the increasingly globalised nature of high-growth entrepreneurial firms. We typically have a vision of a high-growth firm as a born-global that uses digital tools to sell to a global market from day one. However, this is rarely the case. Many high-growth firms depend on their local market for early customers who they work with to build a product that fits their needs, providing a foundation for future growth (Roundy 2018). This is most apparent for manufacturing firms which enter the supply chains of larger local customers. Indeed, Porter (1998a) saw local customers as one of the building blocks of industrial clusters because of their importance as early customers for entrepreneurial firms.

Even in the absence of large anchors or clusters, early customers in the local environment are a crucial tool for opportunity identification. Entrepreneurs find or create opportunities by drawing on their local social networks (Ramos-Rodríguez et al. 2010). While their product may be aimed at national or global markets, an entrepreneur's initial product often serves the needs of local customers since they are the easiest to access. For business-to-business firms, the density of the founders' local contacts and their ability to work closely with local customers may make it easier for entrepreneurs to acquire referred customers who can help them build up legitimacy in the marketplace (Ruokolainen and Igel 2004).

This means that a region's pre-existing industry structure and its business culture affects the ability of entrepreneurs to leverage their local economy for early-stage growth. Branch plant economies with few local supply chains may not be useful customers for early-stage startups if procurement decisions are made elsewhere (Markusen 1996). Similarly, large corporate anchor firms may have a risk-averse organisational culture that discourages working with newer ventures to solve internal problems. Thus, superficially dynamic economies might not create a beneficial environment for high-growth entrepreneurship.

The presence of open markets and early customers, therefore, represents an essential base upon which entrepreneurial ecosystems rest. The local economy is a hugely important contextual factor in how an ecosystem develops and the types of resources and opportunities available to entrepreneurs. An open economy with anchor firms who cultivate local supply chains encourages high-growth entrepreneurship. It removes barriers to growth, which in turn enables other aspects of the ecosystem, such as its culture and networks, and helps build pools of resources like skilled workers and role models. The lack of a sufficient local market may be a major barrier to the creation of a successful ecosystem, as is often the case with ecosystems in emerging economies (see Chapter 5). But the presence of a strong local economy alone does not guarantee the existence of a healthy ecosystem, nor does the absence of a strong economy preclude an ecosystem from developing. Instead, the state of the local economy is one of many contextual factors that constitute the broader ecosystem.

3.3.5 Physical Infrastructure

Physical infrastructure refers to the impact of the built environment on the prospects of high-growth entrepreneurs. This infrastructure includes things like the availability of high-quality office space, intra- and inter-regional transportation like mass transit and airport connections, and digital infrastructure like fibre-optic connections. It can also refer to more specialised infrastructure such as laboratory space or high-performance computers that very advanced firms need in order to develop new products. Such infrastructure may not be the cause of high-growth entrepreneurship, but it is necessary to retain firms as they grow. As with open markets, the World Economic Forum (2013; 2014) has been one of the loudest voices in highlighting the importance of infrastructure.

Geographers and economists have long studied the impact of intra-regional transportation linkages such as highways and

airports on how economies grow. Peripheral places struggle to attract important actors such as investors, even as visitors, and lack of transportation options become a major issue for local entrepreneurs looking to book meetings in other places. While these problems can be ameliorated through things like teleconferencing, they present real barriers to entrepreneurial growth that are often only resolved by relocating to a more globally connected place (Lepawsky et al. 2010).

Office space is another important infrastructural resource. There needs to be a variety of different office space provisions that can meet the needs of scale-up firms, ranging from small co-working spaces for new firms to higher-quality space for rapidly growing companies. Lack of available space may lead to firms either choosing not to grow or relocating elsewhere to access better facilities. This office space must also be affordable and flexible enough for firms that might either quickly grow or just as quickly fail. Accelerators and co-working companies such as WeWork have been instrumental in creating new office space that fits the needs of high-growth firms.

Internet bandwidth is a necessary infrastructure for many high-growth firms. Without high bandwidth, firms will face difficulty innovating new digital products or using internet platforms that have become crucial for marketing and sales. Lack of internet access to economic growth has become a significant barrier to peripheral places that is difficult to overcome (Mack 2012).

Infrastructure is essential, but not deterministic. In some cases, entrepreneurs overcome this with the help of the ecosystem, such as when entrepreneurs sub-lease space from other scale-up firms. But other infrastructural gaps will require public investment to overcome, such as improving mass transit connectivity to increase the available workforce or to invest in new fibre-optic lines to provide high-speed bandwidth. Ambitious, creative entrepreneurs can always find ways around inadequate local infrastructure provision. In the absence of immediate solutions, entrepreneurs are likely to

move elsewhere to places with better infrastructure or simply choose not to grow their firms in order to avoid the challenges of poor infrastructure.

3.4 CONCLUSION: INGREDIENTS AND RECIPES

The research and practice literature has identified a bevy of different actors and factors seen as crucial to the development and reproduction of a strong ecosystem. Such lists are useful starting points to measure and benchmark what ecosystems have and what they are missing. By examining the actors and factors we would expect to find in a healthy ecosystem, we can understand what other places are lacking and understand how their absence affects outcomes like high-growth entrepreneurship and economic growth. This has led to a growth of different lists and diagnostics for ecosystems that aim to help local stakeholders identify missing elements (Hwang and Horowitt 2012; Aspen Network of Development Entrepreneurs 2018; Startup Genome 2018).

However, lists of actors and factors are not a recipe that will always end in success. Inherent in the ecosystem idea is that all of these elements work together both to create ambitious entrepreneurs, support high-growth entrepreneurship, and to reproduce each other over time (Spigel 2017a). The presence of these elements alone does not constitute an ecosystem: they must cohere together to produce a self-sustaining system. There is an emergent process that develops over time through the self-interested and altruistic actions of multiple different actors with their own goals and motivations. One of the most interesting areas of ecosystems research has been histories of ecosystems that show the many different ways that the connection between ecosystem elements emerges and how this has affected its overall operation and performance (Du et al. 2018; Feldman et al. 2005; Mack and Mayer 2016; Thompson et al. 2017).

We must also consider what happens when one or more of these elements are missing or underdeveloped. Ecosystems can and do form in places without particular elements such as anchor firms, supportive public policy, or research universities (Mahroum 2016). But some other elements seem more critical to the formation of a successful ecosystem, such as entrepreneurial culture, skilled workers, and entrepreneurial leaders. However, this may be tautological: if a place has high levels of entrepreneurship, its culture is likely to be entrepreneurial (Stam 2015). An ecosystem does not start with all of these individual elements: some (like an entrepreneurial culture or mentors) are created by acts of successful entrepreneurship. Other elements like incubators, investment or infrastructure can precede high levels of entrepreneurship with sufficient outside support, but they are aided by the legacy of high-growth entrepreneurship as well.

Thus, while looking at the actors and factors of an ecosystem is an important step, it is just the first step. We need to go beyond who and what these factors are to how they work and how they operate in concert to create a self-sustaining system that evolves in response to internal and external changes. This requires looking at the processes and practices that occur in ecosystems and how (and if) these actually support high-growth entrepreneurship.

4. Ecosystem practices and processes

4.1 INTRODUCTION

There is a vibrant literature around the types of localised actors and factors that are theorised to aid the creation and growth of innovative new ventures. But there are many places with many if not all of these actors and factors that nevertheless cannot be described as strong ecosystems. For example, why does Princeton, New Jersey, a place with a world-leading research university, large corporate laboratories, plentiful potential investors, and a highly skilled workforce, still have little high-growth entrepreneurial activity? How can billion-dollar entrepreneurial scale-ups like Red Bull emerge from the global periphery (Thailand) when their founding locations have few of the resources we think are critical to scale-up success? Such cases show that the presence or absence of individual elements alone is not sufficient to understand the existence of a strong ecosystem. To understand the recipes that turn individual ingredients into an effective and resilient ecosystem, we must look at the practices and processes underlying what ecosystems are and how they work.

Ecosystem practices are the activities entrepreneurs and other actors employ as they interact with their ecosystem. Practices such as peer learning (recontextualising the experiences of others to address emergent challenges) and networking (developing and leveraging social connections with other ecosystem actors) are two ways that entrepreneurs are able to

translate the resources in an ecosystem into benefits for their firms. But these practices are not universal and some individuals may be less willing or able to engage in these activities. This means that entrepreneurial ecosystems do not affect everyone in the same way.

Ecosystem processes are the broader social and economic activities that enact, reproduce and transform the ecosystem. These processes take place beyond the level of an individual actor but instead affect broader and less material aspects of the ecosystem such as the local entrepreneurial culture, network structures, or the power of different constituencies in the ecosystem to enact their own vision for the future. But while these processes take place beyond individuals, they are still affected by individual practices and the broader coalitions that emerge amongst ecosystem actors.

Ecosystem practices and processes can be thought of as the glue binding together different ecosystem actors and factors and allows them to produce a cohesive system of support for high-growth entrepreneurship. A practice and process perspective provides a more nuanced approach to studying entrepreneurial ecosystems that allows the field to move beyond simple cataloguing of different actors and factors. This allows us to draw on a varied array of well-defined concepts such as neo-institutional theory (Kuratko et al. 2017) or entrepreneuring (Keating et al. 2014) to analyse different aspects of entrepreneurial ecosystems. This chapter explores different approaches to understanding core ecosystem practices and processes and how they help ground research on ecosystems in existing conceptual frameworks.

4.2 NETWORKING PRACTICES IN ECOSYSTEMS

Networks like those described in the previous chapter are social connections between people that entrepreneurs can use to get the resources they need to start and scale their

firms. Networking is the bundle of practices that individuals enact to build, maintain, and use these connections. The types of networking practices entrepreneurs carry out affects networks characteristics such as density, diversity, and the relative strength of ties. These, in turn, affect the type of entrepreneurial resources they can access through their networks. Networking practices, like all practices, emerge out of a combination of individuals' internal dispositions and psychological traits along with the broader social norms and conditions of their communities (Anderson et al. 2010). Viewing networking as a practice helps explain why the usefulness of entrepreneurial networks may differ between different ecosystems.

Networking practices relate to how network ties are formed, with whom, for what reasons, and how these ties are maintained over time. Such practices can be categorised as: (1) networking broadening in which individuals seek out strangers to add to their networks, (2) network deepening in which they learn more about the competencies and motivations of their contacts and the types of resources they have access to, (3) leveraging of network connections to access advice, support and resources, and (4) maintaining and strengthening these connections over time (Elfring and Hulsink 2007; Vissa 2012; Zheng et al. 2019). These practices are achieved in different ways. Some entrepreneurs might systematically search out for new contacts that they think they need, such as an early-stage entrepreneur seeking out a mentor. Others might scan their pre-existing networks for someone whom they think could provide that same guidance (Hite 2005).

However, research on networking activities has implicitly assumed that they employ these practices in a purposeful manner (Engel et al. 2017). But this is a very anti-social view of the most social of activities. By assuming that networking is always strategic and purposeful, we cast entrepreneurs as economically rational actors who are always calculating the net present value of their social interactions and dropping

those that do not represent value for money (Dodd and Anderson 2007). Entrepreneurs employ networking practices in a state of profound uncertainty over what kinds of support they need and from whom they can get this support. They develop heuristics about what types of networking activities might get them through their current circumstances. This is not to say that entrepreneurs do not engage in purposeful networking but that even strategic networking practices are built on more personal social decisions. For example, many entrepreneurs first draw on their pre-existing social connections when they first start their firm. However, the balancing of how much to draw on these personal contacts for business reasons is a difficult decision.

Networking practices, like the networks they produce, are embedded in local cultural and social contexts. Certain networking practices that are considered normal in one place, such as being strategic in creating or abandoning network connections, might be offensive in other places. But these contexts are not deterministic: individual personality types (introverts versus extroverts), personal situations (e.g. young single people versus entrepreneurs with caring responsibilities), and individual attributes (e.g. gender or age) also affect the types of networking practices entrepreneurs are comfortable engaging in (Anderson et al. 2010; Benschop 2009). Thus, while social structures help determine the types of networking practices that may occur in a region, we should still expect a great deal of heterogeneity within them, providing the seeds for change in developing ecosystems.

Networking practices therefore help constitute ecosystems and are in turn, affected by the ecosystems' nature. They constitute the ecosystems by helping to build the social connections that tie together an entrepreneurial community and mediate the flow of resources and knowledge in the region. As discussed above, there are particular configurations of social networks that are most conducive to high-growth entrepreneurship: dense, trust-based networks between diverse

actors. Practices that encourage this type of network creation, for example by proactively developing new connections, brokering introductions between different people, and being willing to help out others, will be likely associated with the development of stronger ecosystems.

But ecosystems also constitute the types of networking practices that are acceptable within them. The local culture of an ecosystem helps normalise certain networking practices, making them more legitimate and easier to carry out. When successful entrepreneurs build their local networks and act as intermediaries to introduce others, these networking practices will be observed and copied, helping normalising altruistic network-building in the ecosystem. The presence of ecosystem features such as an array of formal and informal networking groups create forums in which entrepreneurs can meet and develop their network (Audretsch et al. 2011a; Shen and Cage 2013). Such events help bring together entrepreneurs and encourage networking practices, over time normalising new ways to leverage personal and business networks to build high-growth ventures (Shen and Cage 2013).

Networking practices illustrate why we would expect to see significant differences between different ecosystems. While networks are crucial aspects of entrepreneurial ecosystems, they are not neutral or inevitable. They are produced by individuals carrying out networking practices based on their own internal logics, which are themselves deeply embedded in localised social norms. Examining how entrepreneurial networking practices emerge and influence the operations of the larger ecosystem is a key way to understand not only how to build effective ecosystems, it also shines light onto deeper questions about the structure and agency in the entrepreneurship process.

4.3 LEARNING PRACTICES IN ECOSYSTEMS

The resources in an ecosystem are often seen as mostly material things like investment capital, university research, or skilled workers. Entrepreneurs use these ecosystem-based resources along with resources obtained elsewhere to identify opportunities, develop their products and services, and otherwise scale their sales and operations. But knowledge is also a fundamental resource for high-growth firms (Alvarez and Busenitz 2001). Most of the management and economic development literature is concerned with technical knowledge, the scientific and technical know-how developed internally or sourced externally that enables firms to develop new products and innovations. Indeed, much of the work on clusters and innovation systems is about how best to produce and circulate cutting-edge technical knowledge (Maskell 2001). But business and entrepreneurial knowledge is just as important (Pugh et al. 2019). This type of knowledge refers to the know-how associated with starting, running, and growing new ventures. This might involve specific skills like how to write a business plan, to more complex processes like how to build trust with customers and anticipate strategic market changes.

Entrepreneurs do not start with this knowledge (Aldrich and Yang 2014). They develop it throughout their lives, from childhood to their work careers. Indeed, entrepreneurs must be effective learners in order to succeed because they are dealing with ambiguous and uncertain situations in which established precedents may not be helpful (Harrison and Leitch 2005). Learning is more than skill development; it involves reflection on personal and business strengths and weaknesses to identify the causes of successes and failures (Cope 2005).

Learning is essential in ecosystems because scale-up firms typically experience similar challenges over their lifecycle. In the beginning, this might be around how entrepreneurs identify opportunities, find co-founders, and get the initial

seed capital they need. As the firm begins to grow, new challenges around marketing, brand development, sales, and managing employees emerge. As firms further scale, founders must grapple with how to manage growth, assemble a board of directors, and potentially how to secure venture capital and achieve a profitable exit. While each firm and industry has its own unique facets, many of these challenges are generic, meaning that most entrepreneurs will face them in one way or another as their firms grow. Indeed, as Autio et al. (2017) argue, digital affordances mean that companies in very different industries have even more in common than even a decade ago. Learning helps entrepreneurs anticipate problems they will likely encounter and develop potential ways to address them.

The circulation of entrepreneurial knowledge in an ecosystem enables founders to anticipate problems they will likely encounter and gives them examples of how others have addressed them. This knowledge constitutes one of the most powerful benefits of being in a strong ecosystem and one that is much harder to provide than other resources such as investment capital or business training.

Entrepreneurs typically learn by either absorbing information from passive sources like books or attending training to develop specific competencies (Henry et al. 2005). This allows them to focus on improving specific skill sets, such as pitching to investors or financial accounting. But entrepreneurs will also learn experientially as well by reflecting on their decisions as they start and grow the firm (Corbett 2005). Through trial and error, they identify techniques that work for given challenges and discard approaches that are not effective (Posen and Chen 2013). Finally, entrepreneurs can also learn vicariously by observing other founders.

This vicarious learning may be one of the most common ways that entrepreneurs learn entrepreneurial skills and internalise the social rules associated with entrepreneurship (Holcomb et al. 2009). This can be done by interacting with

family members and friends who run their own businesses and thereby absorbing lessons about self-efficacy and bricolage (Aldrich and Yang 2014). Others learn through observing more advanced entrepreneurs, trying to identify techniques that have worked for others and using this to develop heuristics about why types of actions might work in different situations (Holcomb et al. 2009). They may also model their behaviour on globally prominent entrepreneurs like Steve Jobs or try to match the performances of entrepreneurs pitching on 'entre-tainment' television shows like Dragons' Den (Swail et al. 2014).

But vicarious learning is usually one-way: an actor observes others and translates and adapt what they see into their own routines (Bresman 2013). Strong, vibrant entrepreneurial ecosystems provide the basis for a different type of learning: Coactive Vicarious Learning (CVL). CVL involves ongoing, discursive interactions between learners and models (for example, a newer entrepreneur and a more experienced one) in which participants discuss the model's experiences, linking it with the learner's own experience to create "emergent, situated understanding of the experience(s)" (Myers 2018, pp. 613–614) This allows learners to do more than copy observed behaviours; their interactions with the more experienced models allows them to understand the logic behind the observed behaviours. This may take the form of an ongoing mentor/mentee relationship, friendship, or more informal conversations between entrepreneurs at meet-ups or cafes. Such interaction allows learners to re-contextualise the experiences of others to develop new solutions to the challenges they encounter. By combining this CVL with their own experience, actors cannot just learn existing solutions but create new, more appropriate practices and reactions.

Learning is a practice rooted in an entrepreneur's own personality and her broader social context as the entrepreneur. The choice to learn a skill by reading about it or taking a training class emerges out of the entrepreneur's choice about

the value of time spent learning as opposed to time spent on the business or other pursuits. Similarly, engaging in CVL depends on a number of networking practices, such as having developed network connections and trust-based relationships with other entrepreneurs and maintaining ongoing relationships. In many contexts, this might involve actively going to meet-ups and other events in the ecosystem to meet other entrepreneurs.

The proximity between entrepreneurs in ecosystems lowers barriers to repeated interactions, meaning that entrepreneurs can more easily find and learn from more experienced model entrepreneurs. But more importantly, the culture of a vibrant entrepreneurial ecosystem should encourage openness to sharing experiences (including failures) and interaction that powers CVL. The cultural structures of a successful ecosystem will help normalise practices like the willingness to be a mentor (Spigel 2017b) or network-building with others in the same industry (Kuhn and Galloway 2013), which in turn enables CVL. In the strongest ecosystems, we would expect that engaging in CVL is a normal, accepted part of the entrepreneurship process, leading to a situation in which entrepreneurial knowledge easily flows through the community. Conversely, CVL will be more difficult in less-developed ecosystems when they have less dense networks between entrepreneurs. Not only will there be fewer opportunities for entrepreneurs to interact and learn from each other, the lower trust in the networks means that each side will be less willing to share their personal experiences. This is particularly true for sharing stories of failures and mistakes, which can be the best learning tools (Bledow et al. 2017).

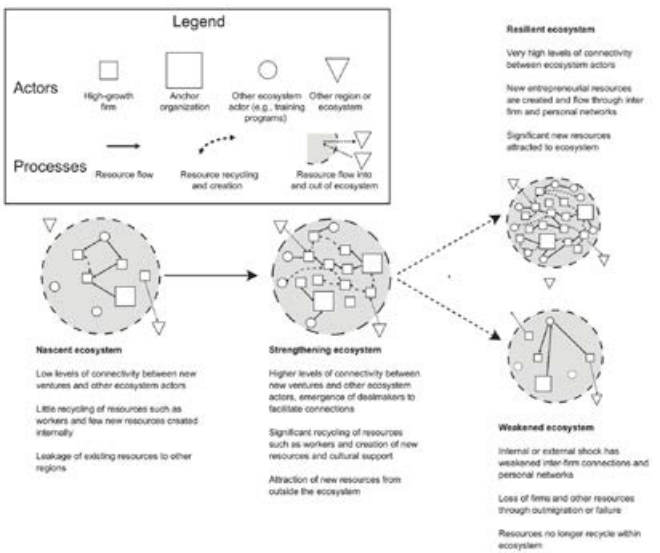
To be clear, there is no hierarchy of learning. All forms of learning enable the entrepreneur to develop new skills and give them new perspectives to solve problems. But while some forms of learning are more or less available to most entrepreneurs (e.g. reading a blog about marketing techniques for or vicariously learning by observing the actions of

a prominent entrepreneur), opportunities for CVL depend on the context of an ecosystem. We would expect that an ecosystem with a supportive, open culture would encourage CVL, making it easier for entrepreneurs to learn about the scale-up entrepreneurship from other entrepreneurs. This, in turn, allows them to learn from their peers and re-contextualise the experiences of others to address an immediate problem. Similarly, in ecosystems with a more closed-off culture, there will be fewer opportunities for network development and sharing, therefore limiting the amount of CVL that occurs. Entrepreneurs in these contexts will need to depend more on other types of learning, with the risk that they may not be able to deal with the issues associated with firm growth.

4.4 PROCESSES OF ECOSYSTEM EMERGENCE AND TRANSFORMATION

The development of entrepreneurial ecosystems is an evolutionary process in which past events influence future actions. Both Brown and Mason (2017) and Spigel and Harrison (2018) suggest development trajectories for entrepreneurial ecosystems in which embryonic (Brown and Mason's terms) or nascent (Spigel and Harrison's terms) ecosystems with low levels of firm creation, connectivity between entrepreneurial actors, and cultural ambition slowly become better functioning ones. These ecosystems can slowly improve the conditions through both concerted work of policymakers and the individual actions of entrepreneurs themselves to become scale-up or strengthening ecosystems (see Figure 4.1). As they develop, the ecosystems strengthen and become better environments for scale-ups as new networks form between entrepreneurs, advisors, and investors and new resources like investment capital and technical knowledge build up in the region. Just as importantly, as the ecosystem strengthens the entrepreneurial actors there begin to develop a collective

identity and language that helps them cooperate, build a community, and lobby for more support (Thompson et al. 2017). This contributes to the creation of a resilient cycle in which the ecosystem enables high-growth entrepreneurship, which in turn helps strengthen the overall ecosystem (Mack and Mayer 2016). But this is an archetypical model: in reality, some ecosystems might strengthen or weaken over time, or the entrepreneurial economy might never cohere into what observers would call a strong ecosystem.



Source: After Spigel and Harrison 2018.

Figure 4.1 The evolution of entrepreneurial ecosystems

Places with an embryonic ecosystem generally stay that way until some triggering event leads to its solidification. This may be a discrete event, such as a large local anchor laying off its workforce which creates a new pool of potential entrepreneurs

(Auerswald and Dani 2017; Feldman et al. 2005). Or it may involve the development of a new policy initiative as well as being the result of a longer-term development, such as the growth of an entrepreneurial venture that helps encourage entrepreneurship within the larger population. It may also spring from a seemingly random event such as a chance meeting between a founder and investor or the decision of a university department to host an event that brings together different actors in the community. Such changes not only lead to an increase in the number of entrepreneurs in a region but also help achieve a shift in consciousness that emphasises the importance of connection and mutual support in the entrepreneurship.

Ecosystem development builds on what already exists in a region. This happens in two ways. First, the initial entrepreneurial activity that sets off the ecosystem building process will likely be related in some way to pre-existing economic specialisations. This creates an arc of development and regions can upgrade their economic activities by taking advantage of existing knowledge bases, infrastructure, and expertise to expand into new, related areas (Boschma 2016). For example, Silicon Valley's digital technology ecosystem developed out of its pre-existing defence-related electronic cluster (Lécuyer 2006) while Austin's ecosystem emerged from a local microelectronics manufacturing cluster (Gibson and Oden 2019). As an entrepreneurial ecosystem begins to develop, new ventures will likely be similar to what already exists in the region because new entrepreneurs are either coming from that sector or observing those existing firms to identify opportunities and potential resources.

Second, pre-existing economic and social trajectories lay down layers of informal institutional and cultural norms that will guide how ecosystems emerge and develop. Informal institutions create shared understandings of how business and entrepreneurial activities operate within a region, which influence how future entrepreneurship is conducted. Such

institutional understandings act to regulate actions, supporting existing regional trajectories (Bathelt and Glücker 2011). In many cases, the institutional norms in an embryonic ecosystem are antithetical to the ecosystem developing further. There may be a lack of trust between entrepreneurial actors, businesses, and government (Williams and Vorley 2018) or a fear of the risks associated with ambitious, high-growth entrepreneurship (Lang et al. 2014). In order for a well-functioning ecosystem to develop, these informal institutions must be overcome or changed to be more supportive of ecosystem activities such as knowledge sharing and cooperation.

Ecosystem development requires institutional entrepreneurs: socially skilled actors who can help forge new understandings of the world (Fligstein and McAdam 2012). These are not limited to firm founders but can be anyone who identifies challenges facing entrepreneurs and builds a community of like-minded people to find collective solutions and create new meanings. This can involve setting up formal groups or informally bringing together a group of people facing similar challenges to create a collective identity and envision a path forward. While the groups or forums these institutional entrepreneurs create are important, their more significant contribution is to shift a place's informal institutional structure towards the trust and cooperation required for strong ecosystems to form. More open norms create the foundation for the more material aspects of an ecosystem such as supportive policies and organisations to build new connections within the high-growth entrepreneurial community.

The processes underlying the emergence and development of entrepreneurial ecosystems are therefore not dependent on new technologies or policies. Rather, an ecosystem develops through processes of social transformation in which networks are built and new institutional and cultural understandings of entrepreneurship are forged. A large part of ecosystem development is creating a supportive cultural environment within which other aspects of the ecosystem can develop. This

is driven not by historical inertia but through the actions of specific institutional entrepreneurs who take an active role in building new communities which eventually shift institutional norms into a more supportive configuration.

4.5 ENTREPRENEURIAL RECYCLING PROCESSES

Firm exit, be it successfully in the form of an acquisition or IPO or unsuccessfully through bankruptcy, are pivotal moments in an ecosystem's development. Success creates new resources in the ecosystem, such as capital in the form of windfall profits for the founders and investors and equips senior employees with knowledge of how to successfully scale a firm. Failure may not create new resources (besides hard-learnt insight about mistakes to avoid in the future), but it frees up labour and capital that can be used by other firms. Mason and Harrison (2006) refer to the flow of resources from exited firms back to the rest of the ecosystem as recycling. Recycling is key for the development and evolution of ecosystems over time. Successful recycling can circulate new resources in an ecosystem to support new rounds of entrepreneurship and cement cultural changes that normalise entrepreneurial growth. On the other hand, lack of recycling can mean that resources such as skilled workers and investment capital leave the region after a success or failure, potentially harming future development.

After a successful acquisition or IPO founders and early employees are able to cash out and use their windfall profits to create new ventures and/or become angel investors for new generations of entrepreneurs (DeTienne 2010). The experience of scaling a firm to the point of exit gives them powerful insight that will aid them in their subsequent ventures and which they can pass on to other entrepreneurs as mentors, advisors or board members (Zahra 2008). Workers leaving a successful firm might use their profits from stock options

as seed capital for new ventures or take what they have learnt from their prior employer as they assume more senior roles at other local firms (Feldman et al. 2019; Zhang 2011). In either case, their association with that successful company gives them entrepreneurial legitimacy that will help their future careers.

Firm failure can also contribute to ecosystem development. The resources of failed and low-growth firms recycle back into the ecosystem to be used by other local firms as the falling firm shrinks and ultimately ceases to exist. Most directly, employees of failed firms will seek employment at other local firms (or leave the region entirely). These workers bring with them potentially valuable know-how and insights that can be useful at other firms. As argued above, because many firms face similar challenges, experience with the problems that lead to the failure of one company can help others avoid or overcome them. Similarly, the founders of these failed firms can learn from their experience, increasing their chances of succeeding in subsequent ventures (Baù et al. 2018). However, the ability of entrepreneurs and workers from failed ventures to recycle back into the ecosystem depends on the presence of a local culture that is tolerant of failure and setbacks. As Bahrami and Evans (1995, p. 81) argue: "the process of 'flexible recycling' is enhanced in the absence of the typical stigma associated with organizational failure."

One of the biggest impacts of recycling from successful and unsuccessful firms is the movement of workers between firms (Spigel and Vinodrai 2020). Workers are an important source of knowledge spillovers within ecosystems, as employees bring with them insights and skills they developed at their previous employer (Wu et al. 2017). The skills developed by working at innovative startups and scale-ups increase the productivity of the firms they subsequently join (Tambe and Hitt 2014). By observing how the firms they worked at grew and what worked (and did not work), they gain a better understanding of the firm growth process that can be applied

elsewhere. Recycling, however, is not limited to the movement of workers between entrepreneurial firms. Workers may move between large anchor firms, universities, or other non-entrepreneurial firms and high-growth new ventures as their career and personal goals shift, and as labour market conditions change. Their experience at larger firms may give them insights into cutting-edge technologies or the connections to access global value chains; attributes which can be very valuable to newer firms that lack these skills in-house.

Recycling is not automatic. Freeing up of capital and people creates the potential for recycling but these resources can also leave the region or simply disappear after firm exits. Employees may migrate to another region after the firm they worked at failed and the capital produced by a successful exit may flow to other places where rates of return might be higher. The quality of life and job opportunities in a region must be sufficient to keep footloose knowledge workers there when they have options to move elsewhere (Florida et al. 2008). For exited founders to re-emerge as investors and mentors, there must already be sufficiently developed pools of younger entrepreneurs for them to support and sufficient social prestige attached to mentorship to make it worth their time and effort. This means that local cultural norms influence recycling. If a place's culture attaches social value to giving back to the local business community by becoming an angel investor, mentor, or dealmaker, these types of activities will be encouraged. Without such social value, successful entrepreneurs might engage in other activities, productive (giving to charity or supporting other community efforts) or unproductive (focusing on their golf game) that do not help the ecosystem develop.

The recycling of entrepreneurial resources back into an ecosystem is, therefore, a socially embedded process. Some places, due to aspects of their local culture, social structures, and community networks, will see more recycling after a successful or unsuccessful exit than other places. Other places

will exhibit less stickiness, meaning that resources and people will flow out of a region after an exit, reducing the capabilities of new rounds of entrepreneurship. This process provides a hint at the temporal dynamics of ecosystems, which are continually changing over time in response to internal and external developments.

4.6 CONCLUSION: ECOSYSTEMS AND PRACTICES AND PROCESSES

It is a mistake to see entrepreneurial ecosystems as a primarily economic phenomenon simply because they are related to a business topic. Entrepreneurial ecosystems are complex social systems that are enacted through self-interested individual action while at the same time developing and evolving through collective action. It is these practices and processes that constitute what an ecosystem is rather than any collection of actors or factors. Indeed, it is these learning and networking practices along with recycling and other development processes that connect individual actors and factors into a more cohesive ecosystem.

This makes understanding ecosystem practices and processes crucial for both ecosystem researchers and ecosystem builders. Researchers need to understand practices and processes because they transcend 'ingredient lists' and help us understand the underlying social organisation of ecosystems. Entrepreneurial ecosystems are interesting examples of the tension between social structure — the influence of social rules and orders — and the individual agency entrepreneurial actors have to enact their own practices based on what they think will work best. Entrepreneurial ecosystems simultaneously influence entrepreneurs but are also created and changed by their actions. A focus on practice and processes also allows us to link to existing research frameworks such as neo-institutional theory (Scott 2013), practice theory (Nicolini 2012), and structuration (Coleman 1990). This helps move

ecosystems research away from thick description towards a more rigorous social theory.

Just as importantly, ecosystem builders need to understand ecosystem practices and processes because it shines light on how ecosystems actually work. Creating resources like groups, investment funds, and educational programs is not enough. Ecosystem development strategies need to fit into existing cultural and institutional frameworks if they are to be effective (Audretsch 2015). Understanding what practices entrepreneurs use in an ecosystem and why they use these practices is crucial for developing new support systems that will work in that particular context. Just as importantly, ecosystem builders must understand the complex social relationships within entrepreneurial ecosystems in order to build a collective vision for what the community really wants.

5. Global entrepreneurial ecosystems

5.1 INTRODUCTION

Entrepreneurship has long been an important economic development mechanism for developing economies in the Global South (McMillan and Woodruff 2002). Productive entrepreneurship produces social goods beyond the profits and employment they create: successful entrepreneurs provide goods and services more efficiently than poorly functioning markets and over time entrepreneurs contribute to the formation of more stable markets and institutions (Baumol 1990; Williams and Vorley 2018). Given this, many governments of developing economies (those with a less-developed industrial base) and emerging economies (those with a more developed industrial base but still low average wages) have embraced policies around supporting more ambitious and high-growth entrepreneurship. This has often included attempts to improve the state of entrepreneurial ecosystems in urban and rural areas. These efforts have attracted substantial interest from the international development agencies of many Western economies such as the United States, where USAID has announced several initiatives with the goal of "catalysing investment, strengthening entrepreneurial ecosystems and working with intermediaries" (USAID 2016) or the United Kingdom, whose international development agency has committed to "catalys[ing] the growth of ecosystems through innovative and investable entrepreneurial ventures" (Department for

International Development 2015, p. 1). Global NGOs like the Gates Foundation have also invested in building a more supportive environment for ambitious entrepreneurship in the developing world (Morris and Török 2018).

The purpose of such efforts is to support high-growth, opportunity-driven entrepreneurship through creating stronger institutional structures for business and through bringing together local and global actors to create the resources needed for firm growth. Support infrastructure like innovation hubs or accelerators have popped up throughout the developing world and attracted attention from visiting entrepreneurial stars like Mark Zuckerberg (Friederici 2019). Building on this interest, several researchers have begun to examine ecosystems in emerging and developing economies. This research has ranged from a broad examination of the strengths of different ecosystem pillars (Arruda et al. 2014; Coduras et al. 2018; Davari and Najmabadi 2018) to individual case studies of ecosystem development in diverse economic settings (Hemmert et al. 2019; Li et al. 2017).

But despite the enthusiasm for building ecosystems in emerging and developing economies, we should be cautious about these efforts. The vast majority of research on entrepreneurial ecosystems comes from the United States and Western Europe. The basic facts of entrepreneurial ecosystems are derived from a limited number of contexts characterised by modern economic systems, strong institutional structures, and fairly homogeneous cultural systems. Even research on ecosystems in non-Western economies such as India (Goswami et al. 2017), Tanzania (Galperin and Melyoki 2018) or China (Du et al. 2018), implicitly assumes ecosystems work as they do in Western contexts.

This raises questions about what can be generalised from entrepreneurial ecosystems theories developed in rich Western contexts to the rest of the word. To what extent do ecosystems work in the same way in different places, and what aspects of ecosystems are related to specific local con-

ditions and histories? The policy optimism in ecosystems in the Global South belies the reality that entrepreneurship in developing and emerging economies works very differently than it does in the Global North countries. We should expect the ways that ecosystems work and the impact that they have will differ as well. This means that not only will ecosystems actors and factors in developing and emerging economies be different from those found in more developed economies, the processes underlying how these support high-growth entrepreneurship, and the economic and social value this type of entrepreneurship creates, will also be different. These differences have major implications for ecosystem policy, calling into question the extent to which ideas created in the context of the rich Anglo-American and European countries can be applied in other contexts and how they should be adopted to new economic and social environments. Studying ecosystems in emerging and developing economies does more than just test how productive ecosystems can be in these contexts. By exploring the nature of ambitious entrepreneurship outside places like America, we get a better insight into what aspects of the process are related to universal aspects of capitalism and which are socially embedded and politically contingent.

5.2 GLOBAL ECOSYSTEMS IN CONTEXT

The economic, cultural, and institutional contexts of emerging and developing economies affect the entrepreneurship process, which in turn changes how entrepreneurial growth impacts society. This includes everything from the availability and type of opportunities entrepreneurs can pursue to the way networks and social capital are used to how wealth is appropriated from entrepreneurial endeavours. Therefore, to understand entrepreneurial ecosystems in the context of less-developed economies we need to understand what these contexts are and how they affect the entrepreneurship process.

There are four clear differences in the relationship between society and entrepreneurship between developed and developing/emerging economies that are important for understanding how entrepreneurial ecosystems work in these contexts: (1) rates of opportunity and necessity entrepreneurship, (2) differing economic structures, (3) differing levels of institutional development, and (4) provision of important entrepreneurial resources. These differences shift the underlying connection between society and entrepreneurship, in turn affecting the ability of supportive entrepreneurial ecosystems to form and more fundamentally, it affects the relationship between entrepreneurial activity and overall societal prosperity.

5.2.1 Opportunity and Necessity Entrepreneurship

The most visible difference in entrepreneurship between developed and developing economies is that while poorer countries often have high rates of entrepreneurship, it is predominately necessity entrepreneurship undertaken due to a lack of other opportunities for employment (Naudé 2009). Necessity-driven entrepreneurs are often in non-tradable sectors such as small-scale retail or services. These businesses have low growth prospects and produce few, if any, knock-on effects for the broader economy. There is a very clear negative correlation between GDP per capita and the total entrepreneurship rate as measured by the Global Entrepreneurship Monitor. There is a similar relationship between GDP per capita and necessity entrepreneurship, with richer countries having substantially less due to both their more dynamic economies and better social safety nets.

Necessity entrepreneurs are often informal, meaning that they are not registered with the government, do not pay taxes, and lack protection of (and from) the law. It is important to note that while the poorest countries do have higher rates of opportunity-driven entrepreneurship than more developed economies, this type of entrepreneurship declines rapidly as

the economy grows, suggesting that entrepreneurs are quickly pushed out of market niches once they are profitable for larger firms.

Because entrepreneurial ecosystems are focused on supporting the scale-up entrepreneurship process, they have minimal impact on low-growth entrepreneurs. While all entrepreneurs will benefit from some outcomes of ecosystem development, such as improved infrastructure and formal institutions, much of the ecosystem does not affect lower-growth entrepreneurship. Necessity entrepreneurs do not need growth capital from local investors (though micro-credit providers may be helpful), skilled employees or research produced by universities, nor are they likely to take advantage of support programs like incubators. While some interventions can help micro-entrepreneurs achieve a measure of growth and help them expand to the point where they may be able to hire a few employees and formalise their operations, it is difficult to convert a micro-enterprise into a high-growth one (Morris and Török 2018). There is little systematic evidence of interventions that can help micro-entrepreneurs develop and implement scalable business models and operations that would allow them to achieve the job-creating growth associated with high-growth entrepreneurship in developed economies.

5.2.2 Differing Economic Structures

Part of the difficulty in the ability of generating and supporting high-growth entrepreneurship in developing and emerging economies comes from the nature of their economic structures. The vast majority of ecosystem research (and entrepreneurship research more generally) takes place in the context of developed, innovation-driven economies. These economies prize new value creation based on technological and market innovation (Baumol et al. 2007). The economies of these countries are large enough so that entrepreneurs can create value and move beyond the immediate market into other

sectors. This creates the potential to create impactful, scalable business models that in turn lead to the job and wealth creation that are the goals of entrepreneurial ecosystems.

The economies of the least developed countries are fundamentally efficiency-based: firms compete on their ability to capture benefits from low labour and input costs in order to compete on price (Acs et al. 2008). Examples of this include low-skill textile manufacturing in Bangladesh or maquiladoras in Latin America of the 1970s and 1980s. Low labour costs, large pools of untapped labour, and a lack of regulations represent opportunities for entrepreneurs. While entrepreneurial growth does create new jobs, the effect on overall social welfare is limited due to the low wages and poor working conditions. This can be a boon to local entrepreneurs who have the initial capital and connections to create businesses that exploit these advantages, but such conditions are not sustainable. Over time, wages increase and firms can lose their cost competitiveness to other, even less-developed economies.

More developed emerging economies — such as those in East Asia, Eastern Europe, and Latin America — eventually lose the ability to compete on low factor costs alone and instead must compete on labour and production efficiency (Acs and Amorós 2008). Such economic systems are even less favourable to entrepreneurs because achieving these efficiencies requires economies of scale gained by large production runs and the use of capital-intensive equipment. This requires substantial up-front capital, managerial ability, and technological skills that are difficult for entrepreneurs to acquire. However, domestic markets are not yet well-developed enough to create many opportunities for new, value-creating solutions that can be exported to other countries. This economic structure depresses high-growth and scalable entrepreneurship in emerging economies by creating new barriers for finding exploitable opportunities (Acs et al. 2008). It is hard for entrepreneurs in these types of economies to find domestic market niches where it is possible to rapidly scale. The lack

of South-to-South markets (for example, a Brazilian company exporting its products to Indonesia) means it is difficult to find international markets that would allow them to achieve these economies of scale.

5.2.3 Resource Constraints

By definition, developing and emerging economies lack the economic resources of more developed economies. This includes financial resources such as bank loans, risk capital, human capital due to the lack of an educated workforce, lack of research universities producing cutting-edge knowledge, and business know-how regarding issues such as internationalisation or product development. These constraints increase the price for entrepreneurial resources and reduce founders' ability to source high-quality advice about managerial practices and growth strategy (Bloom and Chatterji 2009). The extent to which this impacts entrepreneurship varies massively between and within countries. Global cities such as Shanghai, San Paulo or New Delhi have access to many of the same entrepreneurial resources as places like New York City or London (Sassen 1991), but more peripheral places in emerging and developed economies may have very limited access to the resources needed by entrepreneurs.

Lack of easily obtainable resources force entrepreneurs to adopt new strategies and practices of resourcefulness to combine different assets (Welter et al. 2017). This often takes the form of bricolage — making do with whatever is on hand, refusing to accept the limitations imposed by resource constraints, and combining resources together in new ways (Baker and Nelson 2005). But while this type of resource ambidexterity is at the very heart of entrepreneurship, lack of dependable access to resources makes it difficult to plan sustainable business growth. This means that even in places with ample opportunities and skills, entrepreneurial growth is still difficult.

5.2.4 Institutional Development

Finally, poorly developed formal institutional structures are common in emerging and developing economies and hamper high-impact entrepreneurship. Poor rule of law, the inability to enforce ownership rights, and the absence of protection for property rights make it difficult for entrepreneurs to attract investors and retain competitive advantage against copycats (Vorley and Williams 2016; Williams and Vorley 2018). Ineffective governance due to corruption or simple lack of resources makes it hard to provide public support for entrepreneurial training or labour force development. The lack of institutional rules and protections increase the costs of entrepreneurship, making smaller ventures, formal and informal, less competitive versus larger domestic or foreign firms (Brixiova 2011).

This has a number of knock-on effects for the formation and sustainability of an entrepreneurial ecosystem. First, institutionally thin places are likely to have a large pool of unofficial or unregistered firms. This drive to informality may be due to onerous requirements for registering new firms, corruption in governments or a lack of capacity in government agencies. Non-registration means that they do not pay taxes nor are they eligible for public support or even the protection of law. Because of this, these firms have low growth prospects, as growth would attract attention of the state and lead to civil or criminal penalties (La Porta and Shleifer 2014). Even without this threat, lack of legal registration also makes it hard to borrow money from banks or receive public services such as utilities or internet connectivity.

Without a thick network of formal and informal institutions, it is hard for the different elements of a strong entrepreneurial ecosystem to take hold. In such situations, it is likely that even if new support programs were introduced (for example, a new innovation centre financed by an NGO), their effectiveness would be limited because its operations depend on the behav-

iour of entrepreneurs, which is in turn linked to their understanding of the formal institutions about how entrepreneurship works in their context (Friederici 2018).

Given these issues, we cannot expect entrepreneurial ecosystems in developing and emerging economies to work in the same way as ecosystems in more developed economies. The influence of economic structures, resource availability, and institutions make the types of entrepreneurship found there and the entrepreneurship process itself fundamentally different than in the more commonly studied Global North. In the same way that entrepreneurial firms are not smaller-scale versions of big firms, ecosystems in emerging and developing economies are not resource-poor versions of more developed ecosystems in the West. Indeed, it should not be surprising that emerging economies do not produce the same types of high-growth firms as in developed countries because the nature of high-growth entrepreneurial ventures in economies like the US are shaped by its economic and institutional characteristics.

5.3 ECOSYSTEMS IN DEVELOPING ECONOMIES

Our perspective on entrepreneurial ecosystems in factor- and efficiency-driven economies might be improved if we expand our focus to include productive entrepreneurship rather than limit it to high-growth entrepreneurship. Baumol (1990) famously distinguished between productive, unproductive, and destructive entrepreneurship. Productive entrepreneurship is socially beneficial, adding to new economic output of the overall economy. Unproductive entrepreneurship may generate wealth for the entrepreneur but does not spur broader benefits for society, while destructive entrepreneurship actively harms the fabric of society through both legal but rent-seeking entrepreneurship as well as illegal entrepreneurship such as criminal activities. The distribution of entrepreneurial activity

between these three categories depends on a society's institutional framework (Lucas and Fuller 2017). Strong institutions and positive cultural attitudes normalise productive entrepreneurship while other institutional arrangements might make unproductive and destructive entrepreneurship more individually profitable than productive entrepreneurship.

But the most significant differences between ecosystems in developed and those in developing and emerging economies may be the practices and processes that underlie how the ecosystems emerge and function. Practices in more peripheral ecosystems may differ from those in more developed ones due to the lack of formal access to financial and business resources as well as the absence of pre-existing institutional structures. Productive entrepreneurial practices may be expected to aid the entrepreneur through profits as well as produce broader social benefits such as spillover effects or to help build the ecosystem over time. By focusing on ecosystem practices and processes rather than only support structures or pillars, we can develop a more nuanced understanding of the emergence of entrepreneurial ecosystems in emerging economies. In the following sections, we review the recent research on entrepreneurial ecosystems in China, Africa, and in the post-socialist economies of Eastern Europe to better understand what types of ecosystems form in these very distinctive economies and the extent to which policy initiatives and grassroots activities have shifted to work in their unique contexts.

5.3.1 China

The emergence of healthy entrepreneurial ecosystems in Chinese cities such as Shanghai, Beijing, and Hong Kong present interesting examples of ecosystems developing in very different cultural, institutional, and economic conditions than in the US or Western Europe. While China's rapidly growing economy has created many entrepreneurial opportunities, poor institutional structures such as a lack of copyright

protection and low trust between different business actors create barriers for entrepreneurial growth. By studying the emergence, structure, and impact of Chinese ecosystems we can begin to see which aspects of ecosystems are connected with universal aspects of the entrepreneurship process and which are based around locally specific contexts.

The paradox of Chinese ecosystems is that while the country has a generally supportive environment for high-growth entrepreneurship in the sense that entrepreneurs there have access to dynamic domestic markets, investment capital, and a strong educational system, it also suffers from the absence of an underlying supportive, trust-based culture for innovative entrepreneurship (Hemmert et al. 2019). This has led to entrepreneurs struggling to build cohesive networks with each other and underdeveloped pools of mentors and advisors who could help entrepreneurs avoid the challenges associated with growth. While the Chinese notion of *guanxi* — social relationships dominated by reciprocity and favour granting — has been instrumental for building global business networks (Hsu and Saxenian 2000), it has proved harder to construct more informal networks of support and peer learning in Chinese ecosystems. This has made the emergence of entrepreneur-led ecosystems in China difficult.

As a result, ecosystem development in China has been more state-led than ecosystems in the US or Western Europe. National and provincial governments have taken a much more active role in cultivating ecosystems through investments in universities or public-run business support organisations (Chen et al. 2019). In their study of the emergence of Zhongguncun in Beijing, one of China's most vibrant entrepreneurial ecosystems, Du et al. (2018, p. 1166) refer to this as "spontaneous emergence under government guidance." In the case of Zhongguncun, after initial grassroots community building helped transform a small electronics mall in suburban Beijing in the 1980s into a cluster of technology startups in the early 2000s, the national government stepped in to provide

funding for more formal organising and a business support infrastructure (Li et al. 2017).

But while Zhongguncun benefited from being a government showcase for the emergence of Chinese-focused technology entrepreneurship, other Chinese ecosystems have had less success in developing strong formal and informal scale-up support infrastructure. While government support has helped create new incubators and science parks, there are few examples of the collaboration and mutual support that characterise the strongest ecosystems in other countries. The lack of trust within business communities is a significant issue, reducing the capacity of groups of entrepreneurs to come together to work towards creating a mutually beneficial environment (Puffer et al. 2010). As a result, ecosystem development is less organic in China than it is in Western countries, dependent on government direction.

Nevertheless, there is evidence that even in the absence of grassroots community effort, entrepreneurial success still creates the conditions that support future entrepreneurial success. In a country-wide study, Lai and Vonortas (2019) find that high levels of tech-based entrepreneurship at an initial period lead to sustained increases in human capital, business and technology knowledge, and finance, that in turn increase subsequent levels of tech-based entrepreneurship. Thus, there is evidence of the emergence of sustainable entrepreneurial ecosystems in China, even if they look quite different from the originally studied ecosystems in the West. Chinese ecosystems have lower levels of inter-personal networks connecting entrepreneurs and rely much more on state intervention to provide startup and scale-up support, but nevertheless, they still aid the creation and survival of high-growth entrepreneurial ventures.

5.3.2 Africa

Many African countries have counterintuitively high rates of opportunity-driven entrepreneurship. GEM data show that averaged across all years, African countries have the highest rate of entrepreneurs who expect to create more than five jobs over the next five years and the second-highest total entrepreneurship rate, just behind South America. High rates of job-creating entrepreneurship are the result of rapid economic growth in many African countries, caused by increasing global trade, the end of long-lasting conflicts, and the emergence of a new middle class. There is, therefore, the potential for significant entrepreneur-led economic development in Africa based on local entrepreneurs serving local markets. One of the best examples of this is the mobile phone-based payments technology M-Pesa, which now has more than 20 million users across Africa.

When discussing African entrepreneurial ecosystems, it must always be stated that Africa is a continent of fifty-four countries whose economies vary from globally competitive emerging markets (South Africa, Nigeria, Egypt) to amongst the poorest in the world. There is no one typical African ecosystem. Nevertheless, there are common themes in how entrepreneurial ecosystems are forming in various African countries related to their state of economic and institutional development, the types of entrepreneurial resources available, and those that are missing. This creates commonalities over how entrepreneurial ecosystems develop and their contribution to the feasibility of entrepreneurship-led economic growth.

There is substantial excitement for the prospect of entrepreneurial ecosystems in developing African nations, where, "policymakers, corporations, development organisations, consultancies, and the media […] frame domestic digital economies as a significant source of economic transformation and revival" (Friederici 2019, p. 10). Success stories like

Nairobi's emergent Silicon Savannah have helped popularise the notion of African technology entrepreneurship as an economic development path (Bramann 2016). There has been substantial interest in how African nations can technologically leap-frog more developed nations who are locked into existing infrastructure and therefore enable a new generation of African entrepreneurs making products for African markets (Mureithi 2016).

One of the most concrete ecosystem development actions by governments, NGOs, and investors in African countries has been the establishment of technology hubs, with more than 100 established by 2016 (Kelly and Firestone 2016). Technology hubs are not necessarily incubators (which provide subsidised space for entrepreneurs) or accelerators (which are curated, time-limited growth programs). Their closest analogue in the West are co-working spaces. However, technology hubs play an important symbolic role in normalising digital entrepreneurship (Friederici 2019). Their goal is not only to provide suitable office space for digital companies but also foster the creation of a community of digital entrepreneurs and spur the emergence of local entrepreneurial cultures and norms. Their presence helps overcome a lack of pre-existing networks between entrepreneurs and helps create a space for learning and interaction that can support entrepreneurs in a very uncertain environment.

But despite this excitement, there are significant challenges facing entrepreneurial ecosystems in Africa. Most African economies face a shortage of business and technical talent, little financial capital, and limited domestic markets, all of which make it difficult for ambitious entrepreneurship to thrive (Bramann 2016). While many countries in both North and sub-Saharan Africa have generally supportive cultural views of entrepreneurship, governmental and institutional quality are low, with little trust present in business networks (Coduras et al. 2018). This makes it difficult for governments to kickstart ecosystem development as they have else-

where. As Sheriff and Muffatto (2015, p. 42) argue: "actors like members of the private sector, entrepreneurs, public opinion leaders and non-governmental organisations … are rarely engaged in the process of creating entrepreneurial ecosystems."

Lack of capital, both local and through international sources such as outside investors, cuts off growth opportunities even when entrepreneurs have identified a strong opportunity (St-Pierre et al. 2016). Low penetration of universities and other higher educational institutions has made it hard to develop integrated training initiatives to help develop entrepreneurial skills amongst founders (Maas and Herrington 2011). Even technology hubs themselves have underperformed expectations, with many tenants focusing their efforts on less innovative areas such as web-page design which produces little growth potential (Friederici 2018). Due to lack of government funding and inconsistent international grants and investment, support and financial infrastructure is often weak with large gaps in provision. Given these issues, African entrepreneurial ecosystems outside of the most globalised cities like Johannesburg and Cape Town will look very different from their counterparts in the Global North. Ecosystem mapping projects in countries such as Nigeria, Rwanda, and Ghana have found that much of the ecosystem activity takes place outside of digital sectors, a good deal of which occurs outside urban areas (Aspen Network of Development Entrepreneurs 2018).

Entrepreneurial ecosystems can and do exist in Africa, but they are less digital than in other places with instead a greater concentration on agriculture, manufacturing, or resource processing. But this does not mean that such entrepreneurs would not benefit from an ecosystem. Beyond the obvious improvements entrepreneurs in developing African economies would see from improved ecosystem attributes such as better physical infrastructure and stronger governmental institutions, the cultural and social elements of an ecosystem are also

beneficial in circulating entrepreneurial know-how and skills. But it is not clear if the significant differences in resources, institutional quality and structure, and cultural outlooks from the Global North will mean that the development of African entrepreneurial ecosystems will have less immediate economic impact.

5.3.3 Post-Socialist Economies

Since the breakup of the Soviet Union, there has been a sustained research and policy interest in entrepreneurship in the former socialist countries of Eastern and Central Europe. Not only is entrepreneurship seen as an important economic development strategy, entrepreneurship also indicates a cultural shift from a state-directed society and towards a more liberal, market-driven culture (Fritsch and Storey 2014). There has been substantial research and policy interest on how to promote high-growth entrepreneurship in these countries, particularly those like Poland, Hungary, and Romania which have since joined the European Union (Szerb et al. 2018). But while these nations may have larger economies and stronger global trade ties than other economies in Africa or Latin America, they still have weak institutional structures and norms that create challenges for ecosystem development.

Substantial trust and openness between different entrepreneurial actors is a fundamental requirement for creating a cohesive entrepreneurial ecosystem. But the normalisation of corruption as an everyday part of the entrepreneurship process in places like Romania and Bulgaria makes this difficult. While some types of common corruption (e.g. payments to bureaucrats to expedite government approvals) can be seen as simply a stealth tax, larger-scale corruption and patronage enables businesses to engage in rent-seeking monopolistic activity (Vorley and Williams 2016). While resourceful and nimble small-scale entrepreneurs can exist and even thrive in such contexts, this makes growth beyond certain points

particularly difficult to navigate without making concessions to criminal enterprises or corrupt officials (Welter et al. 2017). Lack of institutional trust makes ecosystem development difficult. Pervasive, low-level corruption lowers trust between actors, meaning that it is difficult to build the informal networks that enable entrepreneurial learning and resource sharing (Williams and Vorley 2018).

But even in the face of these issues there has still been large investments from global NGOs and state actors like the European Union to support ecosystem development efforts (Velt et al. 2018). While these policies have increased overall startup rates, their ability to actually contribute towards building sustainable entrepreneurial ecosystems is questionable (Brooks et al. 2019). Many of these programs focus on trendy, cutting-edge industries such as virtual reality or data science and ignore local specialisations which are often in older, more traditional sectors such as agriculture, manufacturing, or outsourced business services. The deliverables and goals of these projects are driven by expansive, Europe-wide economic development strategies like smart specialisation, which may actually undermine grassroots, entrepreneurial-led ecosystems. That is to say, while outside investments in ecosystems brings in new resources such as new funding mechanisms or support organisations, they often lack connections with local cultural norms and communities that allow them to have their intended impact.

But this is not to say that ecosystem development is impossible in post-socialist economies. Cities like Tallinn and Sofia have built strong entrepreneurial ecosystems by combining their historically strong mathematics and engineering educational systems with a new generation of entrepreneurs (Audretsch and Belitski 2017; Velt et al. 2018). Estonia in particular has emerged as one of the most entrepreneurial nations in Europe, with the highest startup rate per capita on the continent and is the birthplace of globally known software firms such as Skype. However, ecosystems in post-Soviet

countries demonstrate the pervasive influence of institutional structure and culture on how ecosystems emerge and operate. Substantial investment in different support programs cannot overcome issues around inter-personal and governmental trust. Top-down ecosystem development efforts are limited in circumstances where they cannot work in conjunction with the more informal efforts of the entrepreneurs themselves.

5.4 POLICY MOBILITY AND THE POLITICAL ECONOMY OF ECOSYSTEMS

Ecosystem development in places like China, Africa, and Eastern Europe show that there is a distinctive political economy to their implementation. Investments in entrepreneurial infrastructure and support are linked with broader economic and political considerations. This is true for emerging and developing countries in the Global South as well in more developed countries such as de-industrialised cities in the United States (Roundy 2017). In these areas, ecosystem development tends to be led in a top-down fashion by different regional, national, and supra-national organisations in order to address problems of resource scarcity and weak institutional foundations. This differs from the 'canonical' view of ecosystems, in which grassroots communities emerge to address overarching challenges through collective action.

But attempts to build ecosystems around the world show the tendency to try to import not just best practices from well-established Western ecosystems but also more fundamental ideas about what ecosystems are and how they work. Despite pleas from Isenberg and Feld about the impossibility of building new Silicon Valleys, examples from around the world show attempts by governments and NGOs to foster the same actors and factors that are found in the most developed ecosystems. While sufficient investment can build new accelerators, technology hubs, or universities, it has proven much

harder to build the relational connections between these ecosystem elements they cohere and allow for the flow of entrepreneurial resources and knowledge that enables high-growth entrepreneurship. There is a need to critically examine how ecosystem ideas move and how this mobility affects their implementation in many different contexts.

The literature on critical policy mobility is a useful framework to understand these issues. Entrepreneurial ecosystems are part of a larger category of 'fast policies,' which are rapidly implemented in places around the world through the observation of best practices, developed in tandem between public and private parties to fast-track decision making. Fast policy employs experimentation and benchmarking to quickly identify and roll out of successful strategies (Jessop 2007). In the case of ecosystems, policy ideas are circulated by global NGOs like the Kauffman Foundation and the World Bank who organise conferences to share best practices, events like the Global Entrepreneurship Congress which brings together interested parties, private consultancies like Startup Genome who produce lists of top global ecosystems, entrepreneur/authors like Daniel Isenberg who publish books and hold classes on the subject, and researchers who publish scholarly books exploring the politics and practice of ecosystems. Together, these actors form an epistemic community who "transfer, emplace, and utilise" knowledge about entrepreneurial ecosystems around the world (McCann 2011, p. 112).

But it is too simplistic to say that ecosystem ideas diffuse out of a few central entrepreneurial nodes in the US to the rest of the world. Policy ideas about ecosystems are translated to new locations by actors with their own agendas, transforming to fit within local contexts and political constraints (Peck and Theodore 2015). The core ideas of ecosystems — that high-growth entrepreneurship matters, that its surrounding community supports this entrepreneurship, and that there are different types of policies and networks that help this form of entrepreneurship — circulate through entrepreneurial

communities in cities all over the world. These ideas are not transferred in a pure form. They mutate according to the needs of the context they are implemented in and the goals and outlooks of the people in charge of implementing them. For example, Chinese ecosystems are largely centred around top-down government intervention, an organisational structure antithetical to Feld's vision of how ecosystems should work. But they still retain the importance of policy, culture, finance, and mentorship.

MIT's Regional Entrepreneurship Acceleration Program (REAP) is a good example of how ecosystem knowledge is produced and diffused. The program admits cohorts of local leaders from around the world, ranging from developed economies in Scotland, Iceland and Australia to emerging and transitional economies like Russia, Ghana, or Peru. There, delegations learn from experienced practitioners and academics who introduce the REAP Framework for ecosystem building as well as through visits to Boston's vibrant ecosystem. Teams develop plans to measure institutional and individual variables in their ecosystem in order to better understand the regional pillars required for ecosystem development. While the knowledge circulated at the program is highly useful, its association with a prestigious university (MIT) based in a world-class entrepreneurial ecosystem (Boston) gives participants power and legitimacy to transform the ideas into action.

The implementation and transformation of ecosystems ideas are influenced by the data available on entrepreneurial activity. Modern policy initiatives prioritise using data to measure effectiveness and impact. Ideally, this is done through carefully controlled experiments to establish the causality of the policy impact, but due to the expense of these efforts measurement is often done through surveys matched against key data points like the number of new firms or attendees at different events. Some aspects of entrepreneurial ecosystems are easily measurable, such as the flow of venture

capital investment, the number of jobs created at firms that receive public support or increases in the overall startup rate. But other aspects of ecosystems are harder to measure, from the density of social networks between entrepreneurs to cultural views of entrepreneurship to the challenges of identifying high-growth firms. Support programs tend to orient themselves towards supporting aspects of an ecosystem that can be easily measured, as this allows these programs to justify their operations and their budget better. More difficult to measure — but arguably more important — aspects of the ecosystem may end up with little support because such support is difficult to justify and validate empirically.

From a policy mobility perspective, we should be asking not just where new ecosystems are being developed, but also how they are being developed, why they are being developed, and by whom. Rather than focusing on how ecosystems in emerging economies or peripheral regions differ from the largest ecosystems on which the theories were developed, we should instead examine how these differences emerge, given both the pre-existing localised economic and social contexts and the power relations underlying the ecosystem emergence. This allows us to 'theorise back,' to use the characteristics of ecosystems in places like China, Latin America, or Eastern Europe to understand and critique the basic ideas and discourses of what entrepreneurial ecosystems are and how they work (Yeung 2007).

5.5 CONCLUSION: ECOSYSTEMS IN PLACE

Entrepreneurial ecosystems outside the traditional context of urbanised Western economies are important for two reasons. First, they help establish if and how ecosystem policies can support larger economic development efforts based on harnessing local entrepreneurial talents. While entrepreneurship is not the only economic development path that can be taken,

a thriving entrepreneurial ecosystem is self-sustaining in a way that other types of aid-based economic development are not. It builds on local specialisations while relying less on difficult to finance infrastructure projects or politically sensitive developments. Second, studies of global ecosystems help explain how ecosystems work. By shifting the focus away from resource-rich, institutionally strong places to less well-endowed places, we begin to see what aspects of entrepreneurial ecosystems are rooted in universal notions of Schumpeterian capitalism and which are derived from contingent and place-specific contexts. For example, networks are crucial for entrepreneurs to gain access to the resources they need to build fast-growing firms. However, different contextual factors such as cultural attitudes and institutional structures affect the ability of entrepreneurs to build networks, and the availability or absence of groups or meeting places affects their ability to build networks. Similarly, the capacity of governments to effectively support entrepreneurial ecosystems through interventions and policies depends on the nature of the context they are being implemented in.

Even though studies of non-Western ecosystems are still in their infancy, we can already see some important points of difference. Research on the challenges of building ecosystems in China and their dependence on top-down state action suggests that strong ecosystems are not an emergent property of lots of high-growth entrepreneurship. High-growth entrepreneurship can exist without it causing an ecosystem similar to what we see in the West to develop. There are cultural and economic structures that are important for bringing together the independent actions of many different actors and orienting them towards a common goal.

But ultimately it is the practices and processes in ecosystems that matter for the actual economic and social impact of ecosystems. When studying ecosystems in emerging and developing economies, it is important to go beyond listing what resources they have and those that are missing and

instead examine how their context impacts what these practices and processes are and how they impact the entrepreneurship process. This means that we should not treat these ecosystems as less-developed versions of stronger Western ecosystems. Instead, they should be treated as their own, unique entities that do not have to be identical to other ecosystems from vastly different contexts. The literature on policy mobility is one framework to understand how ideas about what ecosystems are and how they work mutate as they go between contexts, but other approaches such as the international political economy are also valuable. The key is to understand global ecosystems for what they are rather than expecting that they replicate a model that emerged from particular places at particular moments in history and which should not be seen as being the only model.

6. Critiques of entrepreneurial ecosystems

6.1 INTRODUCTION

We can break the critiques of ecosystems into two categories: critiques of ecosystem theory and critiques of ecosystems themselves. Critiques on ecosystem theory focus on the issues raised throughout this book: the lack of a coherent and logical conceptual framework for ecosystems, the tension between how policymakers and researchers understand ecosystems and the lack of predictive power in existing ecosystem frameworks. These issues are not unique to ecosystems: similar criticisms were levelled at industrial clusters and innovation systems during their ascendance (Markusen 1999; Martin and Sunley 2003). Addressing these critiques creates a research agenda for ecosystems.

The second type of critique is more pernicious: that ecosystems themselves are part of a broader system of neo-liberal capitalism that does not produce the social value it claims to. From this perspective, ecosystems are not the solution to a problem but instead a manifestation of a modern capitalist system that shifts risk towards the individual and forces cities to compete against other places for scarce investment. This perspective critiques ecosystems as just the latest attempt to normalise a particular form of neo-liberal capitalist relations that will do little to achieve the putative goal of increasing overall social welfare.

This chapter reviews both types of critiques. The goal is not to systematically address and dismiss each criticism: many of them are accurate and must be addressed through both more research and engagement. Instead, the goal is to highlight weak points in how ecosystems are theorised and implemented in order to establish a coherent research agenda.

6.2 CONFUSED THEORY

One of the most common issues raised with the current ecosystems literature is that there is no coherent theoretical framework. Entrepreneurial ecosystems are more of a "conceptual umbrella encompassing a variety of different perspectives on the geography of entrepreneurship" than it is a singular framework describing the geography of scale-up entrepreneurship (Spigel 2017a, p. 49). The most obvious manifestation of this confusion is the lack of a shared definition of entrepreneurial ecosystems. Some authors (and this book) use Stam's (2015) definition: the interconnected actors and factors in a place that supports high-growth entrepreneurship. But as was shown in Table 1.1, there are a variety of different definitions. While some of these definitions cluster closely, others differ in important ways, often around the specific geography at play and the outcome of interest. This is particularly true for research on entrepreneurial ecosystems in developing economies, where there is less high-growth entrepreneurship and the absence of institutional and political foundations taken for granted in Western notions of ecosystems. Without a shared definition, it is difficult for the research community to have a coherent conversation about what ecosystems are and why they matter.

Even more concerning than the lack of agreement over what ecosystems are is the lack of agreement over how ecosystems work. Theoretical frameworks abstract away from specific cases and make more generalisable points about how the world works (Shepherd and Suddaby 2016). A theory of

entrepreneurial ecosystems allows us to make predictions about how ecosystems work in different contexts that can be empirically tested. While models for ecosystems exist, such as Isenberg's or Stam's, these describe the actors and factors at play rather than how these combine to support high-growth entrepreneurship. They describe the what and the who but not the how or the why. Without a theory of ecosystems means that it is difficult to go beyond describing the phenomenon. The process and practice perspective of ecosystems discussed in Chapter 4 is a way to move beyond ingredient lists of actors and factors and develop a theory of how ecosystems work. This allows for a more nuanced understanding of not just what ecosystems are but how they work.

However, this does not mean that ecosystems research needs its own bespoke theory to explain the social processes that underlie it and the practices that it leads to. There is much to gain by linking ecosystems perspectives with existing theories of social action and organisational behaviour. A few researchers have applied existing theoretical frameworks like social field (Thompson et al. 2017) or institutional theory (Steinz et al. 2016) to understand ecosystems. But more is needed to translate complex social theories to this new domain. While ecosystems obviously draw on other concepts like clusters and innovation systems (see Chapter 2), these ideas have been applied in an ad hoc manner instead of given a systematic treatment to transpose them to a new conceptual context. But existing theories like neo-institutionalism provide a powerful framework to understand how entrepreneurial communities emerge and create the collective meanings needed to build a thriving ecosystem.

Regardless, if we are talking about developing a novel theory of ecosystems or applying an existing one to ecosystems, a core theoretical framework is necessary to answer important questions such as what happens when an ecosystem is missing attributes or why places can overall have strong ecosystem attributes but still have low levels of high-growth

entrepreneurship. A cohesive theoretical framework is necessary to move beyond descriptions of cases or statistical analysis of a few variables to make more generalisable points about how being near other entrepreneurs and entrepreneurial actors and being embedded in a wider support network supports the scale-up entrepreneurship process. Without a framework, it is hard for research to evolve beyond thick description or overly parsimonious mathematical models.

6.3 LITTLE RELATIONALITY

Numerous critiques of ecosystems note the lack of relationality in current work (Alvedalen and Boschma 2017; Brown and Mawson 2019; Stam 2015). There has been a limited discussion of the connections between different factors and actors in ecosystems and how these relationships produce the effects of an ecosystem. These relations are one of the unique aspects of ecosystems that distinguish it from clusters, innovation systems, and other concepts. While some authors like Spigel (2017a) have explicitly highlighted the relational connections between ecosystem attributes, and indeed used these relationships to explain their reproduction and transformation, this is not the case for all research. As a result, there is a significant body of work on ecosystems that fails to engage with what makes ecosystem research distinctive.

This is the result of two interlocking factors: methods and data. First, many quantitative research methods seek to isolate the effects of individual variables rather than see their connections. This means that most research on ecosystems using regression techniques (e.g. Bruns et al. 2017; Content et al. 2019; Horváth and Rabetino 2018) face difficulty in seeing the connection between variables. While methods such as structural equation models (Audretsch and Belitski 2017) or interaction effects (Vedula and Kim 2019) can identify how some variables relate to each other, they are limited in how they can address a complex system with no clear causal paths.

Second, the relationships themselves are subtle and hard to detect with existing data sources. Ecosystem elements like culture, networks, and informal institutions are difficult to measure, and their impact on other aspects of the ecosystem are hard to observe. The lack of quantifiable data on aspects of ecosystem factors such as culture, networks, and institutions makes it difficult to draw out the connections between them. Even qualitative methods such as retrospective interviews and observations face difficulty identifying the often invisible connections that link ecosystem elements. No matter the approach, it is hard to systematically study how different aspects of ecosystems interact and affect each other.

A similar issue can be seen in the lack of research on the connections between entrepreneurial ecosystems themselves. Ecosystems are often seen as isolated cells, which might receive resources from outside but which lack deeper connections with other regions. But this flies in the face of a long legacy of work on relational economic geography, which has demonstrated the importance of connections between regions as well as the connections within them (Bathelt et al. 2004; Bathelt and Glücker 2011). Every entrepreneurial ecosystem exists within a web of connections to other places, connections that allow for the flow of ideas, resources, and people. We are now beginning to see research that engages explicitly with this notion. For example. Schäfer and Henn (2019) note the circulation of entrepreneurs from Tel Aviv's ecosystem to American ecosystems such as New York City or San Francisco, where they develop their contacts and skills before returning to Israel to continue to grow their ventures. Similarly, ideas flow in from other places, either in practical business wisdom gained by observing globally famous tech firms or in how local norms of entrepreneurship are constructed from observations of global tech centres like Silicon Valley (Spigel 2017a). However, empirical work on ecosystems has tended to treat them as isolated units, ignoring this aspect of relationality.

Addressing this problem requires a complex systems approach. While individual studies may want to focus on a particular aspect of an ecosystem — such as its entrepreneurial culture, the role of a university or anchor firm, or social networks — these must be viewed within the wider context of the ecosystem rather than isolated from it.

6.4 RESEARCH LED BY POLICY

An important critique of the current ecosystems literature is that the research community is merely trying to catch up with policymakers' enthusiasm for the topic, rather than policymakers' decisions guided by rigorous social science research. Of course, the extent to which policy decisions are ever led by academic research is questionable. However, the critique is a useful reminder that researchers and policymakers often have different understandings of what ecosystems are. As a result, it may be that researchers are investigating a very different phenomenon than what policymakers and practitioners are implementing. In this case, the ultimate impact of research on entrepreneurial ecosystems is unclear.

Brown and Mawson (2019) argue that ecosystem policies often focus on startups and very early-stage enterprises rather than the scale-up firms that ecosystems theory says they should focus on. When policymakers think of ecosystems, they tend to think of them as tools to boost the overall startup rate rather than as a way to support a narrower band of high-growth entrepreneurship. Consequently, many ecosystem policies end up being just a re-implementation of older entrepreneurship policies under a new name, ignoring important aspects of ecosystem theory such as the different support needs of high-growth firms and the importance of entrepreneur-led communities. This leads to "profound conceptual ambiguity surrounding [entrepreneurial ecosystems], creating something of a 'comprehension mistake'" (Brown and Mawson 2019, p. 359).

A similar issue is visible in how many new ecosystem development policies are designed and implemented in a top-down manner by state actors rather than being made by and for entrepreneurs. This is often the case in emerging economies like China but can also be seen in many city-regions in the US and UK. In these cases, ecosystem ideas and discourses are used to justify new public intervention to support scale-up entrepreneurship, but the policies are created and implemented without the input of entrepreneurs and other ecosystem actors. This results in policies that only acknowledge public support through formal funding mechanisms and which ignore informal groups and networks (Spigel 2016). We can see this in increasing references to an 'entrepreneurship support ecosystem' rather than entrepreneurial ecosystem (Markley et al. 2015). Often, support ecosystems ignore the informal sources of support that scale-up entrepreneurs see as most important. This is most often a consequence of a lack of institutional capacity to bring together different ecosystem actors and build a collective mission and goal.

These issues raise an important question: what is the point of ecosystems research if it does not influence policy? The reality is that state action alone cannot build ecosystems. Ecosystems develop through the actions of entrepreneurs, investors, business leaders, and government officials. While explicit ecosystem development policies can play a part of this, policies are just one manifestation of an ongoing process of ecosystem development rather than the entire ecosystem. Addressing this critique requires both more engaged scholarship in which we study ecosystems as they exist rather than their platonic ideal. This requires working with a diverse set of ecosystem actors to understand how they view their ecosystem and help them identify the type of entrepreneurial community and environment they want to create. But it also requires more critical scholarship, where researchers question the way things are being done and suggest alternatives.

It may be that the real contribution of ecosystem research is not in precise policy recipes but instead, it can help instil a sense of humility about the impact of purposeful policies. Histories and case studies of ecosystems show that ecosystem formation takes decades and statistical analyses suggest a relatively limited effect from direct interventions. Rather than a single path or model, research has revealed the sheer complexity of ecosystems and the multitude of different forms they can take. While there are some generalisable conclusions — the importance of human capital, networks, and entrepreneurial talent — there is no single path to recommend.

6.5 ECOSYSTEMS AS A NEO-LIBERAL PROJECT

A more fundamental critique of ecosystems is that they are simply a new manifestation of neo-liberalism. Neo-liberalism is a general term for the organisation of society around self-regulating markets and the submission of social goals to economic logic (Peck and Tickell 2002). Neo-liberalism represents a fundamental shift in how society is organised, reducing state welfare systems and redistributive tax systems in order to ensure a flexible, available labour force and increase consumption in order to sustain growth. Discourses of entrepreneurship and the entrepreneur are frequently used to justify these changes: neo-liberal reformers claimed that reducing red tape and taxes would unleash entrepreneurial potential and create wealth. This was as true in the developing world — in which entrepreneurs would help build the economy and more fundamental social institutions — as it was in developed economies where entrepreneurs became a new generation of heroes and leaders.

Neo-liberalism changed how city-regions themselves operate. Economic and social shifts led to a hollowing out of many cities in the US, UK and Europe along with the loss of redistributive taxes which reduced the capacity of cities to

carry out their normal functions such as maintaining infrastructure and providing social services. This led to the growth of urban entrepreneurship, in which cities compete with each other to attract corporate headquarters, R&D sites, and branch offices (Harvey 1989). Competitive battles between cities were waged by investing in quality of life increases, by building the city in which the young creative employees prized by firms would want to live (Florida 2002), as well as by offering incentives to companies to relocate. Because of these incentives, cities have reoriented their goals towards creating better environments for businesses. This contributed to what Rossi and Di Bella (2017) term 'startup urbanism:' an economic development discourse that centres entrepreneurs as the key growth engine of twenty-first-century cities. This discourse is spread and empowered by think tanks and urban planning gurus, giving urban politicians power to implement pro-business policies. Such policies go beyond just startup support, but also include fundamental changes in the urban landscape such as strategic investments in new transportation infrastructure and changes to urban planning regimes and tax policies (McNeill 2016).

Entrepreneurial ecosystems emerged at the confluence of these two inter-related processes. First, neo-liberalism contributed to the importance of high-growth entrepreneurship by reducing barriers to growth and valorising entrepreneurship, while at the same time eliminating other avenues of social reform. Indeed, the structural shifts that have occurred in the labour market have required workers to become more entrepreneurial (Cockayne 2016). Second, entrepreneurial cities reoriented themselves towards startup urbanism as a means of economic development and growth (Fiorentino 2018). Cities need to make themselves attractive to potential entrepreneurs and provide them with the resources they need to start and grow their firms. And when the firms have scaled, they must continue to create a conducive environment for these entrepreneurs in order to retain them, for example through tax

breaks or more favourable zoning laws (McNeill 2016). In this context, entrepreneurs become a key political constituency, able to leverage their social power and potential mobility to obtain new forms of government support unavailable to other parts of the economy.

Under this system, the urban social welfare goals are subsumed to economic development goals centred on entrepreneurial activity. This might be acceptable if entrepreneurial activity was linked to broader public benefits (e.g. Stam 2015), but this is rarely the case. There is little evidence that technology entrepreneurship improves wages for those outside the sector or reduces overall economic inequality (Lee and Clarke 2019; Lee and Rodriguez-Pose 2013). Unlike manufacturing firms, digital and technology firms rarely have local supply chains nor do they have many semi-skilled jobs, which reduces their spillover effects on the larger regional economy. In places like San Francisco and Berlin, technology entrepreneurship is associated with rapid increases in housing prices that spurs gentrification and displaces long-term residents (McNeill 2016). The structural changes wrought by neo-liberalism mean that there is less redistribution of the wealth generated by high-growth entrepreneurship: it is shared by founders and investors with less and less of it being taxed by the government to provide social services.

6.6 CONCLUSION: NEW THINKING ON ECOSYSTEMS

Entrepreneurial ecosystem research will only progress through critique. The rapid growth of the concept within research and policy circles has brought lots of new ideas, but they lack coherence and conceptual rigour. These critiques are not fatal: no theory is born whole but instead requires a period of development and debate until a core framework emerges. The critiques on the ecosystem idea discussed here — the confusing theory, the lack of relationality, and the mismatch between

research and policy — can be addressed through a deeper engagement with what makes ecosystems work and what differentiates them from other entrepreneurship and economic theories. These critiques point to areas where more research and development is needed to refine the idea of concepts and make it more conceptually robust.

The critique of ecosystems themselves as a neo-liberal project is more salient. There must be more critical perspectives that engage with ecosystems as a discourse and political idea as well as a more 'neutral' theory of entrepreneurship. The question of who ecosystems are for and who benefits from them is an important area of inquiry. Similarly, the role of gurus like Brad Feld and Daniel Isenberg in the spread of these ideas must be questioned in the same way the discourses and goals of creative economies were questioned (Peck 2005). Particularly, researchers need to interrogate the relationship between high-growth technology entrepreneurship and the putative goals of ecosystems policy: job creation, expansion of the tax base, and an overall increase in the social welfare of a community. This critical research may lead to an expansion of who ecosystems are for by exploring how social entrepreneurs might benefit from a high-growth ecosystem and if this might lead to a better impact on society.

7. Conclusion

7.1 WHY ECOSYSTEMS MATTER

Entrepreneurial ecosystems are not a new paradigm. Researchers and practitioners have long agreed that some places are better for ambitious entrepreneurs than others. That there are some aspects of places, from culture to infrastructure to demographics, that help firms rapidly scale. This firm growth, in turn, helps create new jobs, builds the local tax base and leads to other socially beneficial outcomes for the broader region. Decades of statistical research on the heterogeneous geography of entrepreneurship as well as dozens of case studies of the histories of successful entrepreneurial places have shown that there are some sets of social, economic, political, and cultural elements that catalyse a particular kind of innovative entrepreneurship. The ground truth suggests that place matters and it has been the goal of researchers and practitioners to generate an understanding of what aspects of a place matter, what aspects can be changed through concerted policy and community effort, and what aspects are more or less fixed. So, by itself, entrepreneurial ecosystems do not say much that is fundamentally new.

What is important about entrepreneurial ecosystems is that they bring together often unconnected strings of research and policy into a more cohesive vision of how entrepreneurs affect and are affected by the place they start and grow their business. Ecosystem ideas draw on insights from fields such as geography, sociology, business research, economics, business history, and psychology. This allows ecosystems to provide

a new way of understanding the complex interactions between people, organisations, institutions, and cultures that promote high-growth entrepreneurship. Rather than seeing one group or element as dominant, the focus is now on how a wide array of actors and factors interact to produce an overall support environment for entrepreneurial innovation and growth.

Within this framework, entrepreneurs are not passive actors soaking up an ecosystem and outputting new ideas, products and jobs. Instead, they are active participants as both users and as its architects. Through their networking and learning practices, they connect to other actors in their ecosystem which allows them to obtain the resources, knowledge and other types of support they require as their firms start and grow. As architects, many entrepreneurs form informal or formal communities that allow them to share insights about the scale-up process and lobby for the types of support that they need.

But entrepreneurs' activities do not arise in a vacuum. Their willingness to engage with their ecosystem — and the willingness of others to engage with them — is affected by the ecosystem's cultural context. Local cultures are not hard and fast rules but a social framework in which ecosystem actors understand their roles and what types of practices, such as mentorship, investing, or mutual support, are seen as a normal part of the entrepreneurship process. It is this interplay of individual agency cast within the larger cultural, economic, and political structures of their local environment that make ecosystems a novel field of research.

Based on this, the book has made three linked arguments. First, ecosystems are distinctive from progenitor concepts such as industrial clusters, regional innovation systems, and entrepreneurial environments. The specific focus of ecosystem theory on high-growth, scale-up firms and its explicit acknowledgement of actors' agency to shape their environment as well as being shaped by it allow it to ask and answer fundamentally new research questions about the entrepreneur-

ship process and set the stage for a reinvigorated debate over the socially embedded nature of entrepreneurship.

Second, the book has argued that research on ecosystems is more than lists of important actors and factors. Defining what entrepreneurial ecosystems are and using detailed case studies and broad statistical analysis to identify the most important people, groups, and regional attributes of successful ecosystems was an important first step. It gives researchers and practitioners a shared language to understand the phenomenon and a starting point to base new research on. But this is just the first of many steps. The most recent research has begun to investigate the practices and processes that constitute ecosystems and deliver the benefits of being located in a strong ecosystem. A focus on process and practice creates opportunities to bring in other theoretical perspectives such as neo-institutionalism and entrepreneurship-as-practice, which enrich our understanding of entrepreneurial ecosystems.

Finally, the book argued that because ecosystems are inherently contextual, ecosystems around the world will not resemble typical model ecosystems like Silicon Valley. While practitioners like Isenberg and Feld have frequently said that places like Silicon Valley cannot be replicated, we must also acknowledge that ecosystems need not look like Silicon Valley or any other US or European model. Each ecosystem is a specific embodiment of its own particular history and context. Very different organisational structures and internal dynamics can lead to similar outcomes due to the political, social, or economic situation on the ground. This is exemplified in the difference between entrepreneurial ecosystems in China, which have developed and operate in a top-down manner through government intervention, and more organic, bottom-up ecosystems (though which still involve public support) found in the US and UK. While the outcomes — support for high-growth entrepreneurship — are similar, the manner in which the ecosystem achieves this differs significantly. The goal of ecosystem development requires careful

consideration of the local context, while still acknowledging the importance of seeing how other places have achieved their goals.

From these arguments, we can draw out why studying and understanding entrepreneurial ecosystems is important for academic researchers, policymakers, and entrepreneurs themselves.

7.2 WHY ECOSYSTEMS MATTER FOR RESEARCHERS

Entrepreneurial ecosystems matter to entrepreneurship and regional economic development researchers because it allows us to ask new questions about the entrepreneurship process, context, and ultimately about modern capitalism itself. By bringing together different viewpoints from several social science disciplines, it provides new perspectives on the socially embedded nature of high-growth entrepreneurship and the relational connections between actors and factors that produce a supportive (or non-supportive) environment. This allows us to see entrepreneurship as more than an economic phenomenon as it is often understood in the literature, but to position it at the intersection of economic, psychological, sociological, geographic, and policy analyses.

By drawing on these viewpoints together, ecosystems open the door to new research questions. These include questions about specific aspects of the entrepreneurship process such as how entrepreneurs' engagement with their ecosystem changes over the course of the firm's lifecycle and how the use of an ecosystem is affected by firm factors like sector or technological sophistication and founder attributes such as gender, ethnicity, and ambition levels. It also allows new questions around regional economies such as the different development paths ecosystems can take and if and how high-growth entrepreneurial activity creates positive externalities for the overall economy or if it leads to increased spatial inequality through

gentrification and displacement. Some of the most interesting new questions invited by ecosystem perspectives are around the complex relationships between ecosystem elements. There are no simple causal paths between localised social structures like culture and the actions of ecosystem actors but rather complex, dynamic inter-connections that change over time and in response to endogenous and exogenous forces.

But answers are also important! By conceptualising high-growth entrepreneurship within its geographic context and as embedded within local networks and social systems, we can make predictions about how and why the entrepreneurship process will differ between places and the implications this has for firm and regional outcomes. There is already a well-established literature based on identifying the most crucial regional actors and factors for promoting specific outcomes like increased high-growth entrepreneurship. As ecosystem research progresses, we will generate even more robust findings about the regional conditions most associated with innovative, high-growth entrepreneurship. This will help identify the generalisable aspects of ecosystems as opposed to those that are contingent and context-dependent.

The entrepreneurial ecosystems literature is part of a broader contextual turn within business studies and the social sciences more widely. This contextual turn offers an opportunity to enrich phenomenon-based topics like entrepreneurship with new perspectives and offers an alternative to positivistic, universal theories of entrepreneurship. It is this opportunity for new perspectives of firm formation and growth that makes ecosystems an exciting development for the advancement of the interdisciplinary entrepreneurship literature.

7.3 WHY ECOSYSTEMS MATTER FOR POLICYMAKERS

Entrepreneurial ecosystems matter to local policymakers because it suggests a path to harnessing the power of entre-

preneurship to support regional economic growth. It is cities rather than nations that have become the primary sources of economic development capacity. Their narrow remit for a single region means that they can build on specific local resources, specialisations, and resources to develop a customised strategy to build new jobs and prosperity (Audretsch 2015). However, traditional strategies such as attracting large international firms through tax incentives or workforce development programs are increasingly precarious. As the bidding process for Amazon HQ2 in 2018 showed, the cost of attracting major employers is increasing beyond what cities can afford (Nager et al. 2019). In Asia, Eastern Europe and Latin America, city leaders realise that they can no longer rely on cheap labour to grow the economy and must find new solutions.

Tapping a place's native entrepreneurial potential is an alternative to expensive smokestack chasing or cluster development strategies. Rather than trying to choose winning companies or sectors to subsidise, local policymakers instead can focus on building the best possible environment for entrepreneurs in the belief that local entrepreneurs will be able to take advantage of this to build innovative new businesses. By focusing attention on high-growth firms, policymakers recognise that such firms are responsible for a great deal of the job creation that occurs in a place. This does not mean that only high-growth firms need support, but rather it acknowledges that high-growth firms need different types of support than lower-growth firms or those whose growth has plateaued.

Entrepreneurial ecosystems are a framework for developing such a supportive environment. By defining the types of actors and factors commonly associated with strong ecosystems, city leaders are able to identify the resources they have and the weaknesses that they will need to address. Ecosystem building does not require the same investments that other types of ecosystem development strategies require: there is rarely the need to build new highways or science parks. Ecosystem

builders focus on bringing together a community of different entrepreneurial actors and helping create bridges between unconnected groups. In some cases, this means encouraging existing constituencies to be more entrepreneurial, for example a chamber of commerce appealing to senior business people to consider becoming angel investors or mentors. Other times it might involve the use of public resources to attract new resources to the region that are missing, such as organising public financing programs or trade missions to new markets. The goal of these actions is not to just check off a list of actors and organisations but to help spur a more entrepreneurial culture and outlook that enables individual actors to create their own initiatives and groups.

It is here where ecosystem approaches diverge from other economic development approaches. The role of the state is to coordinate and facilitate an ecosystem rather than create and control it. Ecosystems show the importance of policy humility. Top-down action from the state alone does not build sustainable ecosystems. Rather, ecosystem builders need to work to cultivate existing communities and help create a common vision and goal amongst the region's pre-existing business networks. Policy must work with the goals of existing stakeholders if it is to be effective. While this does not mean that policy must be passive, it does mean that policy must have a different orientation to outcomes than other economic development strategies.

7.4 WHY ECOSYSTEMS MATTER FOR ENTREPRENEURS

Ecosystems matter to entrepreneurs because they are affected by an ecosystem even if they do not care about what it is or how it works. Entrepreneurs are influenced by their ecosystem regardless of whether they engage with it or not. At the very least, entrepreneurs in weak ecosystems will struggle to get the resources and support they need while it will be more

readily available for entrepreneurs in more munificent ones. Understanding what entrepreneurial ecosystems are and why they matter helps entrepreneurs shift from being affected by it to a position where they can become ecosystem architects and help create a better community to achieve their goals.

The most important element of ecosystem theory for entrepreneurs is learning. By realising that most of the challenges scale-up entrepreneurs face are not new and have been faced by others, entrepreneurs can gain access to a wealth of experience tied up in their local networks. It can help them address the near-universal challenges that all entrepreneurs will face, such as how to hire and fire employees, manage cash flow, and market their company. The knowledge in their network may also help them address challenges specific to their place and market, such as how to appeal to particular local investors or if a support program is worth participating in. But this knowledge is not free-flowing; it is contained within the social networks connecting members of the ecosystem, meaning that both the ecosystem's cultural and social characteristics and an individual entrepreneurs' own predilections will affect how accessible this knowledge is.

Fundamentally, entrepreneurial ecosystems are by and for entrepreneurs. Ecosystem ideas counter the notion that founders are on a solo journey and shows the importance of their connections to their local community and to global markets. This does not mean that strong ecosystems ensure entrepreneurial success, but understanding their position within an ecosystem helps entrepreneurs see the types of support available to them and how these can help their firm. Understanding what ecosystems are and how they work also empowers entrepreneurs to take ownership of their ecosystem. Instead of seeing it as simply a network of support organisations provided by the government, understanding how ecosystems work demonstrates the importance of entrepreneurial leadership to bring together a community and identify the most pressing needs entrepreneurs face. Entrepreneurs can use their

skill at organising resources to address market gaps to create formal or informal groups aimed at addressing these issues and lobbying the state for targeted support.

Understanding ecosystems helps entrepreneurs better understand the value ecosystems have for them as well as understanding the collective value created by continued ecosystem engagement and leadership. The ecosystem cannot exist without entrepreneurs, and it cannot persist without their consent and guidance. This makes awareness of what ecosystems are and how they work crucial for entrepreneurs.

7.5 FUTURE RESEARCH DIRECTIONS

Entrepreneurial ecosystems is a vibrant, interdisciplinary field that holds great promise in what it can contribute to both research on entrepreneurship but also for how we understand modern capitalism and society. Research on entrepreneurial ecosystems has the potential to expand to new conceptual domains. One of the most interesting areas for future work may be social enterprise ecosystems: the extent to which high-impact social entrepreneurs benefit or do not benefit from being in a strong commercial ecosystem. This is an important area of research because it sheds light both on the extent to which ecosystems transcend the different needs of commercial and social entrepreneurs in the same way they can transcend differences between commercial sectors. Such research not only tells us more about ecosystems work, but it also help us better conceptualise the similarities and differences between social enterprises and commercial entrepreneurship.

As argued in Chapter 5, studying ecosystems outside of core urban regions in the Global North is essential for advancing the field. This work must go beyond extending existing models of ecosystems onto emerging and developing economies but rather understanding how entrepreneurs build their community and draw on local support and resources in

these very distinctive contexts. By building new models of ecosystems, we can begin to see which aspects of ecosystems are related to universal aspects of the entrepreneurship process and which are connected with local specificities. This advances the study of entrepreneurial contexts and by theorising back from these new contexts to the core of the Global North, we gain a better understanding of the generalisability of ecosystem theory.

Finally, there is a need for more critical research into ecosystems. Entrepreneurial ecosystems are not neutral nor non-political. They emerged at a particular historical moment and they are supported by many organisations as part of their larger missions to sustain and improve a particular form of entrepreneurial capitalism. Entrepreneurial ecosystems are without a doubt part of a broader neo-liberal mission and must be understood in this context. Critical investigations into ecosystems are important for revealing the power relationships inherent in how they are created and who benefits from entrepreneur-led economic growth. Feminist and post-structural perspectives are also necessary for exploring who does not benefit from ecosystems and identifying how and why entrepreneurs are excluded due to factors like gender, age, and ethnicity. This is a necessary counter-balance for work that treats entrepreneurial ecosystems as an inherently good thing.

Whatever directions the field develops in, the contextual, holistic nature of entrepreneurial ecosystems is a useful addition to how researchers study the entrepreneurship process. While there is undoubtedly the need to build more rigorous conceptual and methodological foundations for ecosystems, it helps us understand the heterogeneous geography of high-growth entrepreneurship and introduces a new way to build strong, sustainable regional economies. This makes ecosystems a worthwhile phenomenon to study.

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